FINAL
TRIM AND STABILITY BOOKLET
OF
KB-23
(YARD NO. CHISTI/010)

SEE LETTER : E-98728-129636

APPROVED
FOR COMPLIANCE WITH 2008 IS CODE REQUIREMENTS

ON BEHALF OF GOVT. OF INDIA

INDIAN REGISTER OF SHIPPING
MUMBAI
DATE: 20 Jun 2020

Rev.	Description	Date	Prepared	Review

Issued for : Approval

Client:	Project:
K.B. SHIPPING & CO.(JAMNAGAR)	**PONTOON (YARD NO.CHISTI/010)**
Builder:	
CHISTI MARINE ENGINEERING WORKS	

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GENERAL PARTICULARS

Name of the Vessel	KB-23
Client	K.B. SHIPPING & CO. (JAMNAGAR)
Builder	CHISTI MARINE ENGINEERING WORKS
Yard No.	CHISTI/010
Type of Vessel	PONTOON
Registry No.	JMR-0011
Class Notation	卐SU, PONTOON, INDIAN COASTAL SERVICE DESCRIPTIVE NOTE “FOR CRANE OPERATION” “DECK STRENGTHENED FOR 10T/M^2”

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	17.000 M
Depth (Mld.)	03.000 M
Draft (Mld.)	2.200 M
Draft (Ext.)	2.212 M
Displacement(Ext.)	1679.61 T
Lightship	389.39 T
Deadweight	1290.22 T
Gross Tonnage	616
Net Tonnage	185

Stability Criteria

REGULATION

Intact Stability Criteria in accordance **IS CODE 2008**

1) The area under a righting lever curve up to the angle of maximum righting lever should not be less than 0.080 meter-radian.

2) The static angle of heel due to a uniformly distributed wind load of 0.54 kPa (wind speed 30 m/s) should not exceed an angle corresponding to half freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.

3) The minimum range of stability should be:
For L <= 100m: 20^0
For L >= 150m: 15^0
For intermediate length by interpolation

Intact stability criteria during cargo lifting

The following intact stability criteria are to be complied with:

- $\theta C \leq 15°$
- $GZC \leq 0,6\ GZMAX$
- $A1 \geq 0.4\ ATOT$

where:

θC : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 1)

GZC, GZMAX: Defined in Fig 1

A1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle equal to the lesser of:

- heeling angle θR of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 1)
- heeling angle θF, corresponding to flooding of unprotected openings

ATOT : Total area, in m.rad, below the righting lever curve.

In the above formula, the heeling arm, corresponding to the cargo lifting, is to be obtained, in m, from the following formula:

$$b=(Pd-Zz)/\Delta$$

where:

P : Cargo lifting mass, in t

d : Transversal distance, in m, of lifting cargo to the longitudinal plane (see Fig 1)

Z: Mass, in t, of ballast used for righting the pontoon, if applicable (see Fig 1)

z : Transversal distance, in m, of the centre of gravity of Z to the longitudinal plane (see Fig 1)

Δ : Displacement, in t, at the loading condition

The above check is to be carried out considering the most unfavourable situations of cargo lifting combined with the lesser initial metacentric height GM.The vertical position of the centre of gravity of cargo lifting is to be assumed in correspondence of the suspension point.

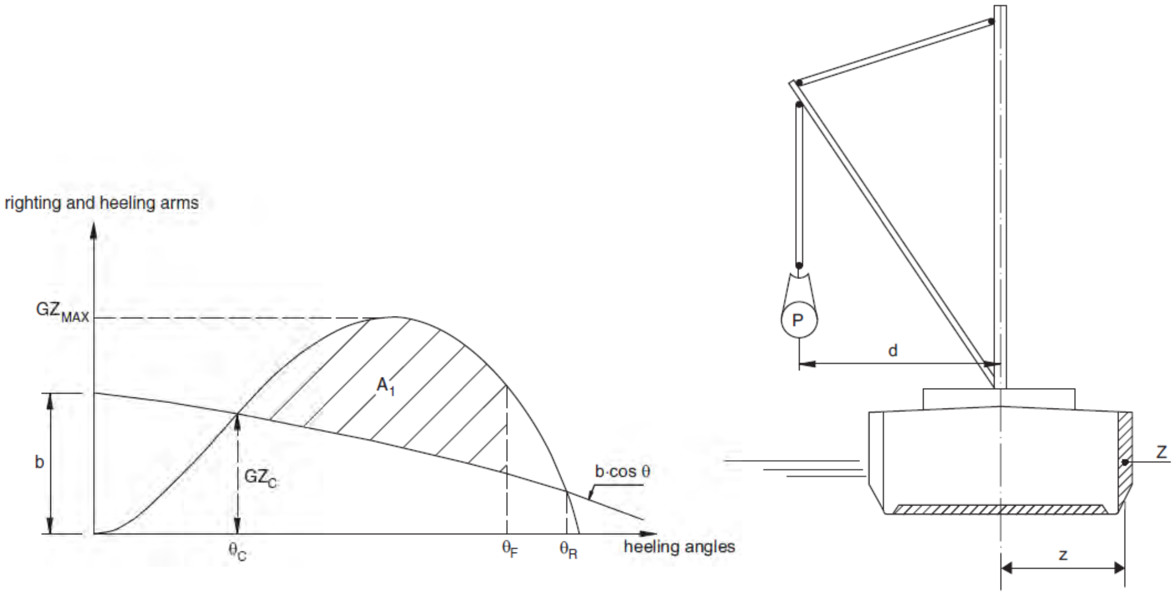


FIGURE 1: CARGO LIFTING

<u>Intact stability criteria in the event of sudden loss of cargo during lifting</u>

This additional requirement is compulsory when counterweights or ballasting of the ship are necessary or when deemed necessary by the Society taking into account the ship dimensions and the weights lifted.

The case of a hypothetical loss of cargo during lifting due to a break of the lifting cable is to be considered.

In this case, the following intact stability criteria are to be compiled with

- $A2 / A1 \geq 1$
- $\theta 2 - \theta 3 \geq 20°$

where:

A1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ1 to the heeling angle θC (see Fig 2)

A2 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle θ2 (see Fig 2)

A3 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle $\theta 3$ (see Fig 2)

$\theta 1$: Heeling angle of equilibrium during lifting (see Fig 2)

$\theta 2$: Heeling angle corresponding to the lesser of θR and θF

θC : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 2)

$\theta 3$: Maximum heeling angle due to roll, at which A3 = A1, to be taken not greater than 30° (angle in correspondence of which the loaded cargo on deck is assumed to shift (see Fig 2)

θR : Heeling angle of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 2).

θF : Heeling angle at which progressive flooding may occur (see Fig 2)

In the above formulae, the heeling arm, induced on the ship by the cargo loss, is to be obtained, in m, from the following formula

$b = Zz \cos\theta / \Delta$

HEELING AND RIGHTING ARMS

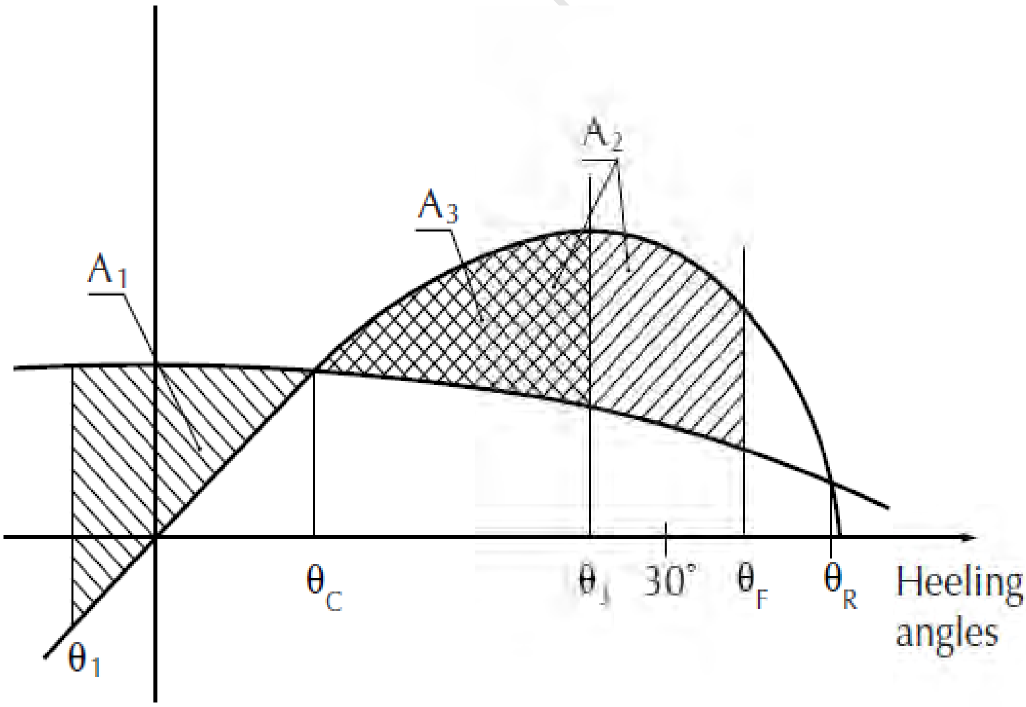


FIGURE 2: CARGO LOSS

Notes to the Master

General precautions against capsizing

1. Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.

2. Loading conditions shown in this booklet are typical of those which may be encountered in service and are worked out to be used for guidance only. Further the booklet contains sufficient data to enable the ship's staff to compute the ship's stability in various loading conditions.

3. Excessive trim by forward is to be avoided in all cases as this may lead to difficulties of maneuverability and course keeping as also infringe the minimum Bow-height requirements.

4. All longitudinal levers, namely LCB, LCG, and LCF are computed with reference to the A.P of Ship.

5. In this booklet hydrostatic particulars and cross curves are given at following trims.

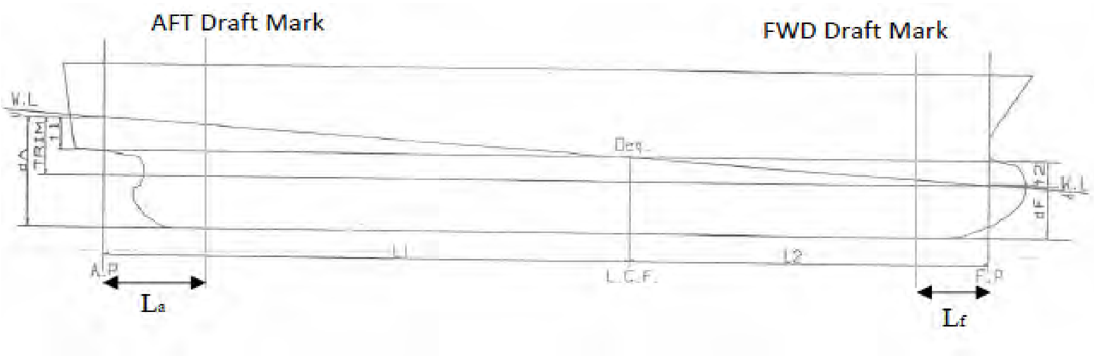
   -1.00m,-0.50m, 0.00m,0.50m, 1.00m

   These trims are the expected operational trims.

Notes on Use of Cross Curves of Stability

The purpose of the Cross-Curves of Stability is to enable Statical Stability curves to be drawn for the ship in any loading condition. Intact stability is then determined on the basis of this Curve. WORKED OUT EXAMPLE demonstrates the use of Cross Curves.

Draft and Trim Calculation



The following data are required for ascertaining trim and draft:

a) Longitudinal Centre of gravity of the ship calculated as follows:

$$LCG = \frac{\text{Summation of longitudinal weight moments}}{\text{Displacement}}$$

b) Longitudinal centre of buoyancy, LCB

c) Displacement, Δ

d) Moment to change trim, MCT1cm

e) Longitudinal centre of floatation, LCF

LCB, LCF and MCT1cm are taken from the hydrostatic tables (trim = 0 m) corresponding to the actual displacement. The trim is calculated according to the following formula:

$$Trim = \frac{\Delta * (LCB-LCG)}{MCT1cm * 100}$$

The Extreme drafts at AP (TAP) and at FP (TFP) are calculated according to the following formulae:

$$T_{AP} = \frac{LCF * Trim}{LBP.} + T\ (extreme)$$

$$T_{FP} = T_{AP} - Trim$$

Where T is the Extreme draft taken from the hydrostatic tables (at trim = 0) corresponding to the actual displacement.

The draft at the Aft and Forward reading marks are as follows:

$$T_a = T_{AP} - \frac{Trim * L_a}{LBP}$$

$$T_f = T_{FP} + \frac{Trim * L_f}{LBP}$$

Where,

T_a = Draft at Aft Draft Mark

T_f = Draft at Fwd Draft Mark

L_a = Distance of Aft Draft Mark from AP and it is consider positive when Aft Draft Mark is Fwd of AP

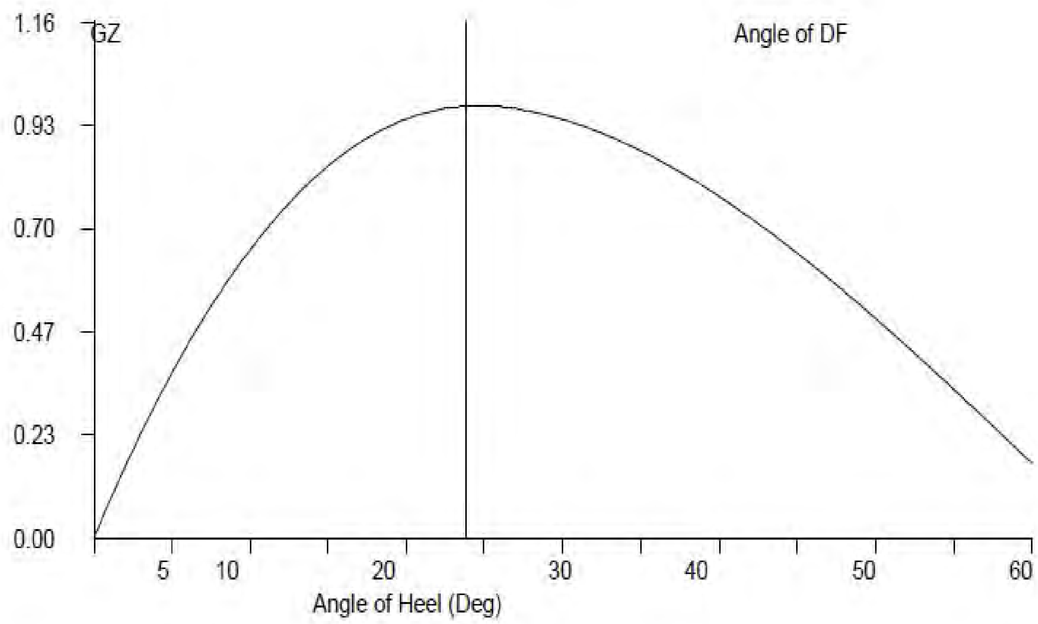
L_f = Distance of Fwd Draft Mark from FP and it is consider positive when Fwd Draft Mark is Aft of FP

GZ Calculation:

1) GZ = KN - KGo Sin (theta)
2) Area under 0-10 = 10*0.01745*[GZ0 + GZ10] / 2
3) Area under 10-20 = 10*0.01745*[GZ10 + GZ20] / 2
4) Area under 20-30 = 10*0.01745*[GZ20 + GZ30] / 2
5) Area under 30-40 = 10*0.01745*[GZ30 + GZ40] / 2
6) Area under 40-50 = 10*0.01745*[GZ40 + GZ50] / 2
7) Area under 50-60 = 10*0.01745*[GZ50 + GZ60] / 2
8) Area upto 40 deg / Angle of DF = sum of 2+3+4+5
9) Max Gz = Read from Gz Curve

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the

Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding & between 30 degrees to 40 degrees/angle of flooding. If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Examples

1 Loading the Vessel

List out the weights, LCG, VCG, TCG and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and VCG of the vessel.

For this worked out Example,

Condition 5: Lightship + Crane + Ballast + Deck Cargo
Towing Condition with Max. draft 2.212m

Displacement	1679.61	Tonnes
LCG	24.598	M Fwd of AP
VCG (KG)	3.873	M above Base Line
TCG	0.000	M Off centerline

2 Free Surface Correction

A tank which is partially filled, adversely affects the stability of the vessel. This is known as Free Surface Effect and is expressed as 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

Free Surface Correction (M) = $\dfrac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$

Free Surface moment for a Tank = Moment of Inertia (M^4) for a Tank x Density of Liquid in Tank

KGo (Fluid) = KG (Solid) + Free Surface Correction.

Hydrostatic Particulars

The trim of a particular loading condition is obtained by

Trim = (LCB-LCG) X displacement / (MCT x 100) + Trim*

Where,

Trim* = Trim at which hydrostatic particulars have been computed.

Condition 5 : Lightship + Crane + Ballast + Deck Cargo
Towing Condition with Max. draft 2.212m

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
01_WBTK.P	39.21	1.025	40.19	2.341	94.08	-4.971	-199.78	1.911	76.80	41.85
01_WBTK.C	39.68	1.025	40.67	2.337	95.05	0.000	0.00	1.901	77.31	41.85
01_WBTK.S	39.21	1.025	40.19	2.341	94.08	4.971	199.78	1.911	76.80	41.85
06_WBTK.C	110.25	1.025	113.01	42.250	4774.67	0.000	0.00	1.500	169.52	78.48
07_WBTK.P	39.21	1.025	40.19	47.659	1915.42	-4.971	-199.78	1.911	76.80	41.85
07_WBTK.C	39.68	1.025	40.67	47.663	1938.45	0.000	0.00	1.901	77.31	41.85
07_WBTK.S	39.21	1.025	40.19	47.659	1915.42	4.971	199.78	1.911	76.80	41.85
TOTAL TANK			355.11	30.490	10827.17	0.000	0.00	1.778	631.35	329.58

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			389.39	25.000	9,734.75	0.000	0.00	3.000	1,168.17
CRANE			150.00	7.500	1,125.00	0.000	0.00	5.200	780.00
DECK CARGO			785.12	25.000	19,628.00	0.000	0.00	5.000	3,925.60
TOTAL FIXED WEIGHT			1,324.51	23.018	30,487.75	0.000	0.00	4.435	5,873.77

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK			355.110	30.490	10827.173	0.000	0.000	1.778	631.355
TOTAL FIXED WEIGHT			1324.510	23.018	30487.750	0.000	0.000	4.435	5873.770
TOTAL WEIGHT			1679.620	24.598	41314.923	0.000	0.000	3.873	6505.125

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Free Surface Correction for VCG = FSM/Displacement

	0.196	m
VCGc	4.069	m

From the MaxVCG table,

Displacement	1679.62	T
Allowable Max VCG	6.365	m

$\boxed{\text{VCGc} < \text{Maxvcg}}$ PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement = 1679.620 T

Draft ,Tm @ 25.0f =	2.212	m
LCB =	25.000	m
LCF =	25.000	m
MCT =	35.880	T-m/cm
KMT =	13.406	m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)
0.188 m,(+) is aft trim,(-) is fwd trim

Determining the Forward perendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.00 (MS),Tmid 2.212 m

Draft @ 50.0f (FP),Tfwd =Tm - Trim/2
2.118 m

Draft@0(AP),Taft =Tm + Trim/2
2.306 m

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

From HYDROSTATICS

Hydrostatic Particulars			**Drafts & Trims**	
MCT	35.880	T-m	Draft (Mean)	2.212 m
LCB	25.000	m	Draft AP (USK)	2.306 m
			Draft FP (USK)	2.118 m
Metacentric Height				
KMt	13.406	m		
KG	3.873	m		
KG0	4.069	m		
GMt	9.337	m	**Trim**	0.188 m

The condition shown above is first worked
using Even Keel hydrostatic perticulars
Program then interpolates the hydrostatic perticulars
from the values available for the displacement

From HYDROSTATICS

Hydrostatic Particulars			**Drafts & Trims**	
MCT	34.580	T-m	Draft (Mean)	2.212 m
LCB	24.587	m	Draft AP (USK)	2.310 m
			Draft FP (USK)	2.114 m
Metacentric Height				
KMt	13.407	m		
KG	3.873	m		
KG0	4.069	m		
GMt	9.338	m	**Trim**	0.195 m

Trim is calculaed as follows:
Trim = (LCB - LCG) X displacement / (MCT X 100) + Trim*
Where Trim* is trim at which hydrostatics were calculated.

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Intact Stability (Primary Criterion)

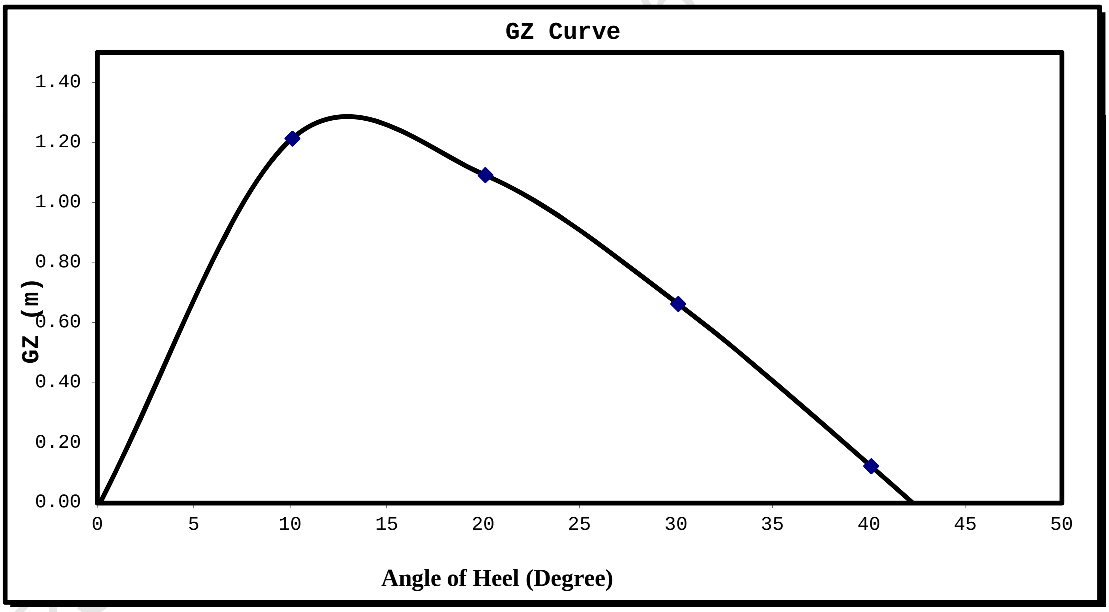
			Minimum Required
Displacement	1679.62	T	
KG0	4.069	M	
GMt	9.338	M	0.150 M

Residual Arm Curve

Angle (Deg)	0	10	20	30	40	50
KN (M)	-0.017	1.928	2.491	2.704	2.744	2.671
GZ (M)	-0.017	1.214	1.092	0.663	0.123	-0.451

	Computed values		Minimum Required	
Residual Area from abs 0 deg to MAxRA	0.1853	m-rad	0.080	m-rad
Angle from Equilibrium to RAZero or flood	42.16	m-rad	20.00	Deg.
Angle from Equilibrium to 50% DK Imm Angle	2.34	m-rad	0.00	Deg.

Plot GZ values against the angles of heel as above.
At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates.Join the point to the origin.
This is the Tangent to the GZ Curve at the Origin.Find the Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding
If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Example Contd.

Mean Draft	2.212	M
LCF from AP	24.977	M
LCB from AP	24.587	M
MCT	34.580	T-M/Cm
Trim	0.195	M
Draft (AP)	2.310	M
Draft (FP)	2.114	M
KG	3.873	M
KGo	4.069	M
GoMt	9.338	M

LOADLINE DRAFT: Check that Mean Draft (Midship) does not exceed the Summer Draft (Ext) for the corresponding season, as per the assigned loadline.

BLANK CALCULATION SHEET

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
TOTAL TANK										

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			389.390	25.000	9,734.750	0.000	0.000	3.000	1,168.170
CRANE									
DECK CARGO									
LOAD LIFTED									
TOTAL FIXED WEIGHT									

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK									
TOTAL FIXED WEIGHT									
TOTAL WEIGHT									

Free Surface Correction for VCG = FSM/Displacement

m

VCGc m

From the MaxVCG table,

Displacement T

Allowable Max VCG m

$\boxed{\text{VCGc < Maxvcg}}$

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement = T

Draft ,Tm @ 25.0f = m

LCB = m

LCF = m

MCT = T-m/cm

KMT = m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)

m,(+) is aft trim,(-) is fwd trim

Determining the Forward perendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.00 (MS),Tmid m

Draft @ 50.0f (FP),Tfwd =Tm - Trim/2

m

Draft@0(AP),Taft =Tm + Trim/2

m

Definitions, Symbols and Sign Conventions:

AP	Fr.0
FP	Fr.100
Midship	Fr.50
Frame Spacing	500mm throughout

List of Symbols

Symbol	Unit	Description
LOA	M	Length Overall
LBP	M	Length between perpendiculars
B (MLD)	M	Moulded Breadth
D (MLD)	M	Moulded Depth
TPCm	T/Cm	Tonnes per 1 Cm immersion
MCTCm	T-M/Cm	Moment to change Trim by 1 Cm
Draft AP (USK)	M	Draft at AP wrt Keel line
Draft FP (USK)	M	Draft at FP wrt Keel line
Draft (Mean)	M	Draft wrt Keel line at Midship
Draft (Mean)	M	[Draft AP(USK) + Draft FP(USK)] x 0.5
VCG, KG	M	Vertical Center of Gravity above Base Line before correcting for free surface
KGo	M	Vertical Center of Gravity above Base Line after correcting for free surface
KB	M	Vertical Center of Buoyancy from Base Line
KMt	M	Transverse Metacenter above Base Line
GoMt	M	Transverse Metacenter above KGo
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
sp.gr.		Specific Gravity
LCG	M	Longitudinal Center of Gravity
LCB	M	Longitudinal Center of Buoyancy
LCF	M	Longitudinal Center of Floatation

Sign Convention

LCB, LCG, LCF	Fwd of AP is denoted by f and aft of AP is denoted by a
TCG	port is negative and stbd is positive
Trim by Aft	Positive
Trim by Ford	Negative

Metric Conversion Table

MULTIPLY BY	TO CONVERT FROM	TO OBTAIN	
0.03937	millimeters	Inches	25.400
0.39370	Centimeters	Inches	2.5400
3.28080	Meters	Feet	0.3050
2.20460	kilograms	Pounds	0.4540
0.00098	kilogram's	tons(2240 lb.)	1016.1
0.98420	Metric tonnes (i.e. tonnes of 1000 kg.)	tons(2240 lb.)	1.0160
2.49980	Metric tonnes per centimeter of immersion	tons per inch (immersion)	0.4000
8.20140	moment to change trim one centimeter one (tonne meter units)	moment to change trim inch(foot ton units)	0.1220
187.976	Meter Radians	Feet degrees	0.0053
	TO OBTAIN	**TO CONVERT FROM**	**MULTIPLY**

Relation between Weight and volume

1000mm cubed	1 Cubic Centimeters
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm.)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg.)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

Angle of Downflooding

1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive downflooding can occur as the vessel heels.
2. If this opening has a weathertight closing device which cannot be kept closed and secured at all times while the vessel is underway (e.g. engine room air supply trunk), or has no weathertight closing device, progressive downflooding will occur when the sill of the opening immerses. This angle is to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve, and the GZ curve terminates at this angle: i.e., the actual angle of flooding becomes the range of the GZ curve.
3. If an opening is fitted with a weathertight closing device which can be kept closed and secured at all times while the vessel is underway, progressive downflooding can only occur through the opening if its closing device is not closed and secured. The angle of immersion of the lower sill off this opening is a `potential' angle of flooding. If the closing appliance is not closed and secured, downflooding will occur when the lower sill of this opening reaches the water level.
   If the lower sill of the opening through which downflooding can occur immerses at 40 degree or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea where this opening is used for access in the normal working of the vessel, an alternate access should be used in heavy weather. In moderate weather, the opening may be used for access, but the closing appliances must be closed and secured again immediately after use.

   As there is no any unprotected opening exists above freeboard deck, hence the downflooding angle is considered at 50 deg.

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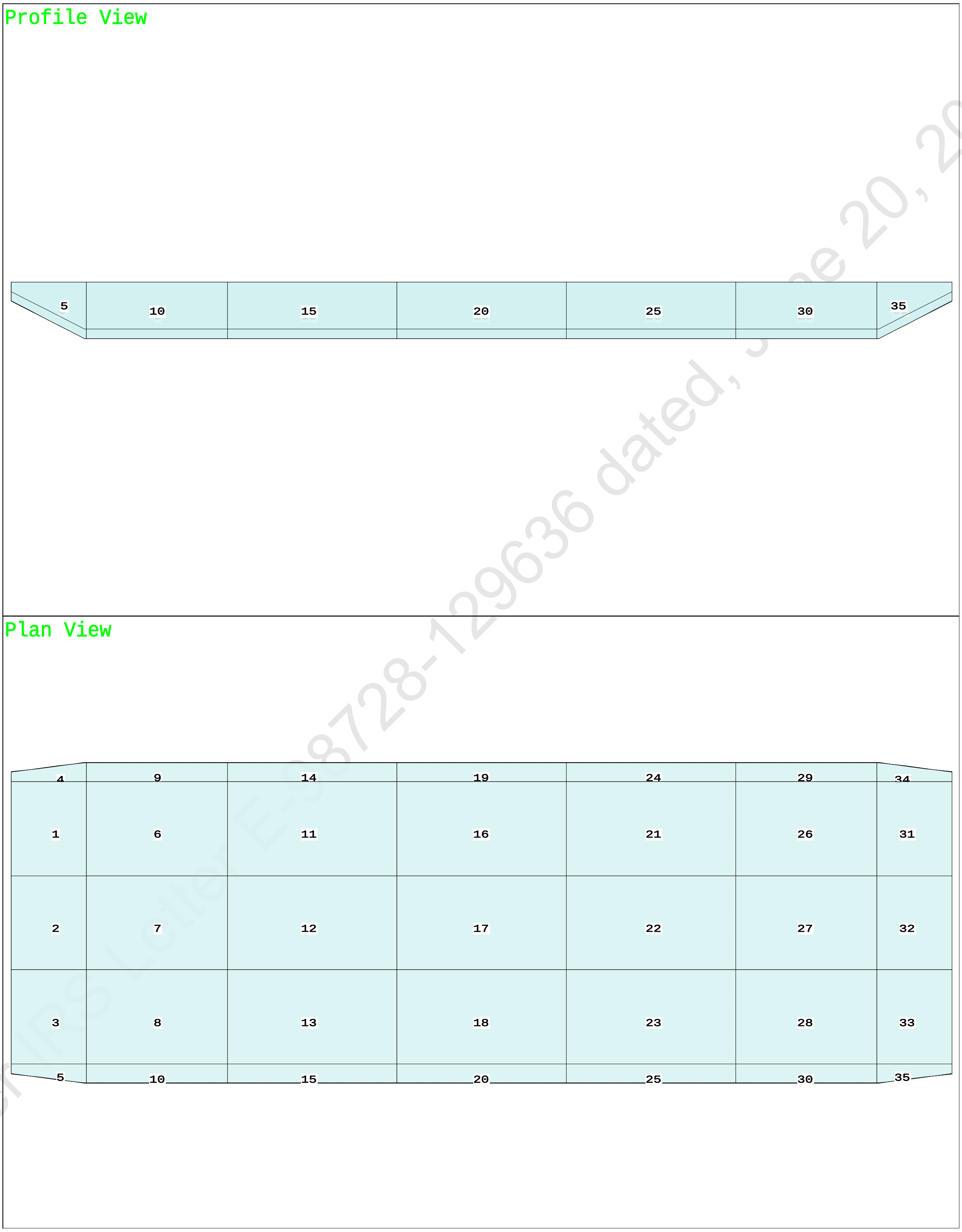
PONTOON

CAPACITY PLAN AND TABLE



Tanks

1 01_WBTK.P
2 01_WBTK.C
3 01_WBTK.S
4 01_VOIDTK.P
5 01_VOIDTK.S
6 02_VOID_CL.P
7 02_WBTK.C
8 02_VOID_CL.S
9 02_VOIDTK.P
10 02_VOIDTK.S
11 03_WBTK.P
12 03_WBTK.C
13 03_WBTK.S
14 03_VOIDTK.P
15 03_VOIDTK.S
16 04_VOID_CL.P
17 04_VOIDTK.C
18 04_VOID_CL.S
19 04_VOIDTK.P
20 04_VOIDTK.S
21 05_WBTK.P
22 05_WBTK.C
23 05_WBTK.S
24 05_VOIDTK.P
25 05_VOIDTK.S
26 06_WBTK.P
27 06_WBTK.C
28 06_WBTK.S
29 06_VOIDTK.P
30 06_VOIDTK.S
31 07_WBTK.P
32 07_WBTK.C
33 07_WBTK.S
34 07_VOIDTK.P
35 07_VOIDTK.S

03-01-20 14:45:57
GHS 16.68C **PONTOON**

TANK STATUS
Trim: zero, Heel: zero

Part	Cu.M.	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
01_WBTK.P	39.2	1.025	40.19	2.341f	4.971p	1.911	41.85*
01_WBTK.C	39.7	1.025	40.67	2.337f	0.000	1.901	41.85*
01_WBTK.S	39.2	1.025	40.19	2.341f	4.971s	1.911	41.85*
01_VOIDTK.P	3.7	1.025	3.82	2.678f	7.879p	2.253	0.18*
01_VOIDTK.S	3.7	1.025	3.82	2.678f	7.879s	2.253	0.18*
02_VOID_CL.P	109.3	1.025	112.06	7.750f	4.980p	1.511	78.48*
02_WBTK.C	110.2	1.025	113.01	7.750f	0.000	1.500	78.48*
02_VOID_CL.S	109.3	1.025	112.06	7.750f	4.980s	1.511	78.48*
02_VOIDTK.P	14.7	1.025	15.07	7.750f	7.958p	1.979	0.63*
02_VOIDTK.S	14.7	1.025	15.07	7.750f	7.958s	1.979	0.63*
03_WBTK.P	131.2	1.025	134.48	16.000f	4.980p	1.511	94.17*
03_WBTK.C	132.3	1.025	135.61	16.000f	0.000	1.500	94.17*
03_WBTK.S	131.2	1.025	134.48	16.000f	4.980s	1.511	94.17*
03_VOIDTK.P	17.6	1.025	18.08	16.000f	7.958p	1.979	0.75*
03_VOIDTK.S	17.6	1.025	18.08	16.000f	7.958s	1.979	0.75*
04_VOID_CL.P	131.2	1.025	134.48	25.000f	4.980p	1.511	94.17*
04_VOIDTK.C	132.3	1.025	135.61	25.000f	0.000	1.500	94.17*
04_VOID_CL.S	131.2	1.025	134.48	25.000f	4.980s	1.511	94.17*
04_VOIDTK.P	17.6	1.025	18.08	25.000f	7.958p	1.979	0.75*
04_VOIDTK.S	17.6	1.025	18.08	25.000f	7.958s	1.979	0.75*
05_WBTK.P	131.2	1.025	134.48	34.000f	4.980p	1.511	94.17*
05_WBTK.C	132.3	1.025	135.61	34.000f	0.000	1.500	94.17*
05_WBTK.S	131.2	1.025	134.48	34.000f	4.980s	1.511	94.17*
05_VOIDTK.P	17.6	1.025	18.08	34.000f	7.958p	1.979	0.75*
05_VOIDTK.S	17.6	1.025	18.08	34.000f	7.958s	1.979	0.75*
06_WBTK.P	109.3	1.025	112.06	42.250f	4.980p	1.511	78.48*
06_WBTK.C	110.2	1.025	113.01	42.250f	0.000	1.500	78.48*
06_WBTK.S	109.3	1.025	112.06	42.250f	4.980s	1.511	78.48*
06_VOIDTK.P	14.7	1.025	15.07	42.250f	7.958p	1.979	0.63*
06_VOIDTK.S	14.7	1.025	15.07	42.250f	7.958s	1.979	0.63*
07_WBTK.P	39.2	1.025	40.19	47.659f	4.971p	1.911	41.85*
07_WBTK.C	39.7	1.025	40.67	47.663f	0.000	1.901	41.85*
07_WBTK.S	39.2	1.025	40.19	47.659f	4.971s	1.911	41.85*
07_VOIDTK.P	3.7	1.025	3.82	47.322f	7.879p	2.253	0.18*
07_VOIDTK.S	3.7	1.025	3.82	47.322f	7.879s	2.253	0.18*
Total Tanks--------->			2,314.06	25.000f	0.000	1.589	1577.26

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

HYDROSTATIC PARTICULARS

03-01-20 15:00:37 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

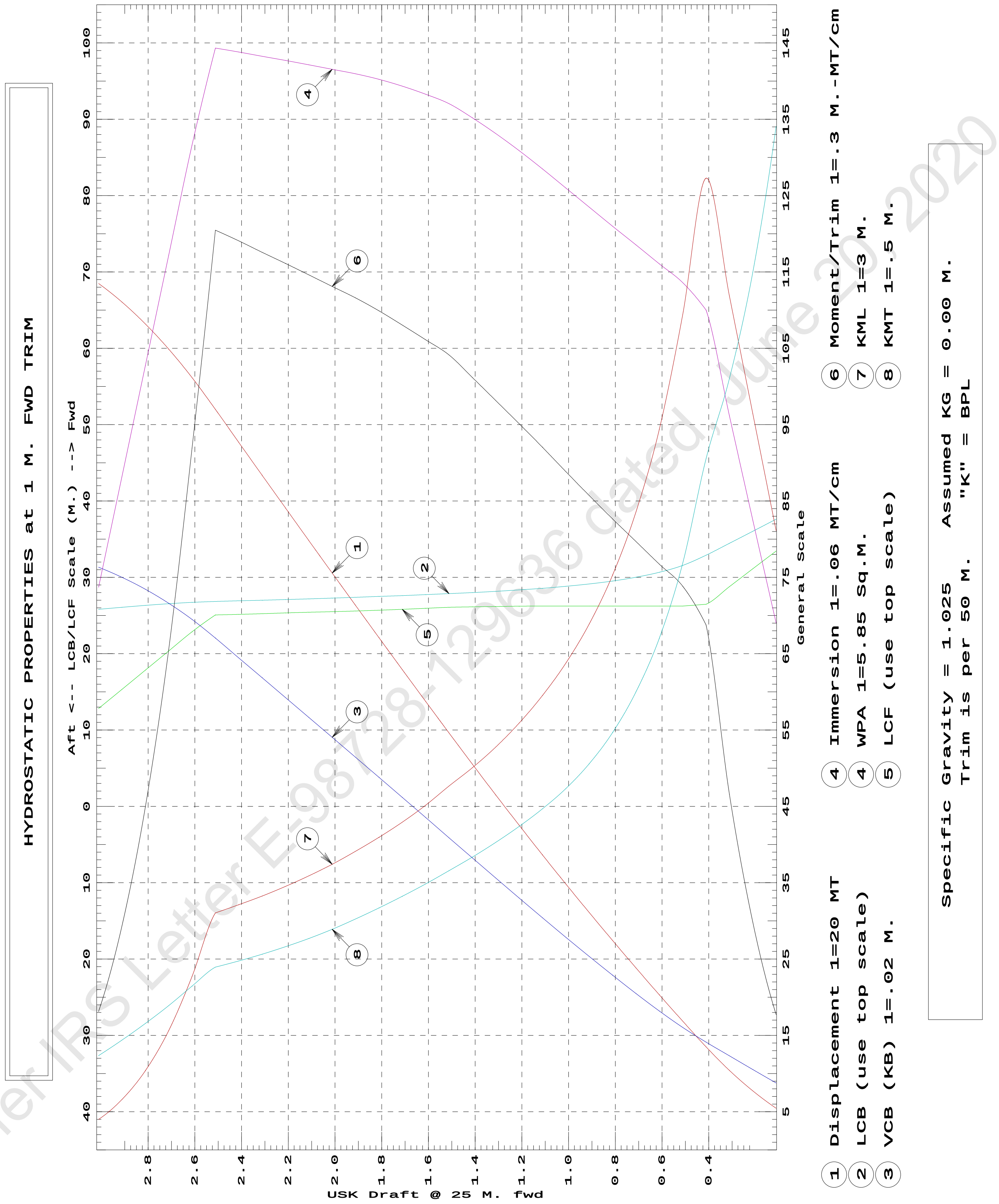
HYDROSTATIC PROPERTIES
Trim: Fwd 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	109.19	37.614f	0.175	4.14	33.447f	5.32	243.33	67.044
0.212	154.63	36.036f	0.211	4.95	31.111f	8.95	289.37	57.467
0.312	208.22	34.462f	0.246	5.77	28.785f	13.98	335.54	50.522
0.412	270.05	32.891f	0.282	6.60	26.469f	20.63	381.85	45.262
0.512	337.43	31.565f	0.321	6.82	26.237f	22.15	328.17	38.417
0.612	406.32	30.662f	0.364	6.96	26.235f	23.02	283.19	33.515
0.712	476.68	30.009f	0.410	7.11	26.232f	23.91	250.72	30.000
0.812	548.53	29.514f	0.458	7.26	26.237f	24.79	225.94	27.341
0.912	621.87	29.128f	0.507	7.41	26.237f	25.71	206.70	25.293
1.012	696.71	28.818f	0.557	7.56	26.237f	26.65	191.25	23.665
1.112	773.05	28.562f	0.608	7.71	26.235f	27.61	178.55	22.341
1.212	850.87	28.349f	0.660	7.85	26.216f	28.54	167.67	21.183
1.312	930.08	28.166f	0.712	7.99	26.180f	29.45	158.28	20.138
1.412	1,010.59	28.006f	0.765	8.11	26.132f	30.35	150.15	19.177
1.512	1,092.29	27.864f	0.818	8.22	26.072f	31.24	142.98	18.285
1.612	1,174.88	27.733f	0.871	8.30	25.937f	31.83	135.45	17.420
1.712	1,258.16	27.610f	0.924	8.36	25.807f	32.41	128.79	16.608
1.812	1,342.03	27.494f	0.976	8.41	25.691f	32.97	122.83	15.847
1.912	1,426.39	27.384f	1.029	8.46	25.592f	33.48	117.34	15.125
2.012	1,511.13	27.281f	1.081	8.49	25.506f	33.93	112.25	14.449
2.112	1,596.21	27.184f	1.133	8.53	25.421f	34.39	107.69	13.848
2.212	1,681.65	27.092f	1.185	8.56	25.331f	34.84	103.55	13.308
2.312	1,767.43	27.004f	1.237	8.60	25.242f	35.27	99.77	12.816
2.412	1,853.55	26.920f	1.289	8.63	25.153f	35.72	96.34	12.374
2.512	1,940.01	26.839f	1.341	8.66	25.071f	36.14	93.13	11.973
2.612	2,023.20	26.727f	1.390	7.90	22.862f	27.58	68.14	10.729
2.712	2,097.83	26.545f	1.432	7.03	20.350f	19.66	46.84	9.448
2.812	2,163.78	26.319f	1.469	6.16	17.838f	13.47	31.12	8.279
2.912	2,221.04	26.069f	1.500	5.29	15.329f	8.80	19.81	7.194
3.012	2,269.60	25.813f	1.526	4.42	12.820f	5.44	11.99	6.178

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

03-01-20 15:00:37 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONTOON



03-01-20 15:00:37 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

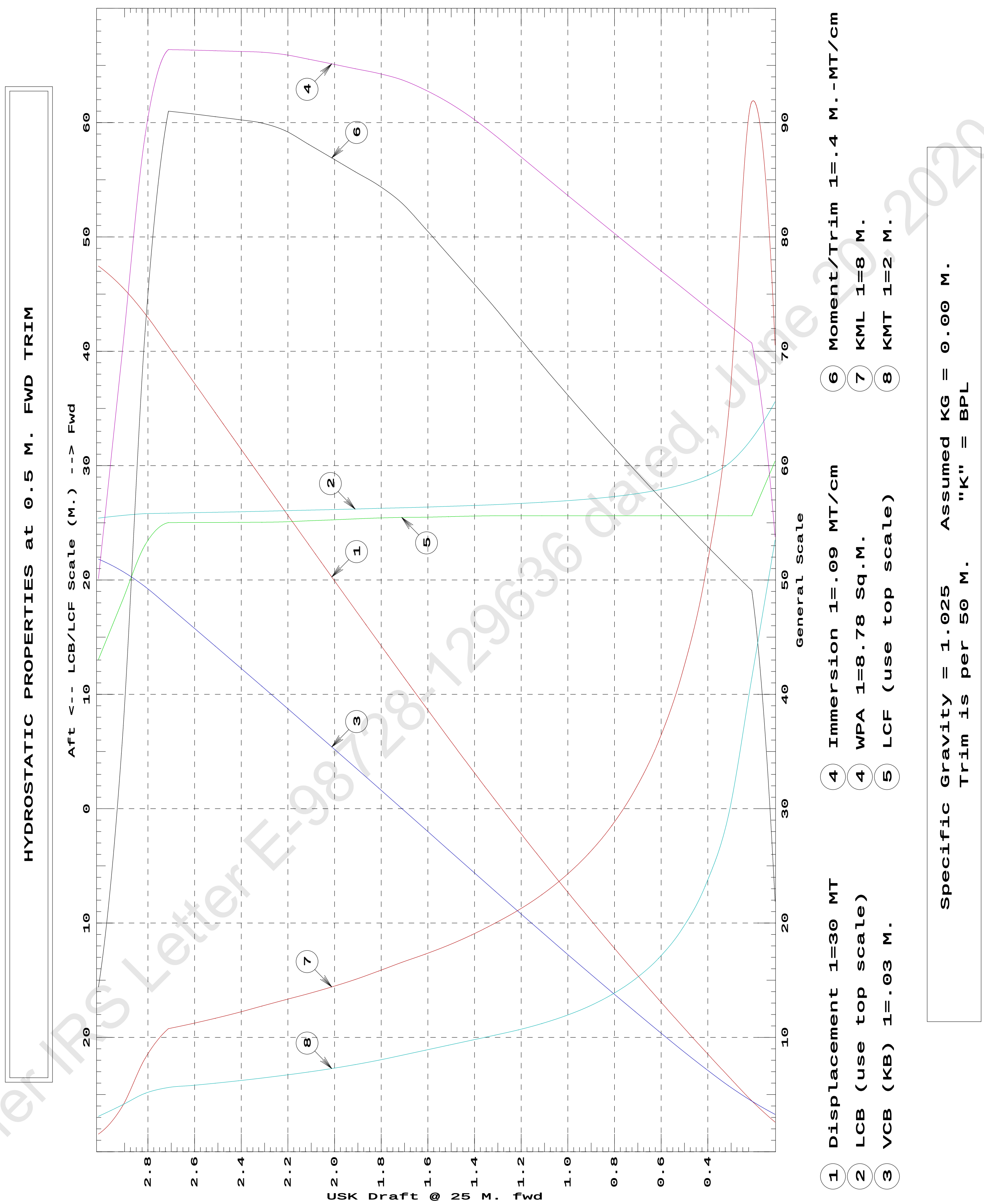
HYDROSTATIC PROPERTIES
Trim: Fwd 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	77.55	35.588f	0.098	4.84	30.395f	8.75	564.32	107.002
0.212	133.62	32.349f	0.133	6.37	25.620f	19.63	734.53	82.843
0.312	197.99	30.161f	0.173	6.51	25.618f	20.43	516.00	58.820
0.412	263.82	29.028f	0.219	6.66	25.616f	21.26	402.89	46.426
0.512	331.12	28.335f	0.267	6.80	25.613f	22.11	333.81	38.890
0.612	399.87	27.868f	0.317	6.95	25.619f	22.95	286.94	33.805
0.712	470.09	27.532f	0.368	7.10	25.618f	23.83	253.42	30.190
0.812	541.80	27.279f	0.419	7.24	25.618f	24.72	228.16	27.489
0.912	615.01	27.081f	0.471	7.39	25.618f	25.64	208.46	25.400
1.012	689.70	26.922f	0.524	7.54	25.618f	26.58	192.68	23.745
1.112	765.90	26.793f	0.577	7.70	25.618f	27.54	179.77	22.406
1.212	843.62	26.685f	0.630	7.85	25.618f	28.52	169.01	21.304
1.312	922.87	26.593f	0.684	8.00	25.617f	29.51	159.90	20.384
1.412	1,003.58	26.514f	0.738	8.14	25.589f	30.45	151.72	19.510
1.512	1,085.59	26.442f	0.792	8.26	25.539f	31.37	144.50	18.632
1.612	1,168.70	26.376f	0.846	8.36	25.490f	32.30	138.19	17.747
1.712	1,252.72	26.316f	0.901	8.44	25.468f	33.19	132.46	16.861
1.812	1,337.38	26.261f	0.955	8.49	25.422f	33.81	126.39	16.002
1.912	1,422.44	26.209f	1.008	8.52	25.339f	34.28	120.50	15.231
2.012	1,507.87	26.157f	1.062	8.56	25.256f	34.77	115.28	14.549
2.112	1,593.69	26.106f	1.115	8.60	25.174f	35.24	110.57	13.940
2.212	1,679.87	26.056f	1.168	8.63	25.087f	35.72	106.31	13.391
2.312	1,766.35	26.007f	1.221	8.65	25.049f	35.99	101.86	12.883
2.412	1,852.92	25.962f	1.274	8.66	25.043f	36.10	97.42	12.413
2.512	1,939.55	25.921f	1.327	8.67	25.034f	36.21	93.34	11.985
2.612	2,026.23	25.883f	1.379	8.67	25.027f	36.31	89.59	11.595
2.712	2,112.95	25.847f	1.431	8.67	25.022f	36.40	86.13	11.239
2.812	2,198.37	25.801f	1.482	8.00	23.069f	28.71	65.30	10.216
2.912	2,269.65	25.640f	1.524	6.26	18.063f	14.12	31.11	8.133
3.012	2,323.50	25.409f	1.555	4.51	13.058f	5.77	12.42	6.201

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

03-01-20 15:00:37 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**



03-01-20 15:00:37 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

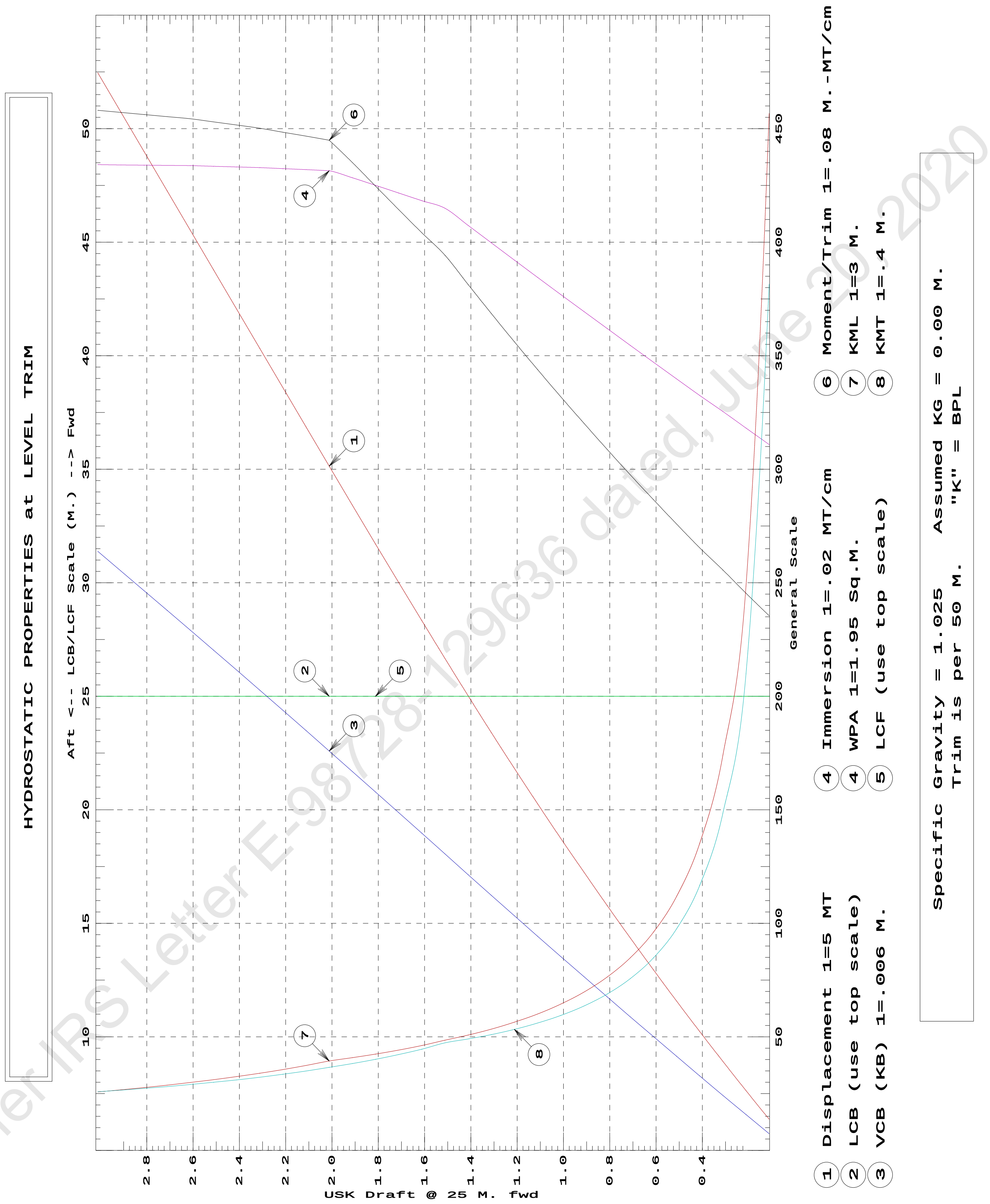
HYDROSTATIC PROPERTIES
No Trim, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	68.74	25.000f	0.044	6.22	25.000f	18.84	1370.12	152.634
0.212	131.64	25.000f	0.095	6.36	25.000f	19.63	745.62	83.943
0.312	195.98	25.000f	0.145	6.51	25.000f	20.45	521.71	59.370
0.412	261.77	25.000f	0.197	6.65	25.000f	21.23	405.56	46.659
0.512	329.01	25.000f	0.249	6.80	25.000f	22.07	335.42	39.017
0.612	397.71	25.000f	0.301	6.94	25.000f	22.93	288.26	33.906
0.712	467.91	25.000f	0.353	7.09	25.000f	23.81	254.38	30.260
0.812	539.56	25.000f	0.406	7.24	25.000f	24.70	228.91	27.538
0.912	612.71	25.000f	0.459	7.39	25.000f	25.62	209.06	25.437
1.012	687.37	25.000f	0.512	7.54	25.000f	26.56	193.17	23.772
1.112	763.54	25.000f	0.566	7.69	25.000f	27.51	180.17	22.425
1.212	841.20	25.000f	0.620	7.84	25.000f	28.49	169.35	21.319
1.312	920.39	25.000f	0.674	7.99	25.000f	29.49	160.20	20.398
1.412	1,001.12	25.000f	0.729	8.15	25.000f	30.51	152.38	19.622
1.512	1,083.39	25.000f	0.783	8.30	25.000f	31.54	145.56	18.939
1.612	1,166.71	25.000f	0.838	8.37	25.000f	32.33	138.57	17.832
1.712	1,250.72	25.000f	0.893	8.43	25.000f	33.15	132.52	16.875
1.812	1,335.41	25.000f	0.947	8.50	25.000f	33.98	127.21	16.037
1.912	1,420.78	25.000f	1.001	8.57	25.000f	34.81	122.50	15.292
2.012	1,506.83	25.000f	1.055	8.63	25.000f	35.59	118.11	14.630
2.112	1,593.18	25.000f	1.109	8.64	25.000f	35.74	112.16	13.984
2.212	1,679.61	25.000f	1.163	8.65	25.000f	35.88	106.80	13.406
2.312	1,766.14	25.000f	1.216	8.66	25.000f	36.01	101.96	12.890
2.412	1,852.74	25.000f	1.269	8.66	25.000f	36.13	97.50	12.418
2.512	1,939.40	25.000f	1.322	8.67	25.000f	36.24	93.43	11.990
2.612	2,026.11	25.000f	1.375	8.67	25.000f	36.35	89.70	11.604
2.712	2,112.87	25.000f	1.427	8.68	25.000f	36.42	86.20	11.243
2.812	2,199.65	25.000f	1.479	8.68	25.000f	36.50	82.96	10.914
2.912	2,286.44	25.000f	1.531	8.68	25.000f	36.57	79.97	10.613
3.012	2,373.26	25.000f	1.583	8.68	25.000f	36.65	77.21	10.339

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

03-01-20 15:00:37 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**



03-01-20 15:00:37 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

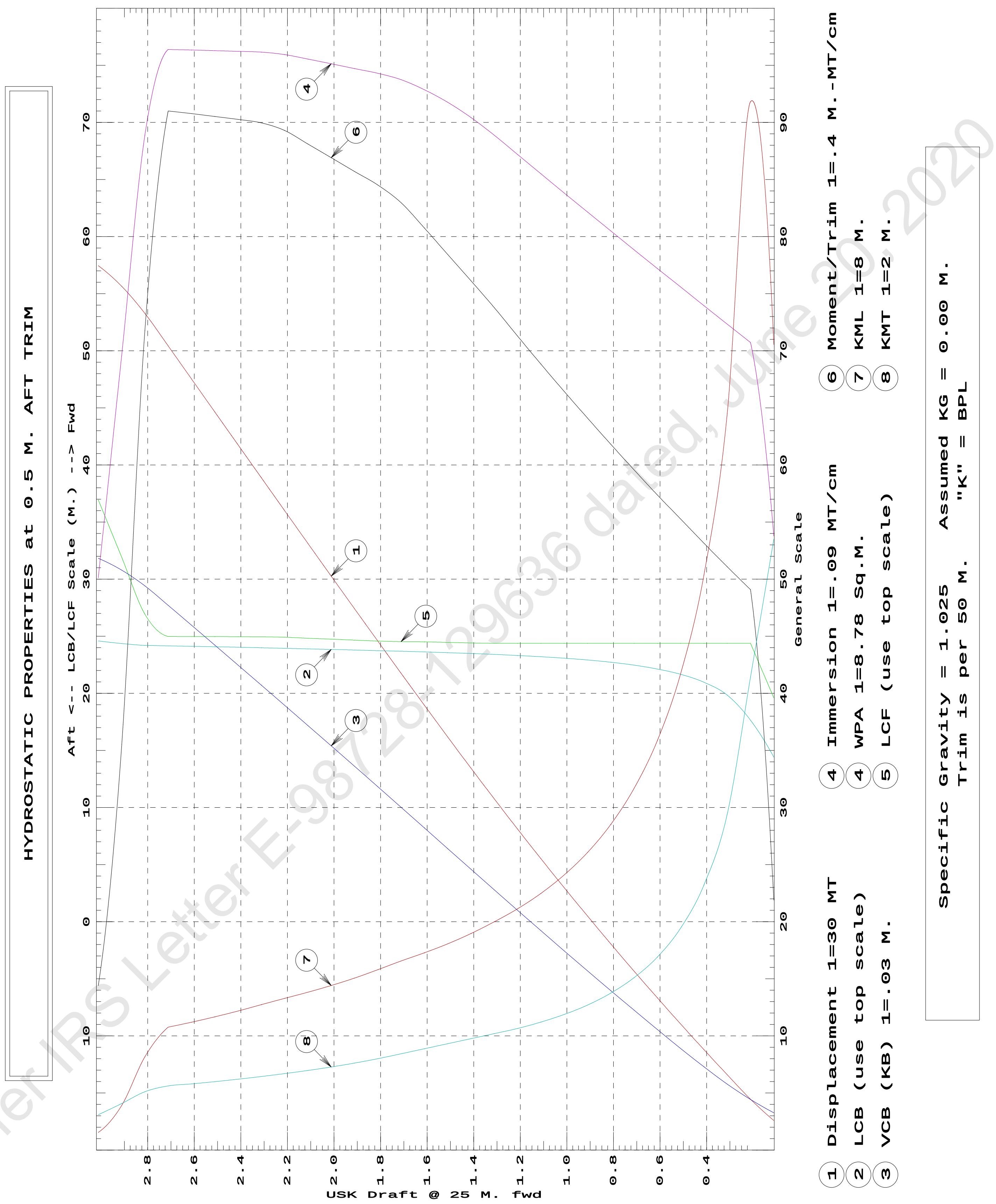
HYDROSTATIC PROPERTIES
Trim: Aft 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	77.55	14.412f	0.098	4.84	19.605f	8.75	564.33	106.997
0.212	133.62	17.651f	0.133	6.37	24.380f	19.63	734.53	82.841
0.312	197.99	19.839f	0.173	6.51	24.382f	20.43	516.00	58.819
0.412	263.82	20.972f	0.219	6.66	24.384f	21.26	402.89	46.426
0.512	331.12	21.665f	0.267	6.80	24.387f	22.11	333.82	38.890
0.612	399.87	22.132f	0.317	6.95	24.381f	22.95	286.94	33.804
0.712	470.09	22.468f	0.368	7.10	24.382f	23.83	253.42	30.190
0.812	541.80	22.721f	0.419	7.24	24.382f	24.72	228.16	27.488
0.912	615.01	22.919f	0.471	7.39	24.382f	25.64	208.46	25.400
1.012	689.70	23.078f	0.524	7.54	24.382f	26.58	192.68	23.745
1.112	765.90	23.207f	0.577	7.70	24.382f	27.54	179.77	22.405
1.212	843.62	23.315f	0.630	7.85	24.382f	28.52	169.01	21.304
1.312	922.87	23.407f	0.684	8.00	24.383f	29.51	159.90	20.383
1.412	1,003.58	23.487f	0.738	8.14	24.411f	30.45	151.72	19.510
1.512	1,085.59	23.558f	0.792	8.26	24.461f	31.37	144.50	18.632
1.612	1,168.70	23.624f	0.846	8.36	24.510f	32.30	138.19	17.747
1.712	1,252.72	23.684f	0.901	8.44	24.532f	33.19	132.46	16.861
1.812	1,337.38	23.739f	0.955	8.49	24.578f	33.81	126.39	16.003
1.912	1,422.43	23.791f	1.008	8.52	24.661f	34.28	120.50	15.231
2.012	1,507.87	23.843f	1.062	8.56	24.744f	34.77	115.28	14.549
2.112	1,593.69	23.894f	1.115	8.60	24.826f	35.24	110.57	13.940
2.212	1,679.87	23.944f	1.168	8.63	24.913f	35.72	106.31	13.391
2.312	1,766.35	23.993f	1.221	8.65	24.951f	35.99	101.86	12.883
2.412	1,852.92	24.038f	1.274	8.66	24.957f	36.10	97.42	12.413
2.512	1,939.55	24.079f	1.327	8.67	24.966f	36.21	93.34	11.984
2.612	2,026.23	24.117f	1.379	8.67	24.973f	36.31	89.59	11.595
2.712	2,112.95	24.153f	1.431	8.67	24.978f	36.40	86.13	11.239
2.812	2,198.37	24.199f	1.482	8.00	26.931f	28.71	65.30	10.215
2.912	2,269.65	24.360f	1.524	6.26	31.937f	14.12	31.11	8.132
3.012	2,323.50	24.591f	1.555	4.51	36.942f	5.77	12.42	6.200

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

03-01-20 15:00:37 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**



03-01-20 15:00:37 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

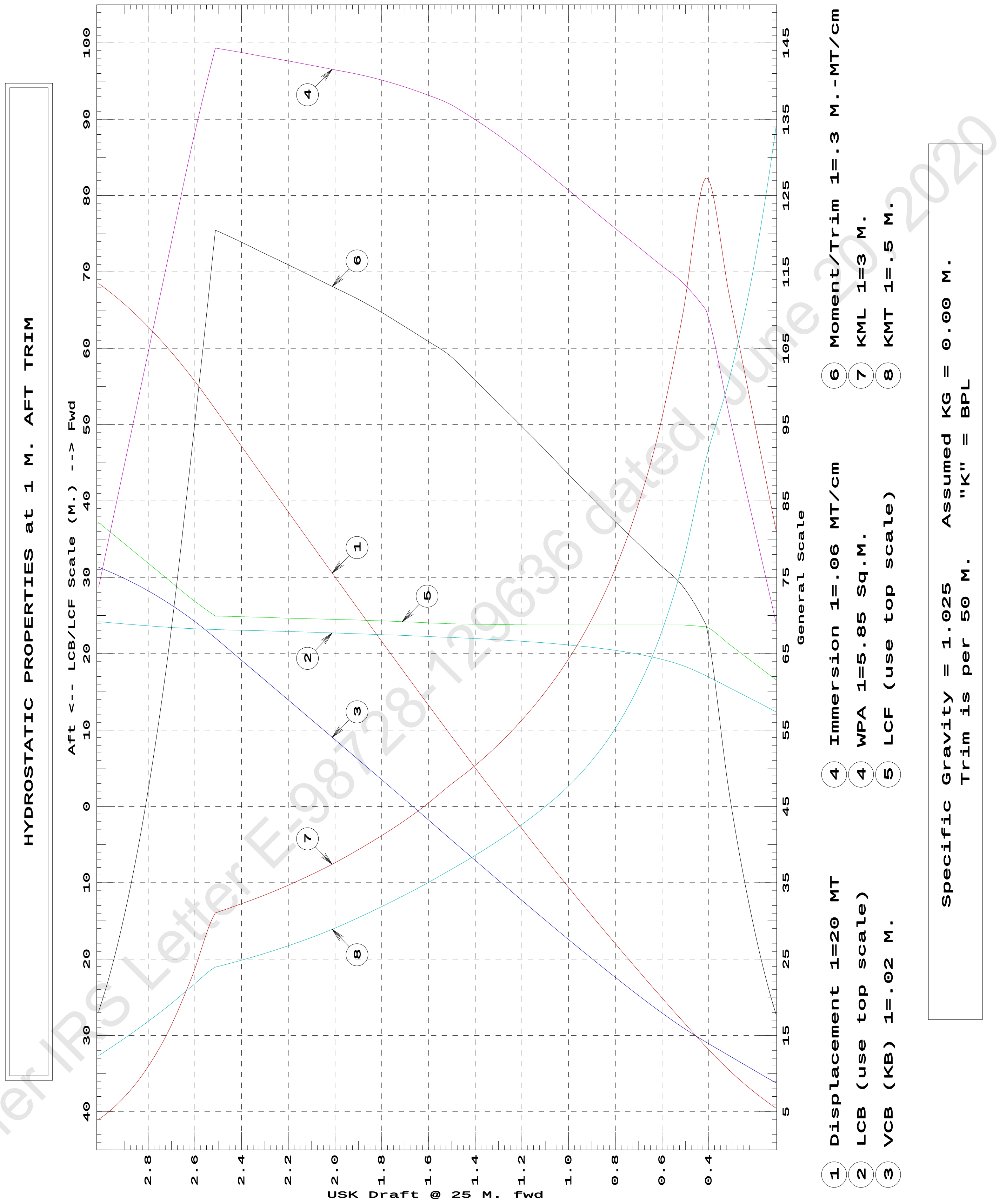
HYDROSTATIC PROPERTIES
Trim: Aft 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	109.19	12.386f	0.175	4.14	16.553f	5.32	243.33	67.041
0.212	154.63	13.964f	0.211	4.95	18.889f	8.95	289.37	57.466
0.312	208.22	15.538f	0.246	5.77	21.215f	13.98	335.54	50.520
0.412	270.05	17.109f	0.282	6.60	23.531f	20.63	381.85	45.260
0.512	337.43	18.435f	0.321	6.82	23.763f	22.15	328.17	38.416
0.612	406.32	19.338f	0.364	6.96	23.765f	23.02	283.19	33.515
0.712	476.68	19.991f	0.410	7.11	23.768f	23.91	250.72	30.000
0.812	548.53	20.486f	0.458	7.26	23.763f	24.79	225.94	27.340
0.912	621.87	20.872f	0.507	7.41	23.763f	25.71	206.70	25.292
1.012	696.71	21.182f	0.557	7.56	23.763f	26.65	191.25	23.665
1.112	773.05	21.438f	0.608	7.71	23.765f	27.61	178.55	22.340
1.212	850.87	21.651f	0.660	7.85	23.784f	28.54	167.67	21.183
1.312	930.08	21.834f	0.712	7.99	23.820f	29.45	158.28	20.138
1.412	1,010.59	21.994f	0.765	8.11	23.868f	30.35	150.15	19.177
1.512	1,092.29	22.136f	0.818	8.22	23.928f	31.24	142.98	18.285
1.612	1,174.88	22.267f	0.871	8.30	24.063f	31.83	135.45	17.420
1.712	1,258.16	22.390f	0.924	8.36	24.193f	32.41	128.79	16.608
1.812	1,342.03	22.506f	0.976	8.41	24.309f	32.97	122.83	15.847
1.912	1,426.39	22.616f	1.029	8.46	24.408f	33.48	117.34	15.124
2.012	1,511.13	22.719f	1.081	8.49	24.494f	33.93	112.25	14.449
2.112	1,596.21	22.816f	1.133	8.53	24.579f	34.39	107.69	13.848
2.212	1,681.65	22.908f	1.185	8.56	24.669f	34.84	103.55	13.307
2.312	1,767.43	22.996f	1.237	8.60	24.758f	35.27	99.77	12.816
2.412	1,853.55	23.080f	1.289	8.63	24.847f	35.72	96.34	12.374
2.512	1,940.01	23.161f	1.341	8.66	24.929f	36.14	93.13	11.972
2.612	2,023.20	23.273f	1.390	7.90	27.138f	27.58	68.14	10.728
2.712	2,097.83	23.455f	1.432	7.03	29.650f	19.66	46.84	9.447
2.812	2,163.78	23.681f	1.469	6.16	32.162f	13.47	31.12	8.278
2.912	2,221.04	23.931f	1.500	5.29	34.671f	8.80	19.81	7.193
3.012	2,269.60	24.187f	1.526	4.42	37.180f	5.44	11.99	6.178

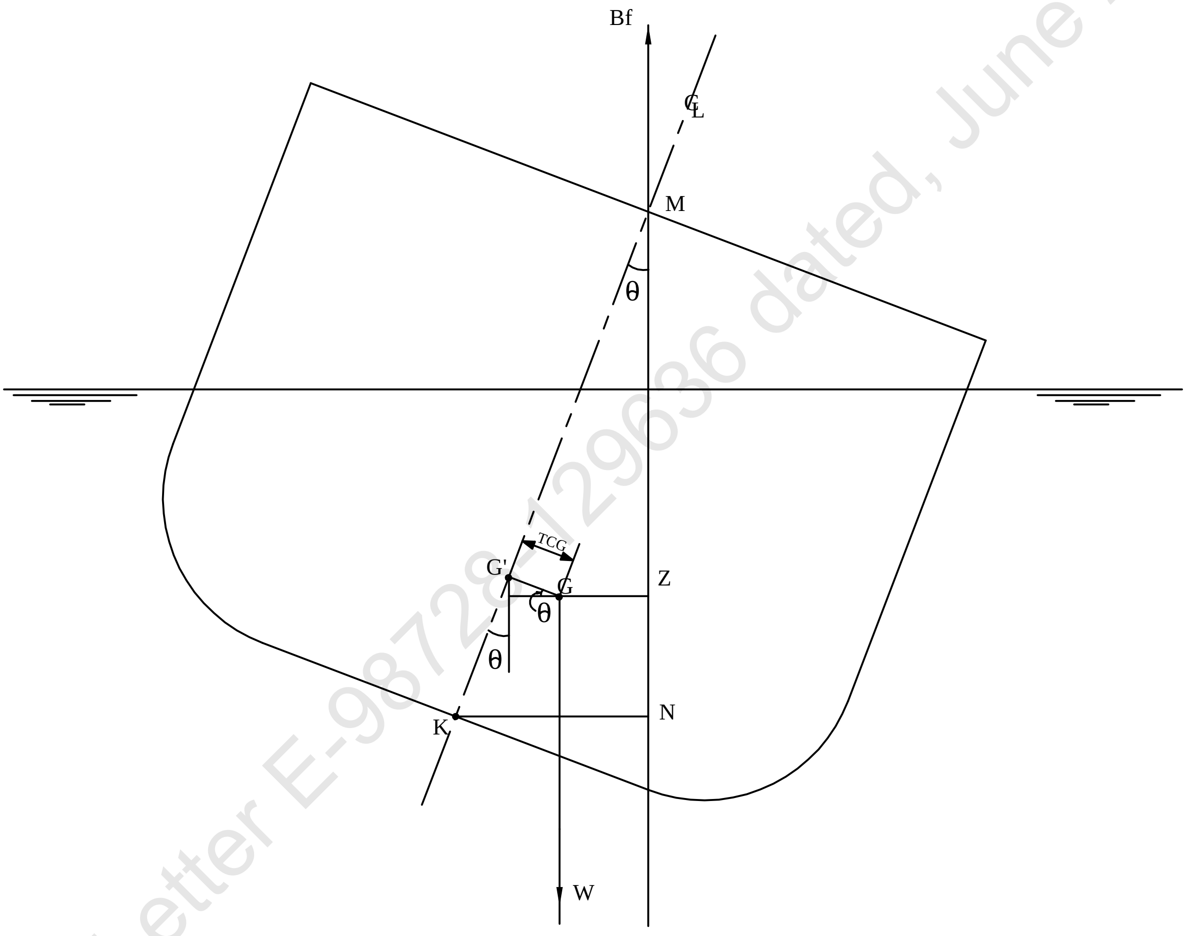
Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

03-01-20 15:00:37 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONTOON



KN CONCEPT

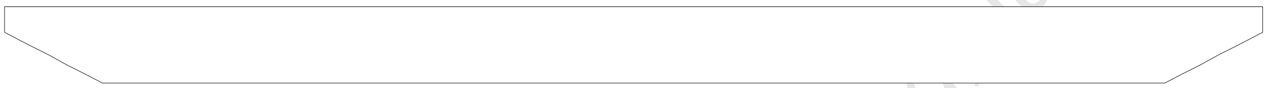


$$GZ = KN - KG' \sin\theta - TCG \cos\theta$$

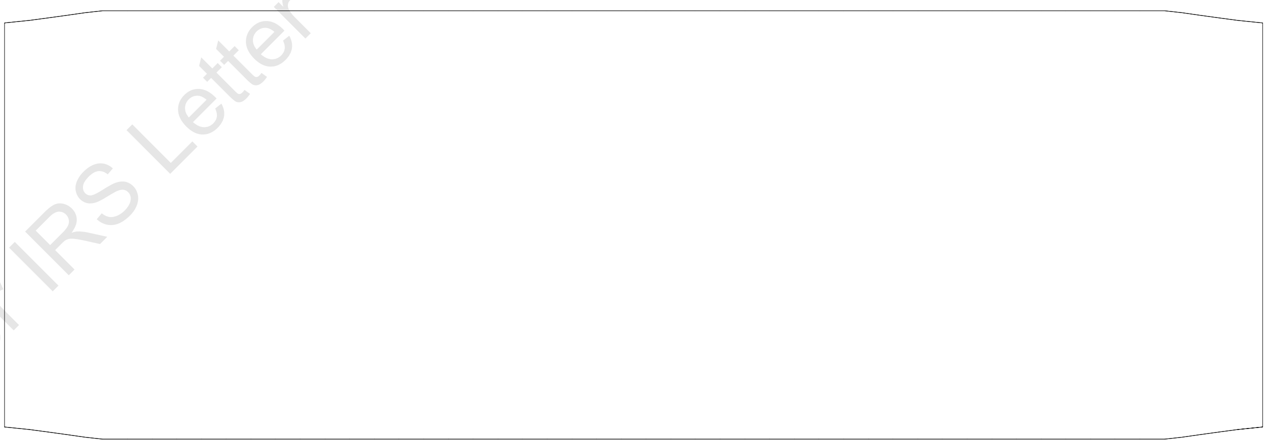
03-01-20 15:13:53 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

VOLUME USED FOR KN COMPUTATION

Profile View



Plan View



CROSS CURVE TABLES

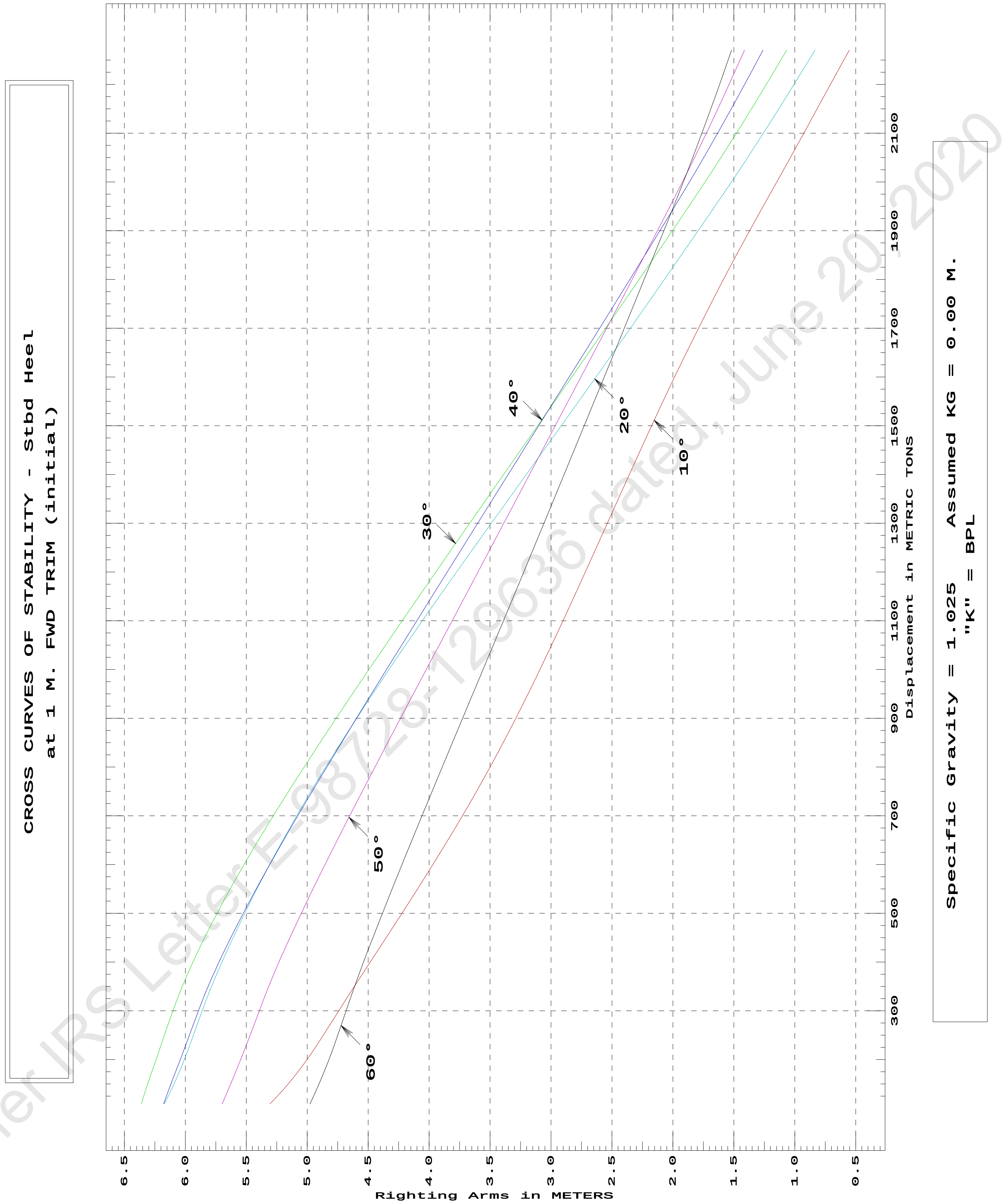
03-01-20 15:03:51 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement	Heel Angles in Degrees					
METRIC TONS	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
109.19	5.307s	6.174s	6.361s	6.178s	5.699s	4.978s
154.63	5.140s	6.085s	6.303s	6.112s	5.621s	4.896s
208.22	4.979s	5.995s	6.229s	6.029s	5.533s	4.810s
270.05	4.818s	5.906s	6.145s	5.937s	5.438s	4.722s
337.43	4.647s	5.805s	6.046s	5.834s	5.337s	4.630s
406.32	4.467s	5.688s	5.926s	5.712s	5.221s	4.529s
476.68	4.284s	5.558s	5.789s	5.573s	5.093s	4.420s
548.53	4.099s	5.416s	5.635s	5.421s	4.956s	4.306s
621.87	3.916s	5.261s	5.467s	5.258s	4.810s	4.187s
696.71	3.735s	5.094s	5.286s	5.085s	4.658s	4.064s
773.05	3.559s	4.914s	5.095s	4.904s	4.501s	3.937s
850.87	3.391s	4.721s	4.893s	4.716s	4.338s	3.808s
930.08	3.228s	4.516s	4.682s	4.522s	4.171s	3.675s
1,010.59	3.071s	4.301s	4.465s	4.323s	4.001s	3.540s
1,092.29	2.916s	4.077s	4.242s	4.120s	3.828s	3.403s
1,174.88	2.764s	3.846s	4.014s	3.914s	3.652s	3.265s
1,258.16	2.613s	3.609s	3.783s	3.706s	3.476s	3.127s
1,342.03	2.462s	3.369s	3.550s	3.497s	3.299s	2.988s
1,426.39	2.310s	3.127s	3.315s	3.286s	3.121s	2.849s
1,511.13	2.154s	2.884s	3.078s	3.074s	2.942s	2.710s
1,596.21	1.994s	2.642s	2.841s	2.860s	2.763s	2.570s
1,681.65	1.827s	2.401s	2.603s	2.647s	2.583s	2.429s
1,767.43	1.653s	2.161s	2.366s	2.434s	2.403s	2.290s
1,853.55	1.472s	1.921s	2.131s	2.221s	2.223s	2.150s
1,940.01	1.282s	1.683s	1.896s	2.009s	2.044s	2.010s
2,023.20	1.096s	1.458s	1.675s	1.810s	1.876s	1.879s
2,097.83	0.932s	1.263s	1.485s	1.638s	1.731s	1.766s
2,163.78	0.788s	1.097s	1.322s	1.491s	1.607s	1.670s
2,221.04	0.662s	0.955s	1.184s	1.367s	1.502s	1.588s
2,269.60	0.554s	0.835s	1.069s	1.263s	1.414s	1.520s

Distances in METERS.---Specific Gravity = 1.025.-------------

03-01-20 15:03:51 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONTOON



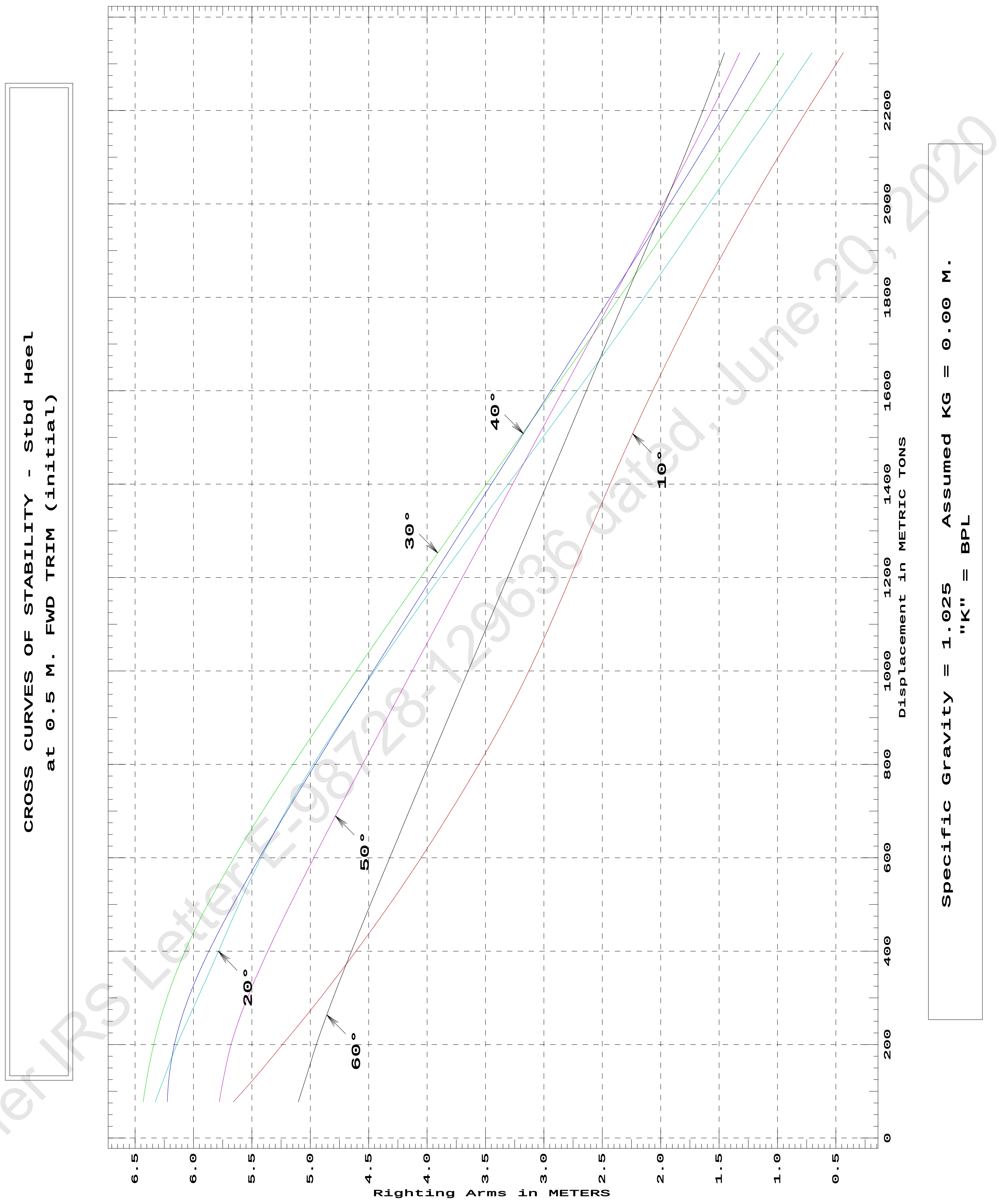
03-01-20 15:03:51 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
77.55	5.658s	6.327s	6.430s	6.222s	5.779s	5.102s
133.62	5.459s	6.244s	6.394s	6.207s	5.738s	5.031s
197.99	5.246s	6.145s	6.343s	6.165s	5.680s	4.951s
263.82	5.029s	6.029s	6.272s	6.093s	5.595s	4.858s
331.12	4.817s	5.907s	6.183s	5.993s	5.487s	4.756s
399.87	4.610s	5.786s	6.073s	5.868s	5.364s	4.648s
470.09	4.406s	5.664s	5.940s	5.725s	5.230s	4.534s
541.80	4.207s	5.534s	5.787s	5.568s	5.088s	4.417s
615.01	4.012s	5.391s	5.617s	5.400s	4.939s	4.295s
689.70	3.823s	5.232s	5.433s	5.224s	4.784s	4.171s
765.90	3.637s	5.056s	5.238s	5.040s	4.625s	4.043s
843.62	3.456s	4.864s	5.034s	4.850s	4.461s	3.912s
922.87	3.283s	4.658s	4.821s	4.654s	4.292s	3.779s
1,003.58	3.121s	4.440s	4.601s	4.453s	4.120s	3.643s
1,085.59	2.970s	4.213s	4.374s	4.247s	3.944s	3.504s
1,168.70	2.826s	3.978s	4.143s	4.037s	3.766s	3.364s
1,252.72	2.685s	3.737s	3.908s	3.825s	3.585s	3.222s
1,337.38	2.543s	3.490s	3.669s	3.610s	3.403s	3.079s
1,422.44	2.395s	3.239s	3.428s	3.393s	3.220s	2.936s
1,507.87	2.241s	2.986s	3.185s	3.175s	3.036s	2.792s
1,593.69	2.080s	2.734s	2.940s	2.957s	2.851s	2.647s
1,679.87	1.912s	2.483s	2.694s	2.737s	2.666s	2.502s
1,766.35	1.738s	2.235s	2.448s	2.515s	2.480s	2.357s
1,852.92	1.556s	1.993s	2.205s	2.295s	2.293s	2.212s
1,939.55	1.367s	1.753s	1.963s	2.076s	2.107s	2.066s
2,026.23	1.170s	1.515s	1.726s	1.859s	1.922s	1.921s
2,112.95	0.964s	1.277s	1.493s	1.646s	1.741s	1.779s
2,198.37	0.752s	1.042s	1.265s	1.439s	1.565s	1.642s
2,269.65	0.573s	0.849s	1.081s	1.274s	1.426s	1.533s
2,323.50	0.437s	0.705s	0.945s	1.152s	1.324s	1.454s

Distances in METERS.---Specific Gravity = 1.025.-------------

03-01-20 15:03:51 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONTOON



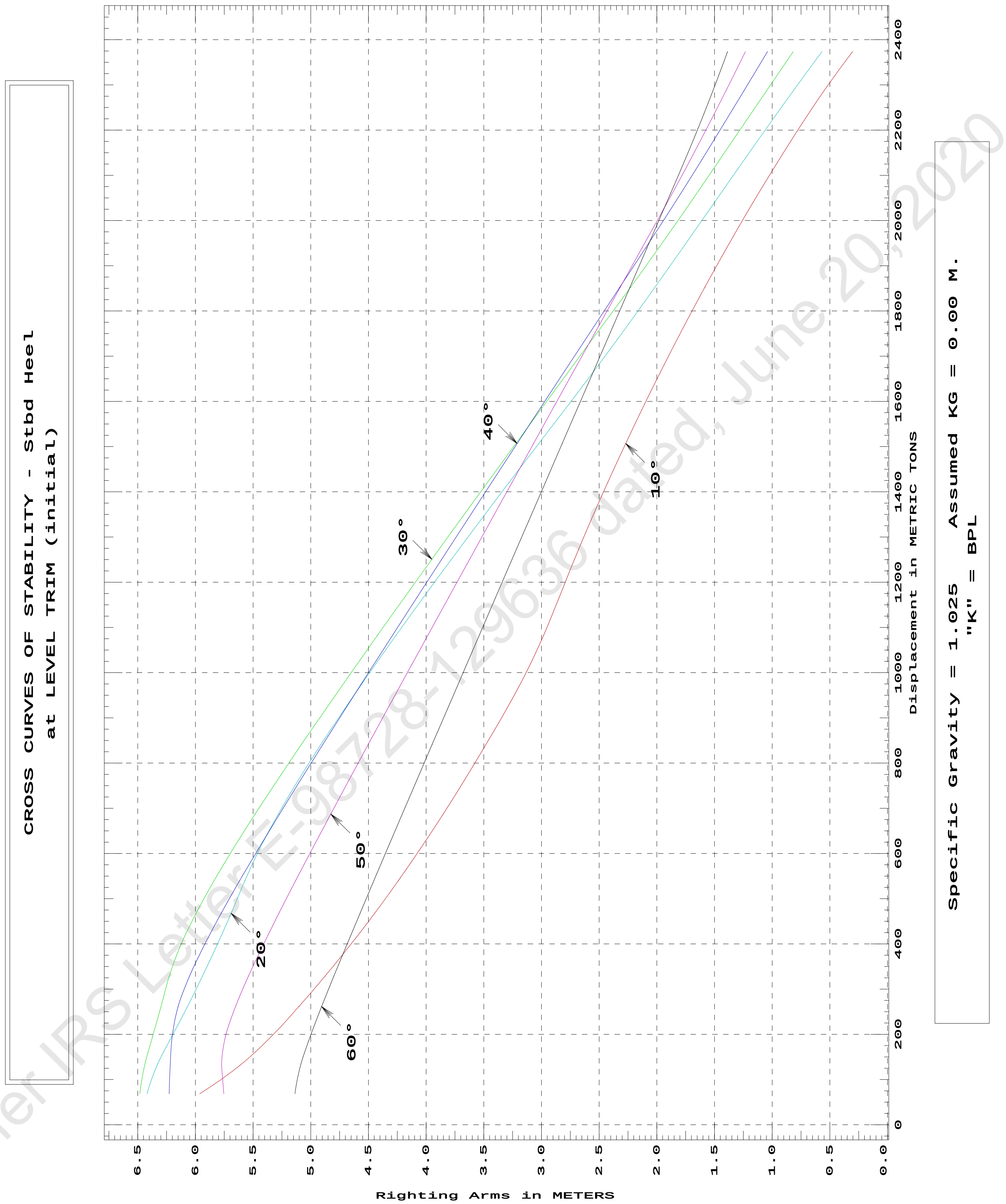
03-01-20 15:03:51 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
68.74	5.964s	6.419s	6.482s	6.228s	5.754s	5.138s
131.64	5.614s	6.332s	6.440s	6.219s	5.774s	5.089s
195.98	5.340s	6.203s	6.370s	6.201s	5.739s	5.004s
261.77	5.103s	6.069s	6.298s	6.149s	5.652s	4.906s
329.01	4.877s	5.939s	6.227s	6.051s	5.540s	4.800s
397.71	4.658s	5.812s	6.127s	5.923s	5.413s	4.688s
467.91	4.446s	5.690s	5.995s	5.776s	5.276s	4.573s
539.56	4.242s	5.572s	5.839s	5.617s	5.132s	4.454s
612.71	4.045s	5.439s	5.668s	5.448s	4.982s	4.332s
687.37	3.852s	5.282s	5.483s	5.271s	4.827s	4.207s
763.54	3.665s	5.105s	5.286s	5.086s	4.666s	4.079s
841.20	3.481s	4.912s	5.081s	4.895s	4.502s	3.948s
920.39	3.301s	4.705s	4.867s	4.699s	4.333s	3.814s
1,001.12	3.131s	4.488s	4.647s	4.497s	4.160s	3.678s
1,083.39	2.981s	4.259s	4.419s	4.290s	3.984s	3.538s
1,166.71	2.847s	4.023s	4.187s	4.079s	3.804s	3.397s
1,250.72	2.714s	3.781s	3.950s	3.865s	3.623s	3.255s
1,335.41	2.574s	3.533s	3.711s	3.649s	3.440s	3.111s
1,420.78	2.426s	3.281s	3.468s	3.431s	3.255s	2.967s
1,506.83	2.270s	3.024s	3.222s	3.211s	3.069s	2.820s
1,593.18	2.108s	2.764s	2.975s	2.989s	2.881s	2.674s
1,679.61	1.940s	2.508s	2.726s	2.767s	2.694s	2.527s
1,766.14	1.765s	2.259s	2.477s	2.544s	2.506s	2.380s
1,852.74	1.584s	2.016s	2.227s	2.321s	2.317s	2.233s
1,939.40	1.395s	1.776s	1.984s	2.097s	2.129s	2.086s
2,026.11	1.198s	1.538s	1.746s	1.877s	1.940s	1.938s
2,112.87	0.993s	1.301s	1.513s	1.662s	1.755s	1.791s
2,199.65	0.779s	1.062s	1.282s	1.454s	1.576s	1.650s
2,286.44	0.553s	0.819s	1.051s	1.246s	1.403s	1.515s
2,373.26	0.304s	0.570s	0.818s	1.041s	1.233s	1.387s

Distances in METERS.---Specific Gravity = 1.025.-------------

03-01-20 15:03:51 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONTOON



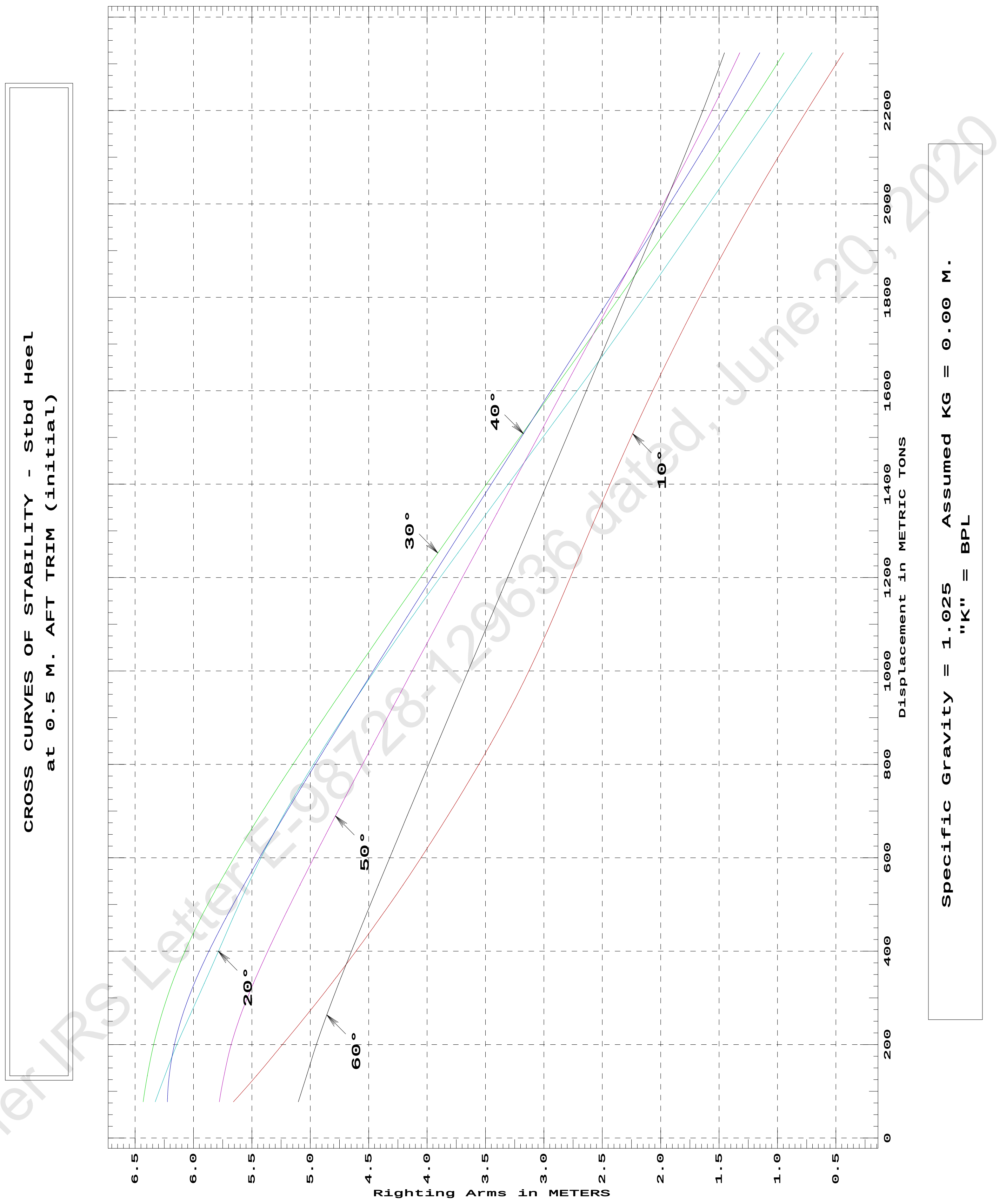
03-01-20 15:03:51 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
77.55	5.658s	6.327s	6.430s	6.222s	5.778s	5.102s
133.62	5.460s	6.244s	6.394s	6.207s	5.738s	5.031s
197.99	5.246s	6.145s	6.343s	6.165s	5.680s	4.951s
263.82	5.029s	6.029s	6.272s	6.093s	5.595s	4.858s
331.12	4.817s	5.907s	6.183s	5.993s	5.487s	4.756s
399.87	4.610s	5.786s	6.073s	5.868s	5.364s	4.648s
470.09	4.406s	5.664s	5.940s	5.725s	5.230s	4.534s
541.80	4.207s	5.534s	5.787s	5.568s	5.088s	4.417s
615.01	4.013s	5.392s	5.617s	5.400s	4.939s	4.295s
689.70	3.823s	5.232s	5.433s	5.224s	4.784s	4.171s
765.90	3.637s	5.056s	5.238s	5.040s	4.625s	4.043s
843.62	3.456s	4.864s	5.034s	4.850s	4.460s	3.912s
922.87	3.283s	4.658s	4.821s	4.654s	4.292s	3.779s
1,003.58	3.121s	4.440s	4.601s	4.453s	4.120s	3.642s
1,085.59	2.970s	4.213s	4.374s	4.247s	3.944s	3.504s
1,168.70	2.826s	3.978s	4.143s	4.037s	3.766s	3.364s
1,252.72	2.686s	3.737s	3.908s	3.824s	3.585s	3.222s
1,337.38	2.542s	3.490s	3.669s	3.609s	3.403s	3.079s
1,422.43	2.395s	3.239s	3.428s	3.393s	3.220s	2.936s
1,507.87	2.240s	2.987s	3.185s	3.175s	3.036s	2.792s
1,593.69	2.080s	2.734s	2.940s	2.957s	2.851s	2.647s
1,679.87	1.912s	2.483s	2.695s	2.737s	2.666s	2.502s
1,766.35	1.737s	2.235s	2.448s	2.516s	2.479s	2.357s
1,852.92	1.556s	1.993s	2.205s	2.295s	2.293s	2.211s
1,939.55	1.367s	1.753s	1.963s	2.076s	2.107s	2.066s
2,026.23	1.170s	1.515s	1.726s	1.859s	1.922s	1.921s
2,112.95	0.964s	1.277s	1.493s	1.646s	1.741s	1.779s
2,198.37	0.752s	1.042s	1.265s	1.439s	1.565s	1.642s
2,269.65	0.573s	0.849s	1.081s	1.274s	1.426s	1.533s
2,323.50	0.437s	0.706s	0.945s	1.152s	1.324s	1.454s

Distances in METERS.---Specific Gravity = 1.025.-------------

03-01-20 15:03:51 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONTOON



03-01-20 15:03:51 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
109.19	5.307s	6.174s	6.361s	6.178s	5.698s	4.978s
154.63	5.140s	6.085s	6.303s	6.112s	5.621s	4.895s
208.22	4.979s	5.996s	6.229s	6.029s	5.533s	4.810s
270.05	4.818s	5.906s	6.145s	5.937s	5.439s	4.722s
337.43	4.647s	5.805s	6.046s	5.834s	5.337s	4.630s
406.32	4.467s	5.688s	5.926s	5.712s	5.221s	4.528s
476.68	4.284s	5.558s	5.789s	5.573s	5.093s	4.420s
548.53	4.099s	5.416s	5.635s	5.421s	4.955s	4.306s
621.87	3.916s	5.261s	5.467s	5.258s	4.810s	4.187s
696.71	3.735s	5.094s	5.286s	5.085s	4.658s	4.064s
773.05	3.559s	4.914s	5.094s	4.904s	4.500s	3.937s
850.87	3.391s	4.721s	4.892s	4.716s	4.338s	3.807s
930.08	3.229s	4.516s	4.682s	4.522s	4.171s	3.675s
1,010.59	3.071s	4.301s	4.464s	4.323s	4.001s	3.540s
1,092.29	2.916s	4.077s	4.241s	4.120s	3.827s	3.403s
1,174.88	2.764s	3.846s	4.014s	3.914s	3.652s	3.265s
1,258.16	2.613s	3.609s	3.783s	3.706s	3.476s	3.127s
1,342.03	2.462s	3.369s	3.550s	3.496s	3.299s	2.988s
1,426.39	2.310s	3.127s	3.315s	3.285s	3.121s	2.849s
1,511.13	2.154s	2.884s	3.078s	3.073s	2.942s	2.709s
1,596.21	1.994s	2.642s	2.840s	2.860s	2.762s	2.570s
1,681.65	1.827s	2.401s	2.603s	2.647s	2.583s	2.430s
1,767.43	1.653s	2.161s	2.366s	2.434s	2.403s	2.290s
1,853.55	1.472s	1.922s	2.131s	2.221s	2.223s	2.150s
1,940.01	1.282s	1.683s	1.896s	2.009s	2.044s	2.010s
2,023.20	1.096s	1.458s	1.675s	1.810s	1.875s	1.879s
2,097.83	0.932s	1.263s	1.485s	1.638s	1.731s	1.766s
2,163.78	0.788s	1.097s	1.322s	1.491s	1.606s	1.669s
2,221.04	0.662s	0.955s	1.184s	1.366s	1.501s	1.587s
2,269.60	0.554s	0.835s	1.069s	1.262s	1.413s	1.520s

Distances in METERS.---Specific Gravity = 1.025.-------------

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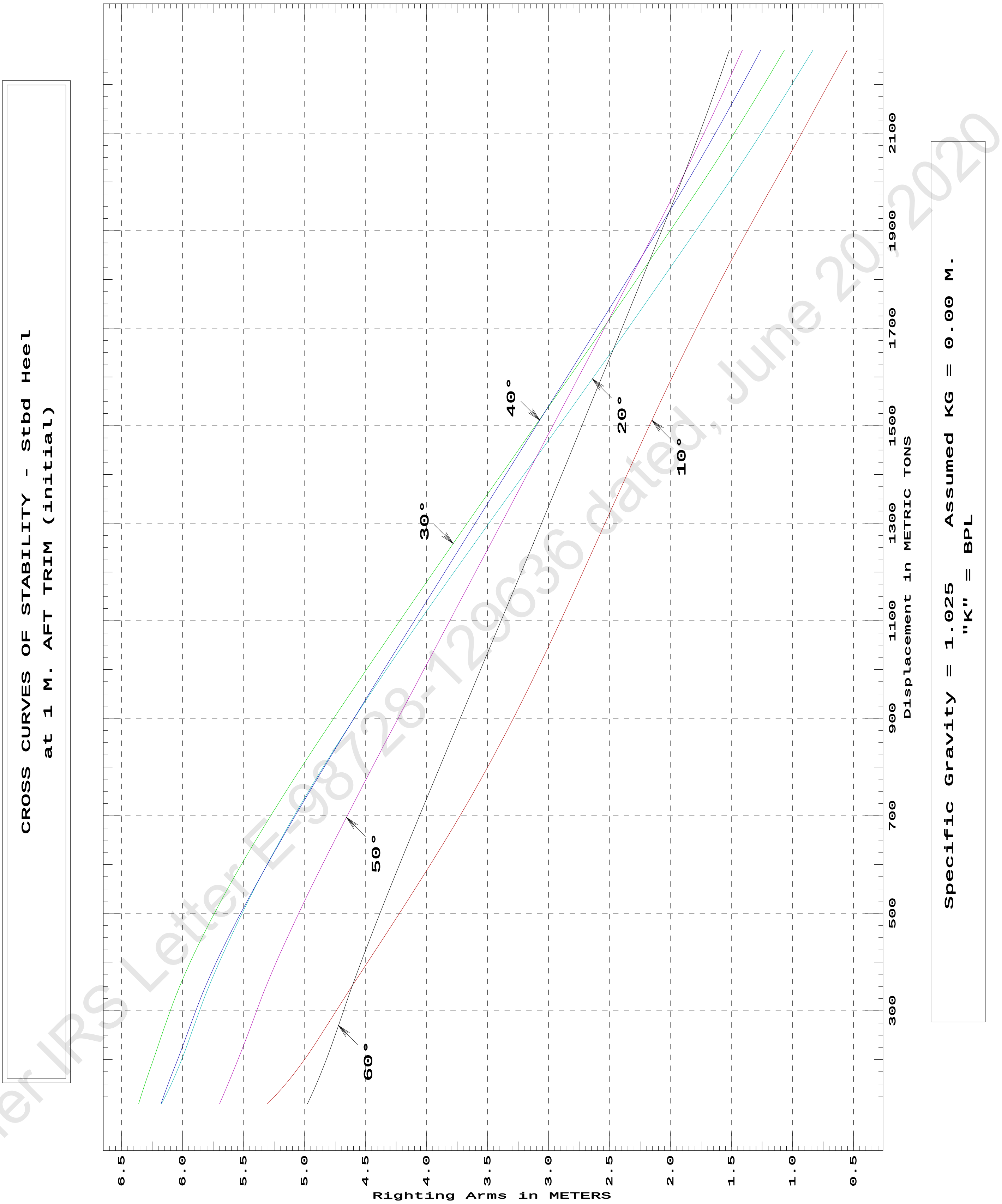
GHS 16.68C **PONTOON**



DEADWEIGHT TABLE

Draft M	Disp T	Dwt T	
0.612	397.71	8.32	
0.712	467.91	78.52	
0.812	539.56	150.17	
0.912	612.71	223.32	
1.012	687.37	297.98	
1.112	763.54	374.15	
1.212	841.20	451.81	
1.312	920.39	531.00	
1.412	1001.12	611.73	
1.512	1083.39	694.00	
1.612	1166.71	777.32	
1.712	1250.72	861.33	
1.812	1335.41	946.02	
1.912	1420.78	1031.39	
2.012	1506.83	1117.44	
2.112	1593.18	1203.79	
2.212	**1679.61**	**1290.22**	**Full Load Displacement**

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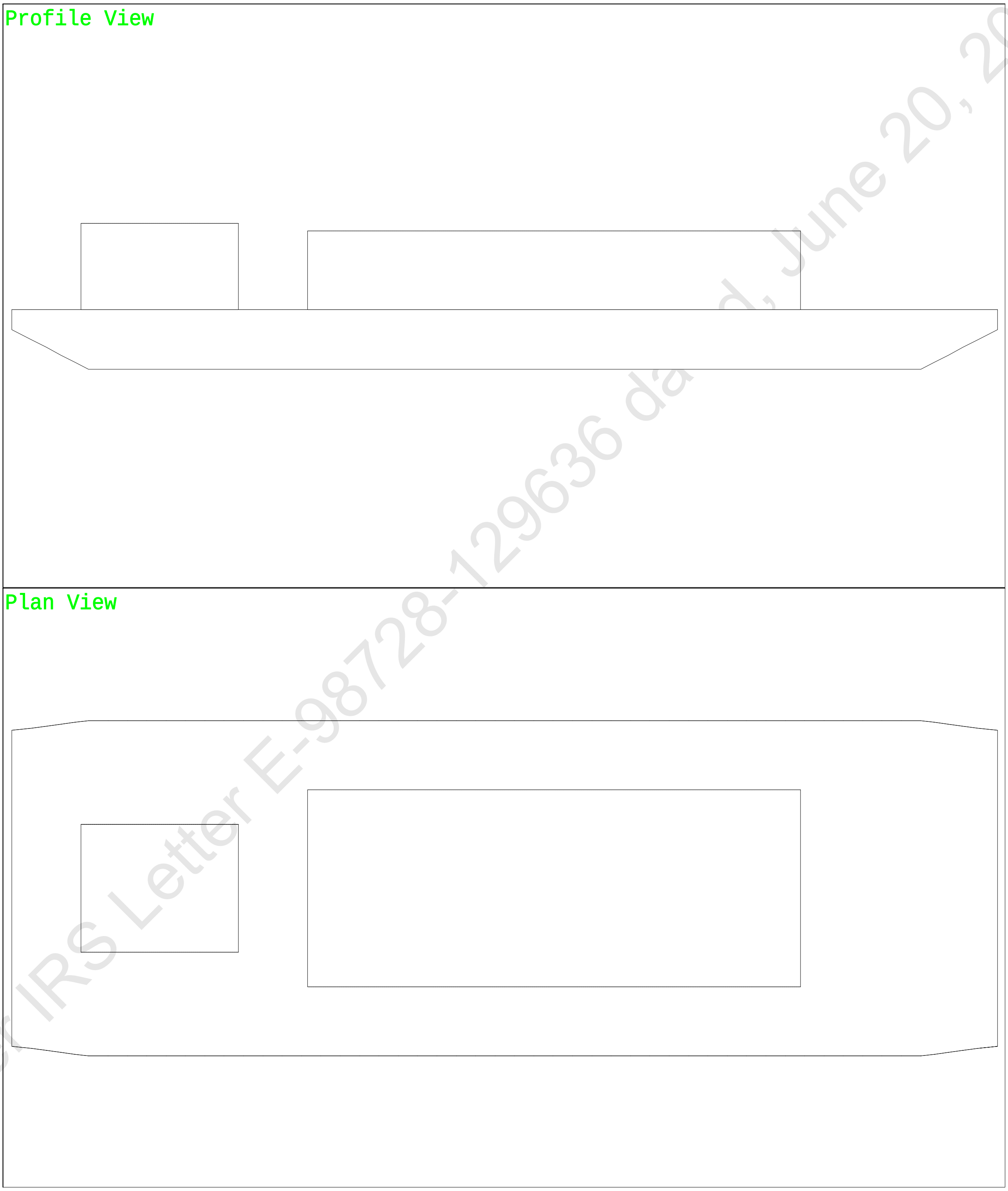
GHS 16.68C **PONTOON**

USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

MAX. PERMISSIBLE VCG CALCULATION



NOTE : THE MAX VCG VALUES ARE VERIFIED WITH RESPECT TO THE WIND PROFILE PRESENTED ABOVE,WITH THE DECK CARGO BEING STACKED TO A HEIGHT OF 4M ABOVE MAIN DECK.

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GHS 16.68C **PONTOON**

MAXIMUM VCG vs. DISPLACEMENT
Heeling moment is present from: wind
Trim = Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	16.149	79%	0d	8d
500.00	15.658	71%	0d	7d
600.00	15.086	60%	0d	6d
700.00	14.454	54%	0d	5d
800.00	13.763	50%	0d	5d
900.00	13.109	40%	0d	4d
1,000.00	12.336	33%	0d	4d
1,100.00	11.600	23%	0d	3d
1,200.00	10.808	15%	0d	3d
1,300.00	10.003	5%	0d	2d
1,400.00	9.103	3%	0d	2d
1,500.00	7.989	0%	1d	2d
1,600.00	6.940	0%	2d	1d
1,700.00	5.820	0%	5d	1d

Heeling moment is present from: wind
Trim = Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	16.420	90%	0d	9d
500.00	15.957	65%	0d	8d
600.00	15.442	56%	0d	7d
700.00	14.835	46%	0d	6d
800.00	14.162	41%	0d	6d
900.00	13.514	42%	0d	5d
1,000.00	12.751	37%	0d	5d
1,100.00	11.932	31%	0d	4d
1,200.00	11.184	18%	0d	4d
1,300.00	10.362	6%	0d	3d
1,400.00	9.426	0%	0d	3d
1,500.00	8.206	0%	2d	3d
1,600.00	7.085	0%	3d	2d
1,700.00	6.104	0%	4d	2d

Heeling moment is present from: wind
Trim = zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	16.487	95%	0d	12d
500.00	16.037	71%	0d	10d
600.00	15.560	49%	0d	9d
700.00	14.976	43%	0d	8d
800.00	14.299	40%	0d	7d
900.00	13.558	45%	0d	6d
1,000.00	12.887	40%	0d	5d
1,100.00	12.071	32%	0d	5d
1,200.00	11.293	19%	0d	5d
1,300.00	10.450	5%	0d	4d

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GHS 16.68C **PONTOON**

1,400.00	9.337	0%	1d	4d
1,500.00	8.702	0%	0d	3d
1,600.00	7.133	0%	3d	3d
1,700.00	6.168	1%	4d	3d

Heeling moment is present from: wind
Trim = Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	16.421	90%	0d	9d
500.00	15.957	65%	0d	8d
600.00	15.442	55%	0d	7d
700.00	14.836	45%	0d	6d
800.00	14.162	45%	0d	6d
900.00	13.514	42%	0d	5d
1,000.00	12.751	37%	0d	5d
1,100.00	11.932	31%	0d	4d
1,200.00	11.184	19%	0d	4d
1,300.00	10.362	6%	0d	3d
1,400.00	9.426	0%	0d	3d
1,500.00	8.202	0%	2d	3d
1,600.00	7.083	0%	3d	2d
1,700.00	6.103	0%	4d	2d

Heeling moment is present from: wind
Trim = Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	16.150	79%	0d	8d
500.00	15.659	71%	0d	7d
600.00	15.086	60%	0d	6d
700.00	14.454	54%	0d	5d
800.00	13.762	50%	0d	5d
900.00	13.110	40%	0d	4d
1,000.00	12.337	33%	0d	4d
1,100.00	11.601	23%	0d	3d
1,200.00	10.813	15%	0d	3d
1,300.00	10.001	7%	0d	2d
1,400.00	9.099	0%	0d	2d
1,500.00	8.002	0%	1d	2d
1,600.00	6.933	0%	2d	1d
1,700.00	5.833	0%	5d	1d

Distances in METERS.---Specific Gravity = 1.025.---d = degrees.

LIM	IS CODE 2008 CRITERION		Min/Max
(1)	Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad
(2)	Angle from Equilibrium to RAzero or Flood	>	20.00 deg
(3)	Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg

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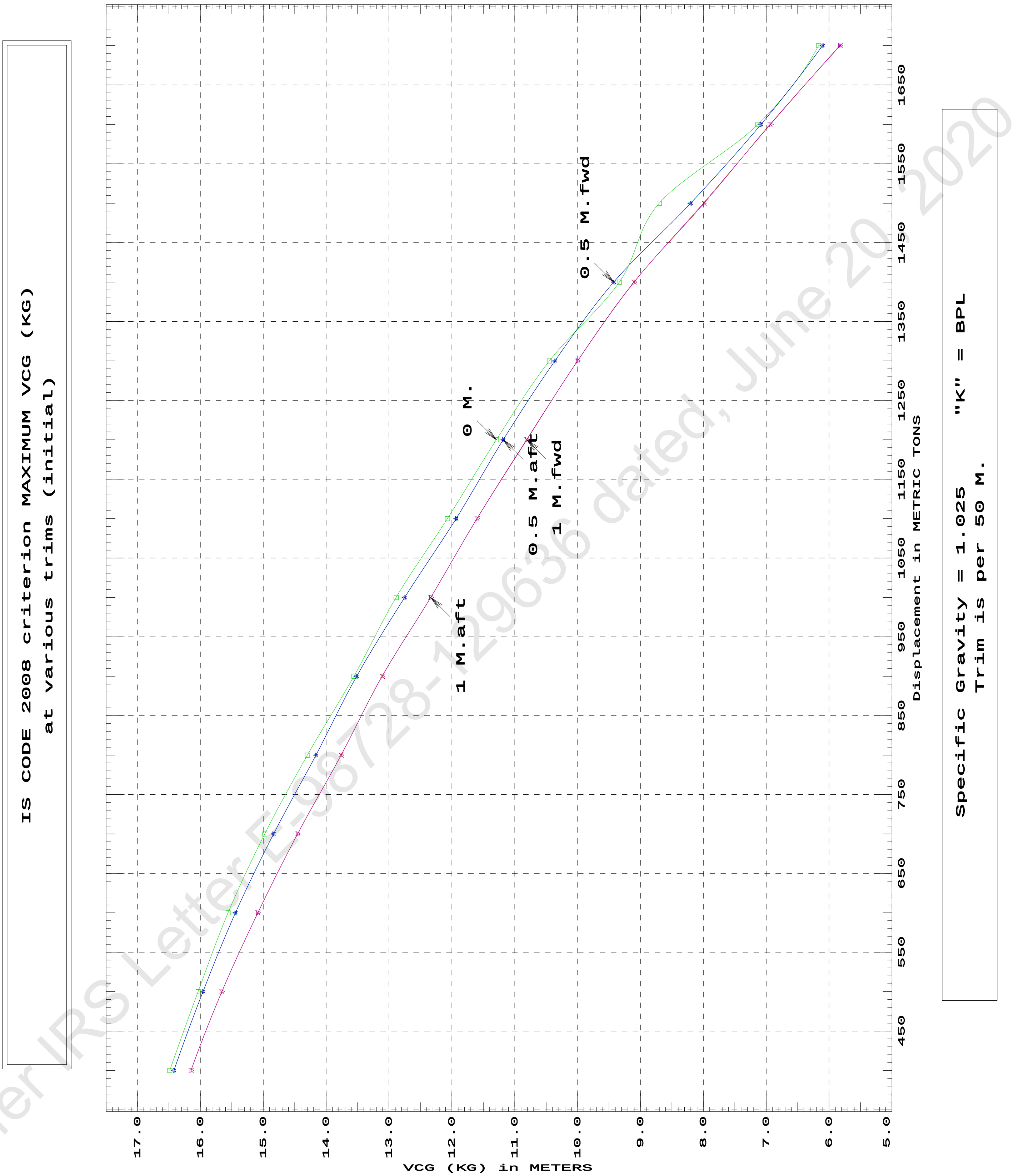
GHS 16.68C **PONTOON**



LOADING CONDITIONS

SUMMARY OF CONDITIONS					
CONDITION NO.	1	2	3	4	5
CONDITION	Condition 00 : Lightship Condition	Condition 1 : Lightship + Crane + Ballast	Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Aft	Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Stbd	Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Fwd
TOTAL TANK LOAD (T)	0.000	121.050	193.380	193.380	193.380
DWT CONST.(T)	0.000	0.000	0.000	0.000	0.000
DEADWEIGHT (T)	0.000	0.000	0.000	0.000	0.000
LIGHT WEIGHT (T)	389.390	389.390	389.390	389.390	389.390
DISPLACEMENT (T)	389.390	660.460	1679.600	1679.600	1679.610
FWD DRAFT (M)	0.600	0.999	2.158	2.248	2.338
AFT DRAFT (M)	0.600	0.953	2.266	2.176	2.086
MEAN DRAFT (M)	0.600	0.976	2.212	2.212	2.212
TRIM (M)	0.000	0.046f	0.108a	0.072f	0.252f
L.C.G. (FROM AP) (M)	25.000f	25.179f	24.781f	25.147f	25.514f
L.C.B. (FROM AP) (M)	25.000f	25.181f	24.773f	25.153f	25.533f
M.C.T. (T-CM)	22.590	25.780	34.190	34.190	34.190
KG (M)	3.000	3.300	5.033	5.033	5.033
GM_T (M)	31.428	20.837	8.277	8.277	8.279
MAX. GZ (M)	4.788	4.123	1.033	1.033	1.026
	Note:For details of deadweight constant refer respective loading condition				

SUMMARY OF CONDITIONS					
CONDITION NO.	6	7	8	9	10
CONDITION	Condition 5 : Lightship + Crane + Ballast + Deck Cargo Towing Condition with Max. draft 2.212m	Condition 6 : Lightship + Max.Deck Cargo Max. deck cargo loading condition	Condition 7 : Lightship + Load at Aft + Ballast Crane Lifting 36.6T @ 18m Radius Aft	Condition 8 : Lightship + Load at Stbd + Ballast‼Crane Lifting 36.6T @ 18m Radius Stbd	Condition 9 : Lightship + Load at Fwd + Ballast Crane Lifting 36.60T @ 18m Radius Fwd
TOTAL TANK LOAD (T)	355.100	0.000	234.050	234.050	234.050
DWT CONST.(T)	0.000	0.000	0.000	0.000	0.000
DEADWEIGHT (T)	0.000	0.000	0.000	0.000	0.000
LIGHT WEIGHT (T)	389.390	389.390	389.390	389.390	389.390
DISPLACEMENT (T)	1679.610	1679.610	1679.610	1679.610	1679.610
FWD DRAFT (M)	2.114	2.212	2.067	2.164	2.261
AFT DRAFT (M)	2.310	2.212	2.356	2.260	2.163
MEAN DRAFT (M)	2.212	2.212	2.212	2.212	2.212
TRIM (M)	0.195a	0.000	0.289a	0.095a	0.099f
L.C.G. (FROM AP) (M)	24.598f	25.000f	24.415f	24.808f	25.200f
L.C.B. (FROM AP) (M)	24.587f	25.000f	24.389f	24.799f	25.209f
M.C.T. (T-CM)	34.580	34.350	33.970	33.970	33.970
KG (M)	3.873	4.536	5.676	5.676	5.676
GM_T (M)	9.338	8.870	7.610	7.609	7.609
MAX. GZ (M)	1.240	1.149	0.907	0.916	0.915
	Note:For details of deadweight constant refer respective loading condition				

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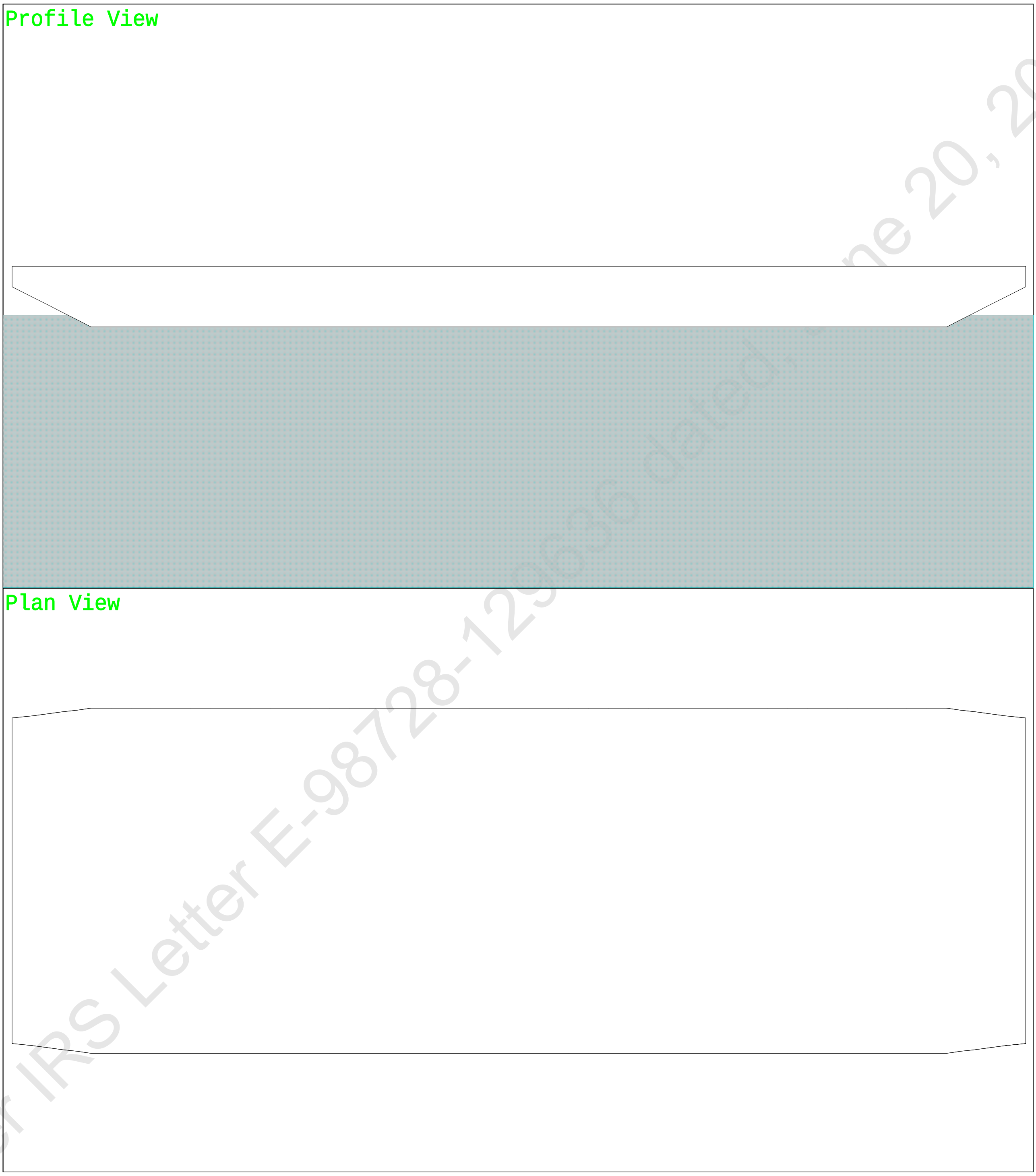
GHS 16.68C

PONTOON

Condition 00 : Lightship

Lightship condition

Condition Graphic - Draft: 0.600 Trim: zero Heel: zero



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GHS 16.68C                          PONTOON
                               Condition 00 : Lightship
                                  Lightship condition

                 WEIGHT and DISPLACEMENT and WATERPLANE STATUS
            USK draft: 0.600 @ 50.00f, 0.600 @ 25.00f, 0.600 @ 0.00
                             Trim: zero,  Heel: zero
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG-------FSM
WEIGHT                              389.39  25.000f  0.000   3.000
               Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG
  Total Tanks--------->               0.00                                0.00
                                 Displ(MT)----LCB-----TCB-----VCB
HULL                     1.025      389.39  25.000f  0.000   0.295
-----------------------------------------------------------------------------
              Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         675.7  25.000f  0.000    292.79  34.133
                                     MT/cm--------m.-MT/cm-------GML-----GMT
                                      6.93           22.59    290.08  31.428
Distances in METERS.-----------------------------------------Moments in m.-MT.

                                 SUMMARY OF LOADING
         0.0 Cu.M.   (0%) SALT WATER          0.0 Cu.M.   (0%) VOID

DRAFT AT FP (M) =    0.600
DRAFT AT MS (M) =    0.600
DRAFT AT AP (M) =    0.600
DRAFT AT FWD MARK (M) =    0.600
DRAFT AT AFT MARK (M) =    0.600
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GHS 16.68C

PONTOON

Condition 00 : Lightship

Lightship condition

HYDROSTATIC PROPERTIES

No Trim, No Heel, VCG = 3.000

	LCF	Displacement	Buoyancy-Ctr.		Weight/		Moment/		
Draft	Weight(MT)	LCB	VCB	cm	LCF	cm trim	GML	GMT
0.600	389.39	25.000f	0.295	6.93	25.000f	22.59	290.08	31.428

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 0.600 @ 50.00f, 0.600 @ 25.00f, 0.600 @ 0.00

Trim: zero, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	117.2	1.236	1.533	0.05140	9.24

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

LCG = 25.000f TCG = 0.000 VCG = 3.000

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
0.588	0.00	0.00	389.31	0.000	-0.024	0.0000	12.829
0.588	0.00	0.04s	389.34	0.000	0.000	-0.0000	12.790
0.549	0.00	5.04s	389.30	0.000	2.746	0.1198	7.790
0.330	0.00	10.04s	389.31	0.000	4.145	0.4322	2.790
0.144	0.00	12.83s	389.39	0.000	4.468	0.6432	0.000
-0.023	0.00	15.04s	389.34	0.000	4.623	0.8186	-2.210
-0.446	0.00	20.04s	389.39	0.000	4.778	1.2326	-7.210
-0.623	0.00	21.99s	389.39	0.000	4.788	1.3953	-9.159
-0.912	0.00	25.04s	389.40	0.000	4.764	1.6499	-12.210
-1.395	0.00	30.04s	389.59	0.000	4.615	2.0602	-17.210
-1.869	0.00	35.04s	389.34	0.000	4.342	2.4519	-22.210
-2.328	0.00	40.04s	389.54	0.000	3.984	2.8158	-27.210
-2.770	0.00	45.04s	389.53	0.000	3.566	3.1457	-32.210
-3.191	0.00	50.04s	389.52	0.000	3.103	3.4370	-37.210
-3.587	0.00	55.04s	389.51	0.000	2.604	3.6863	-42.210
-3.958	0.00	60.04s	389.28	0.000	2.076	3.8907	-47.210

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 9.24 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained
(1)	Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	1.3953 P
(2)	Angle from Equilibrium to RAzero or Flood	>	20.00	deg	LARGE
(3)	Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	12.79 P

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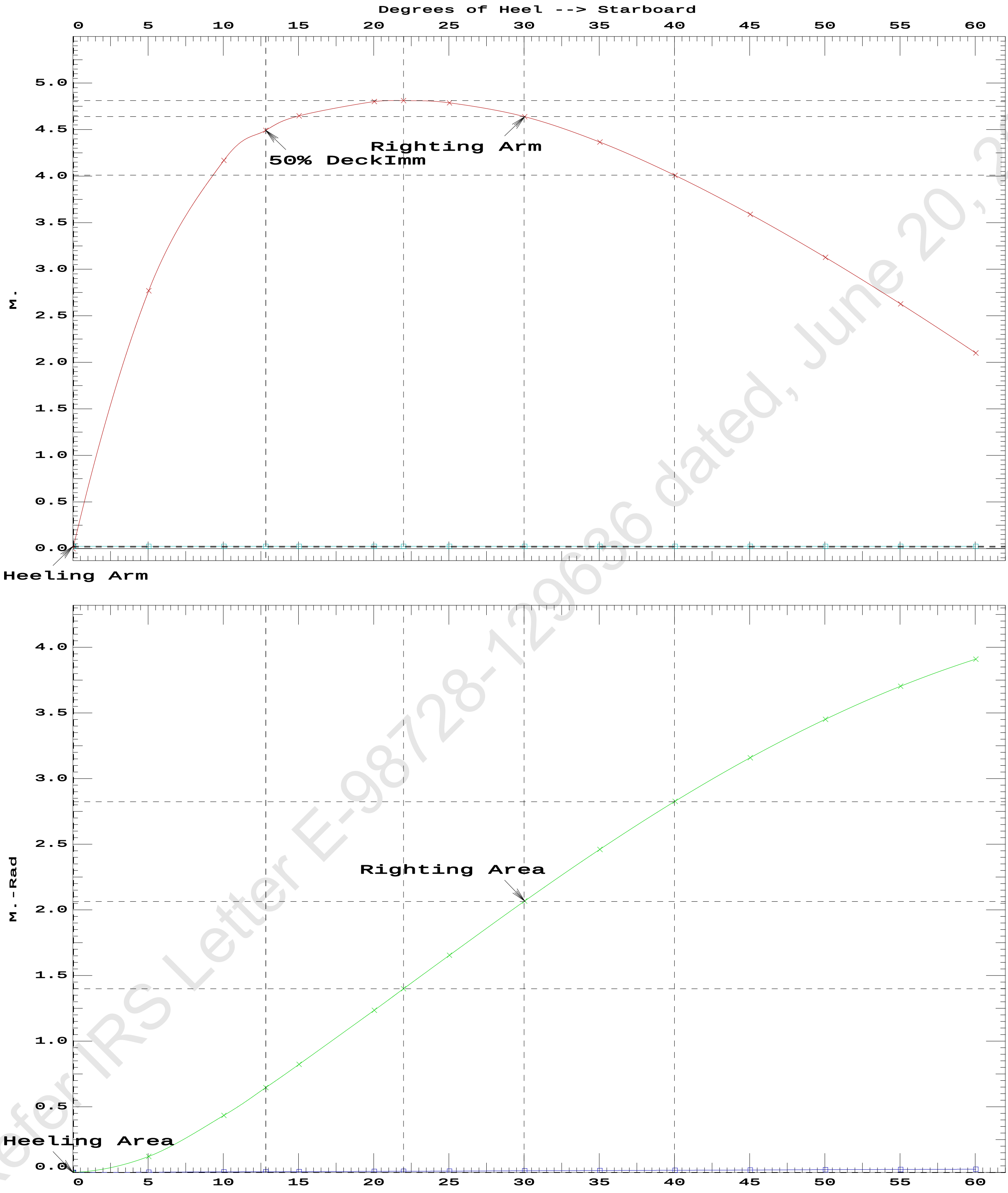
GHS 16.68C

PONTOON

Condition 00 : Lightship

Lightship condition



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GHS 16.68C PONTOON

Condition 1 : Lightship + Crane + Ballast

Condition Graphic - Draft: 0.999 @ 50.000f, 0.953 @ 0.000 Heel: zero

Tanks 31 07_WBTK.P 32 07_WBTK.C 33 07_WBTK.S

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GHS 16.68C

PONTOON

Condition 1 : Lightship + Crane + Ballast

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 0.999 @ 50.00f, 0.976 @ 25.00f, 0.953 @ 0.00
Trim: Fwd 0.046/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			389.39	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Total Fixed--------->			539.39	20.133f	0.000	3.612	
07_WBTK.P	1.000	1.025	40.19	47.659f	4.971p	1.911	41.85*
07_WBTK.C	1.000	1.025	40.67	47.663f	0.000	1.901	41.85*
07_WBTK.S	1.000	1.025	40.19	47.659f	4.971s	1.911	41.85*
Total Tanks--------->			121.05	47.660f	0.000	1.908	125.56
Total Weight-------->			660.44	25.179f	0.000	3.300	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	660.46	25.181f	0.000	0.493	
Righting Arms:				0.000	0.000		

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	730.3	25.056f	0.000	197.99	23.643
		MT/cm		m.-MT/cm	GML	GMT
		7.49		25.78	195.18	20.837

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

118.1 Cu.M. (7%) SALT WATER   0.0 Cu.M. (0%) VOID

150.00 MT Crane

DRAFT AT FP (M) = 0.999
DRAFT AT MS (M) = 0.976
DRAFT AT AP (M) = 0.953
DRAFT AT FWD MARK (M) = 0.995
DRAFT AT AFT MARK (M) = 0.957

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GHS 16.68C

PONTOON

Condition 1 : Lightship + Crane + Ballast

HYDROSTATIC PROPERTIES
Trim: Fwd 0.046/50.000, No Heel, VCG = 3.300

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
0.976	660.46	25.181f	0.493	7.49	25.056f	25.78	195.18	20.837

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 125.6 m.-MT

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 0.999 @ 50.00f, 0.976 @ 25.00f, 0.953 @ 0.00
Trim: Fwd 0.046/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	100.2	1.038	1.518	0.05140	7.82
CRANE.C	34.4	4.244	4.725	0.05140	8.35
Total wind heeling moment to starboard------------->					16.17

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.179f TCG = 0.000 VCG = 3.300
Free Surface Adjustment: 0.190
Adjusted CG: LCG = 25.178f TCG = 0.000 VCG = 3.490

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel--->	Res. Area	50% DeckImm Angle
0.941	0.05f	0.00	660.46	0.000	-0.024	0.0000	8.184
0.941	0.05f	0.06s	660.42	0.000	0.000	-0.0000	8.122
0.902	0.05f	5.06s	660.45	0.000	1.821	0.0795	3.122
0.845	0.05f	8.18s	660.44	0.000	2.867	0.2075	0.000
0.782	0.06f	10.06s	660.43	0.000	3.302	0.3086	-1.878
0.517	0.07f	15.06s	660.43	0.000	3.925	0.6290	-6.878
0.174	0.09f	20.06s	660.48	0.000	4.122	0.9832	-11.878
0.160	0.09f	20.26s	660.45	0.000	4.123	0.9972	-12.073
-0.178	0.11f	25.06s	660.43	0.000	4.022	1.3407	-16.878
-0.529	0.13f	30.06s	660.44	0.000	3.777	1.6820	-21.878
-0.877	0.15f	35.06s	660.44	0.000	3.447	1.9978	-26.878
-1.217	0.17f	40.06s	660.43	0.000	3.062	2.2822	-31.878
-1.549	0.19f	45.06s	660.43	0.000	2.636	2.5312	-36.878
-1.869	0.21f	50.06s	660.43	0.000	2.179	2.7415	-41.878
-2.175	0.23f	55.06s	660.44	0.000	1.698	2.9109	-46.878
-2.465	0.24f	60.06s	660.43	0.000	1.199	3.0374	-51.878

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 125.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 16.17 (constant)

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GHS 16.68C **PONTOON**

Condition 1 : Lightship + Crane + Ballast

```
LIM------------CRITERIA:IS CODE 2008PART B CHAPTER----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.9972 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg      LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     8.12 P
-----------------------------------------------------------------------------
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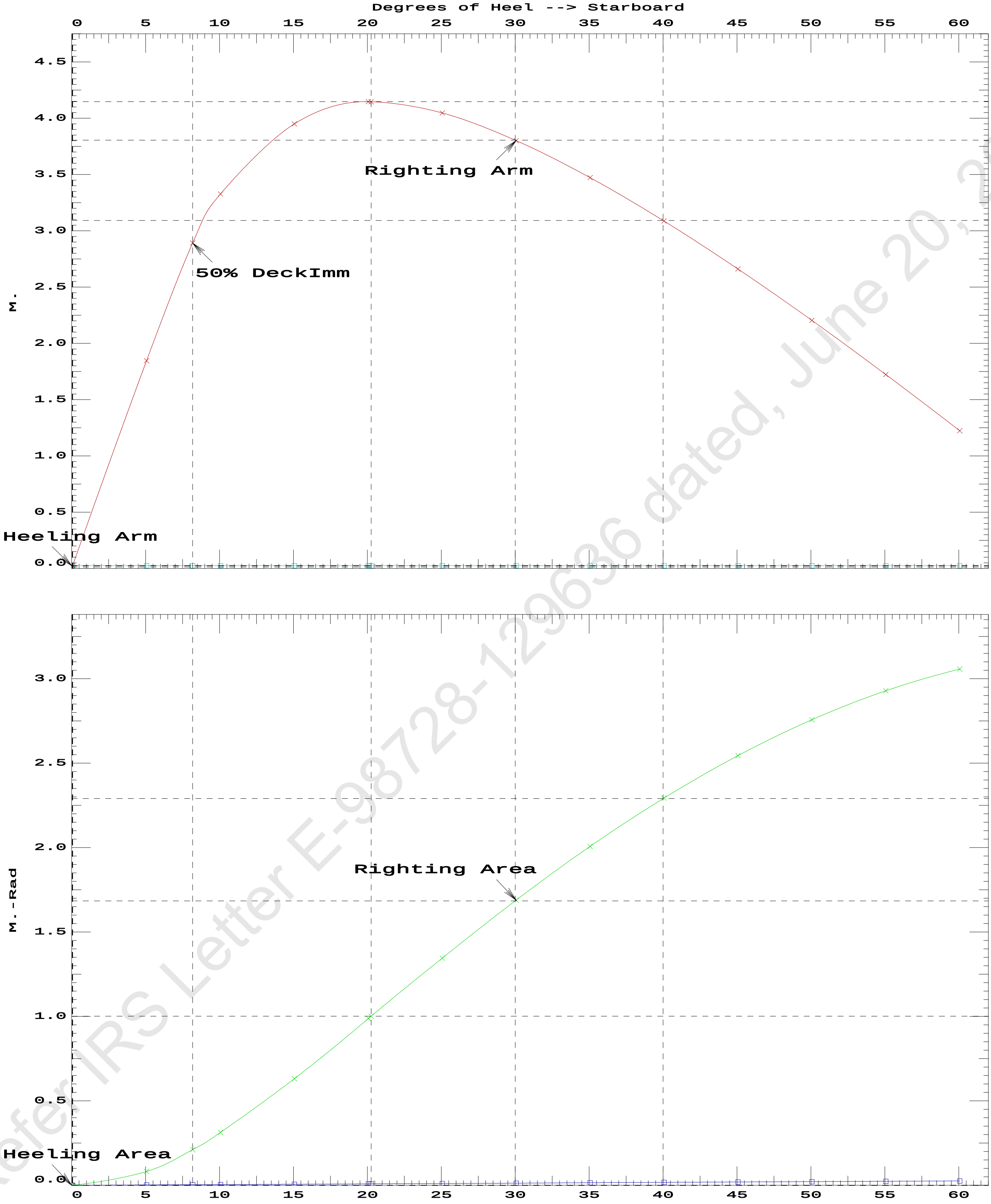
GHS 16.68C PONTOON

Condition 1 : Lightship + Crane + Ballast



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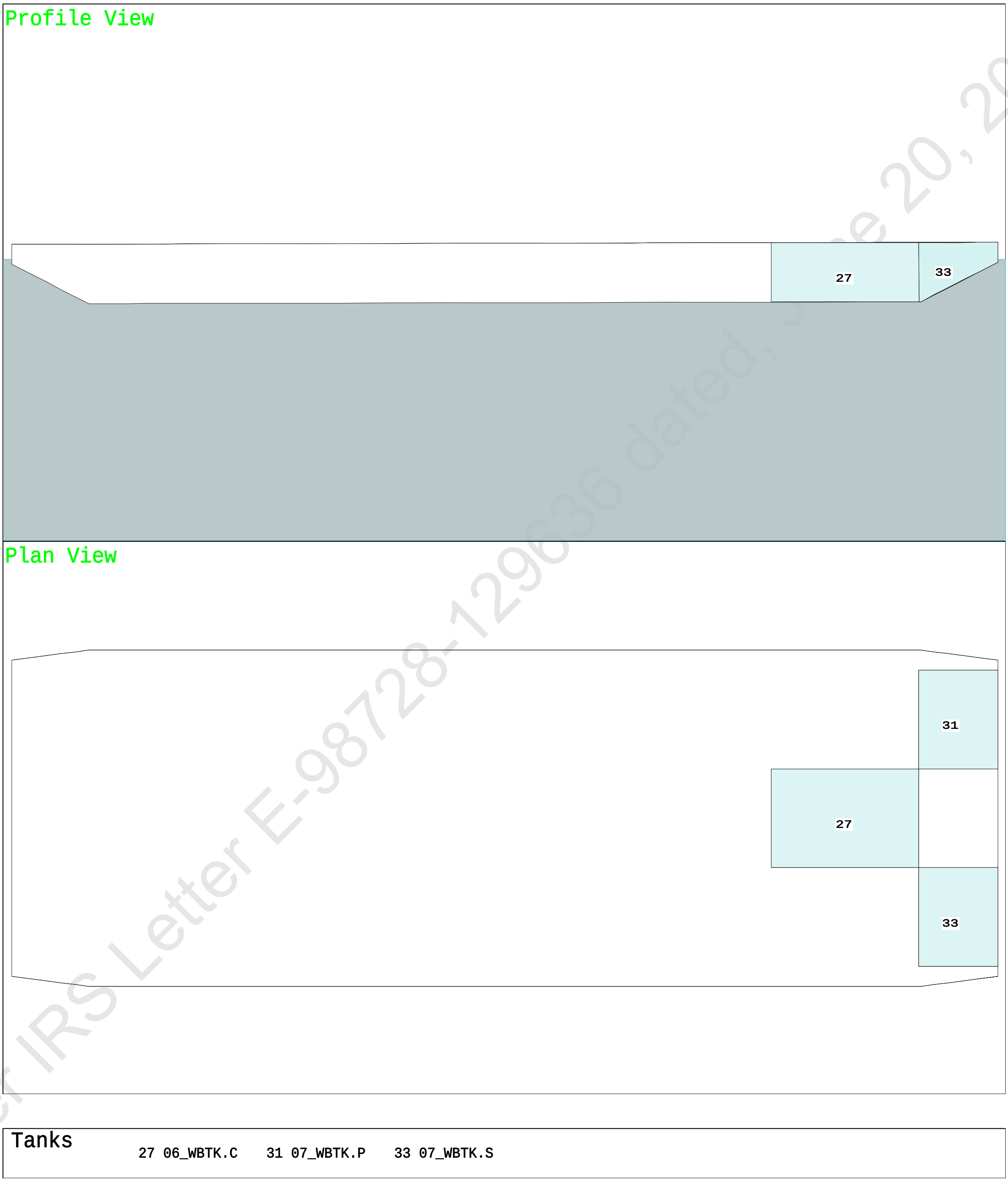
GHS 16.68C PONTOON

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Aft

Condition Graphic - Draft: 2.158 @ 50.000f, 2.266 @ 0.000 Heel: zero



27-01-20 14:18:42 Cybermarine Technologies PTE. Ltd.
GHS 16.68C

PONTOON

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.158 @ 50.00f, 2.212 @ 25.00f, 2.266 @ 0.00
Trim: Aft 0.108/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			389.39	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	4.500a	0.000	33.218	
Deck Cargo			895.54	25.000f	0.000	5.000	
Total Fixed--------->			1,486.23	22.216f	0.000	5.470	
06_WBTK.C	1.000	1.025	113.01	42.250f	0.000	1.500	78.48*
07_WBTK.P	1.000	1.025	40.19	47.659f	4.971p	1.911	41.85*
07_WBTK.S	1.000	1.025	40.19	47.659f	4.971s	1.911	41.85*
Total Tanks--------->			193.38	44.498f	0.000	1.671	162.18
Total Weight-------->			1,679.61	24.781f	0.000	5.033	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,679.60	24.773f	0.000	1.163

Righting Arms: 0.000 0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	843.7	24.988f	0.000	105.64	12.147

MT/cm	m.-MT/cm	GML	GMT
8.65	34.19	101.77	8.277

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

188.7 Cu.M. (11%) SALT WATER 0.0 Cu.M. (0%) VOID

150.00 MT Crane 51.30 MT Crane Load
895.54 MT Deck Cargo

DRAFT AT FP (M) = 2.158
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.266
DRAFT AT FWD MARK (M) = 2.167
DRAFT AT AFT MARK (M) = 2.257

HYDROSTATIC PROPERTIES
Trim: Aft 0.108/50.000, No Heel, VCG = 5.033

Draft	LCF Displacement Weight(MT)	LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.212	1,679.60	24.773f	1.163	8.65	24.988f	34.19	101.77	8.277

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 162.2 m.-MT

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GHS 16.68C **PONTOON**

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Aft

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 2.158 @ 50.00f, 2.212 @ 25.00f, 2.266 @ 0.00
Trim: Aft 0.108/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.406	1.478	0.05140	3.08
CRANE.C	34.4	2.955	4.027	0.05140	7.12
DECKCARGO.C	98.0	2.805	3.878	0.05140	19.53
Total wind heeling moment to starboard------------->					29.73

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.781f TCG = 0.000 VCG = 5.033
Free Surface Adjustment: 0.097
Adjusted CG: LCG = 24.781f TCG = 0.000 VCG = 5.129

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
2.254	0.12a	0.00	1,679.6	0.000	-0.017	0.0000	2.575
2.254	0.12a	0.12s	1,679.6	0.000	0.000	-0.0000	2.455
2.251	0.12a	2.57s	1,679.6	0.000	0.357	0.0076	0.000
2.243	0.13a	5.12s	1,679.6	0.000	0.723	0.0317	-2.545
2.323	0.19a	10.12s	1,679.6	0.000	1.031	0.1134	-7.545
2.335	0.19a	10.51s	1,679.6	0.000	1.033	0.1204	-7.936
2.507	0.24a	15.12s	1,679.6	0.000	0.944	0.2021	-12.545
2.730	0.30a	20.12s	1,679.6	0.000	0.729	0.2760	-17.545
2.968	0.36a	25.12s	1,679.6	0.000	0.453	0.3280	-22.545
3.196	0.44a	30.12s	1,679.6	0.000	0.134	0.3539	-27.545
3.281	0.47a	32.12s	1,679.6	0.000	0.000	0.3563	-29.541
3.401	0.51a	35.12s	1,679.6	0.000	-0.207	0.3509	-32.545
3.581	0.59a	40.12s	1,679.6	0.000	-0.558	0.3176	-37.545
3.733	0.66a	45.12s	1,679.6	0.000	-0.912	0.2535	-42.545
3.857	0.72a	50.12s	1,679.6	0.000	-1.263	0.1586	-47.545
3.951	0.78a	55.12s	1,679.6	0.000	-1.607	0.0334	-52.545
4.014	0.83a	60.12s	1,679.6	0.000	-1.941	-0.1215	-57.545

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 162.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 29.73 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.1204	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	32.00	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.45	P

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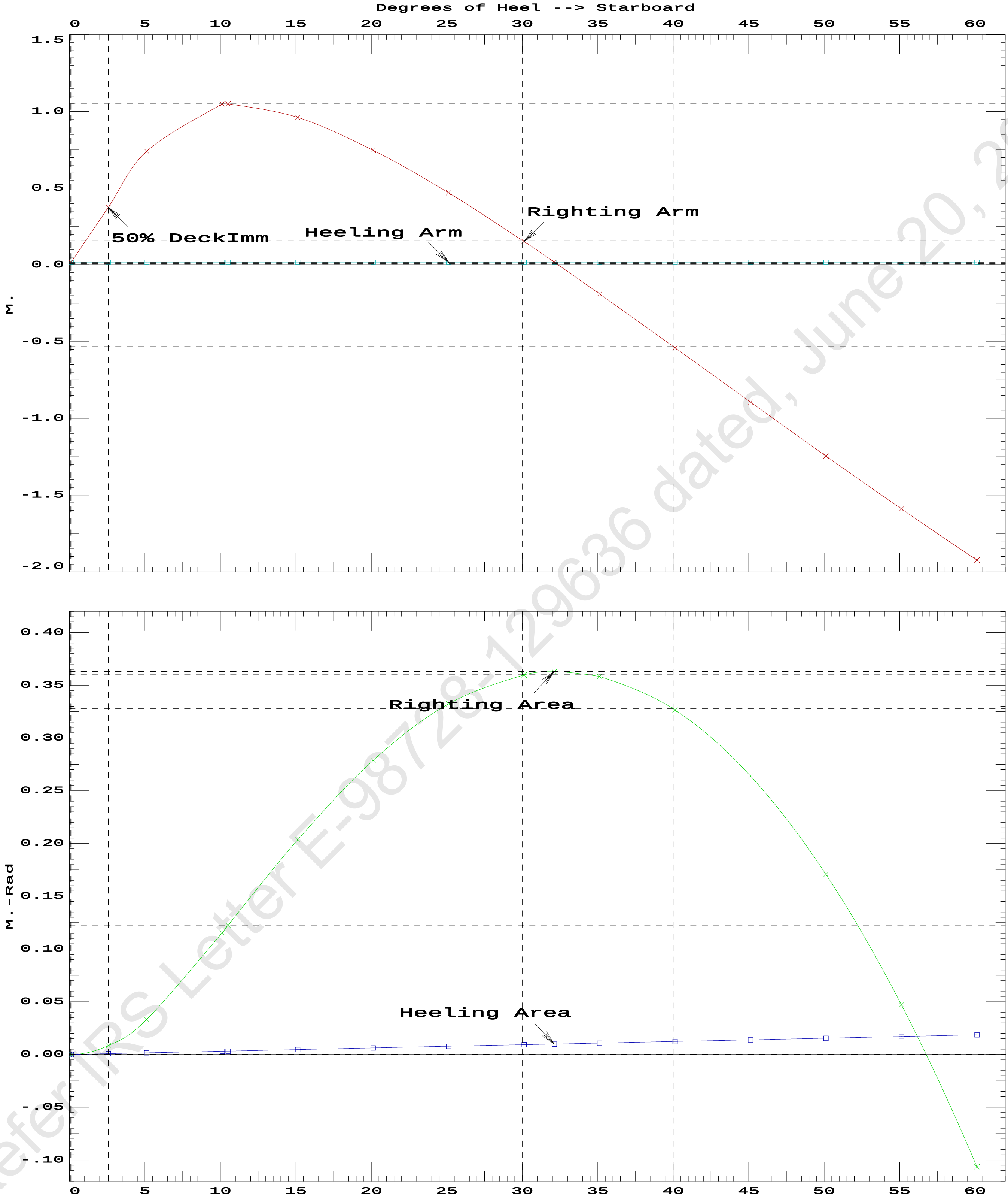
GHS 16.68C
PONTOON

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo

Crane Lifting 51.30T @ 12m Radius Aft



27-01-20 14:18:42 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.781f TCG = 0.000 VCG = 5.033
Free Surface Adjustment: 0.097
Adjusted CG: LCG = 24.781f TCG = 0.000 VCG = 5.129

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.254	0.12a	0.00s	1,679.6	0.000	0.000	0.0000
2.243	0.13a	5.00s	1,679.6	0.000	0.724	0.0316
2.320	0.18a	10.00s	1,679.6	0.000	1.048	0.1142
2.336	0.19a	10.54s	1,679.6	0.000	1.049	0.1239
2.502	0.24a	15.00s	1,679.6	0.000	0.965	0.2044
2.725	0.30a	20.00s	1,679.6	0.000	0.752	0.2809
2.963	0.36a	25.00s	1,679.6	0.000	0.477	0.3350
3.191	0.44a	30.00s	1,679.6	0.000	0.159	0.3630
3.291	0.47a	32.36s	1,679.6	0.000	0.000	0.3663
3.397	0.51a	35.00s	1,679.6	0.000	-0.182	0.3622
3.577	0.59a	40.00s	1,679.6	0.000	-0.533	0.3311
3.730	0.66a	45.00s	1,679.6	0.000	-0.886	0.2692
3.855	0.72a	50.00s	1,679.6	0.000	-1.238	0.1765
3.949	0.78a	55.00s	1,679.6	0.000	-1.582	0.0534
4.013	0.82a	60.00s	1,679.6	0.000	-1.917	-0.0993

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 162.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	176264 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		1090.43 P

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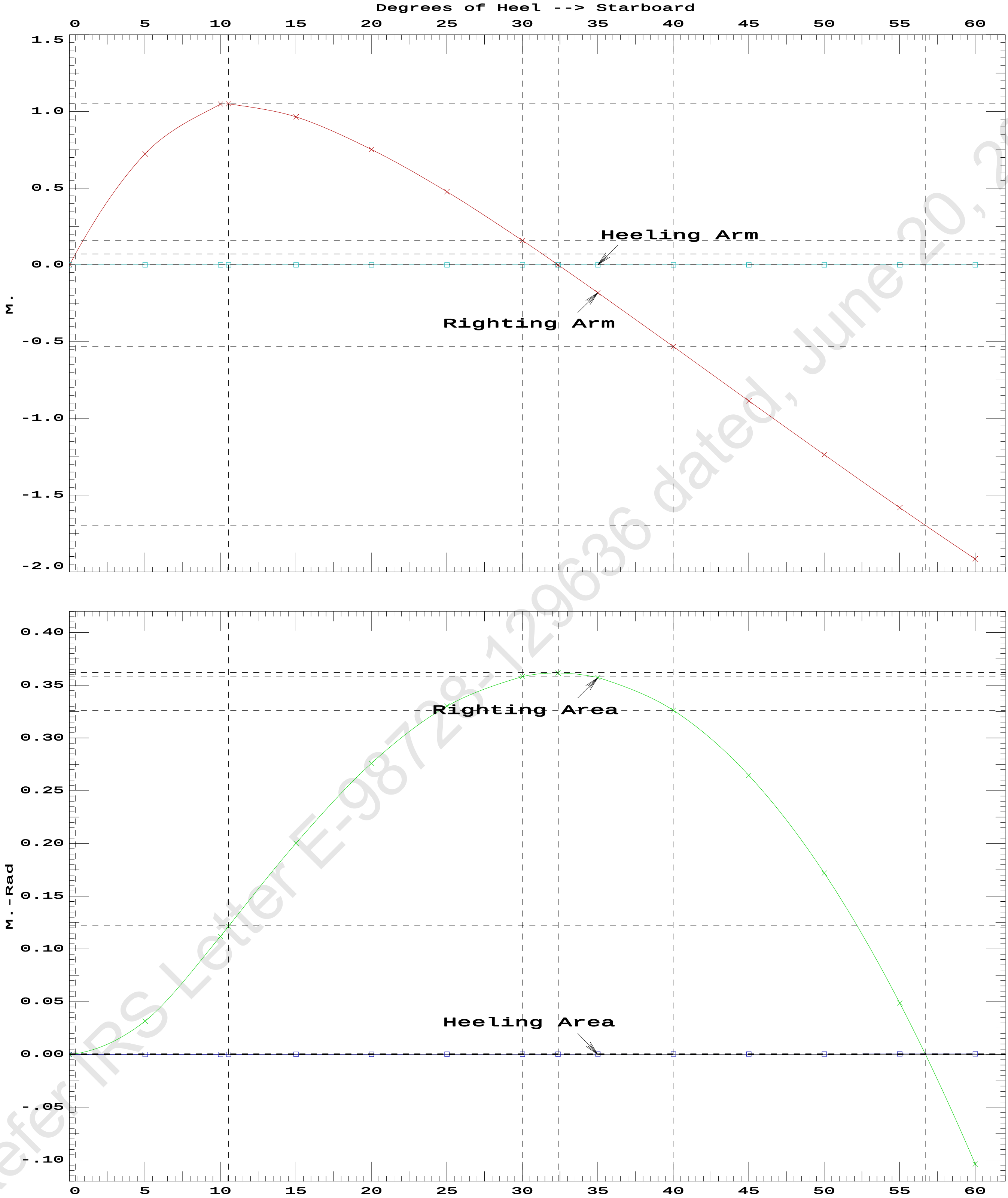
GHS 16.68C PONTOON

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo

Crane Lifting 51.30T @ 12m Radius Aft



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GHS 16.68C **PONTOON**

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo

Crane Lifting 51.30T @ 12m Radius Aft

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.158 @ 50.00f, 2.212 @ 25.00f, 2.266 @ 0.00

Trim: Aft 0.108/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			389.39	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	4.500a	0.000	33.218	
Deck Cargo			895.54	25.000f	0.000	5.000	
Total Fixed--------->			1,486.23	22.216f	0.000	5.470	
06_WBTK.C	1.000	1.025	113.01	42.250f	0.000	1.500	78.48*
07_WBTK.P	1.000	1.025	40.19	47.659f	4.971p	1.911	41.85*
07_WBTK.S	1.000	1.025	40.19	47.659f	4.971s	1.911	41.85*
Total Tanks--------->			193.38	44.498f	0.000	1.671	162.18
Total Weight-------->			1,679.61	24.781f	0.000	5.033	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,679.61	24.773f	0.001	1.163	

Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	843.7	24.988f	0.000	105.64	12.147
		MT/cm		m.-MT/cm	GML	GMT
		8.65		34.19	101.77	8.277

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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GHS 16.68C                            PONTOON

                          After hook load drop
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 2.319 @ 50.00f, 2.153 @ 25.00f, 1.986 @ 0.00
                  Trim: Fwd 0.333/50.000,  Heel: 0.00 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           389.39  25.000f  0.000   3.000
Crane                                150.00   7.500f  0.000   5.200
Deck Cargo                           895.54  25.000f  0.000   5.000
  Total Fixed--------->            1,434.93  23.171f  0.000   4.478
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.C      1.000    1.025        113.01  42.250f  0.000   1.500     78.48*
07_WBTK.P      1.000    1.025         40.19  47.659f  4.971p  1.911     41.85*
07_WBTK.S      1.000    1.025         40.19  47.659f  4.971s  1.911     41.85*
  Total Tanks--------->              193.38  44.498f  0.000   1.671    162.18
  Total Weight-------->            1,628.31  25.703f  0.000   4.145
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,628.30  25.724f  0.002s  1.134
------------------------------------------------------------------------------
               Righting Arms:                 0.000   0.002s
               External Arms:                 0.000   0.001
      Residual Righting Arms:                 0.000   0.001s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         836.1  25.247f  0.030s   106.05  12.449
                                      MT/cm
                                       8.57
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

                Theta3 is 0.07 degrees, as demonstrated below.
```

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GHS 16.68C **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 23.171f TCG = 0.000 VCG = 4.478

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.974	0.38f	0.00	1,628.3	0.000	-0.001	0.0000
1.974	0.38f	0.07p	1,628.4	0.000	0.014	0.0000
1.957	0.39f	5.00p	1,628.3	0.000	0.836	0.0366
1.921	0.55f	10.00p	1,628.3	0.000	1.309	0.1354
1.954	0.66f	13.35p	1,628.3	0.000	1.356	0.2144
1.981	0.72f	15.00p	1,628.4	0.000	1.345	0.2533
2.086	0.89f	20.00p	1,628.3	0.000	1.230	0.3674
2.181	1.09f	25.00p	1,628.4	0.000	1.033	0.4667
2.250	1.32f	30.00p	1,628.3	0.000	0.786	0.5464

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1434.93 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 0.1	>	1.000	LARGE

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GHS 16.68C **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 23.171f TCG = 0.000 VCG = 4.478

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.974	0.38f	0.00	1,628.3	0.000	-0.001	0.0000
1.974	0.38f	0.07p	1,628.4	0.000	0.014	0.0000
1.957	0.39f	5.00p	1,628.3	0.000	0.836	0.0366
1.921	0.55f	10.00p	1,628.3	0.000	1.309	0.1354
1.954	0.66f	13.35p	1,628.3	0.000	1.356	0.2144
1.981	0.72f	15.00p	1,628.4	0.000	1.345	0.2533
2.086	0.89f	20.00p	1,628.3	0.000	1.230	0.3674
2.181	1.09f	25.00p	1,628.4	0.000	1.033	0.4667
2.250	1.32f	30.00p	1,628.3	0.000	0.786	0.5464
2.298	1.54f	35.00p	1,628.3	0.000	0.511	0.6032
2.327	1.76f	40.00p	1,628.3	0.000	0.220	0.6353
2.336	1.92f	43.68p	1,628.3	0.000	0.000	0.6424
2.336	1.97f	45.00p	1,628.3	0.000	-0.079	0.6415

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1434.93 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	43.61 P

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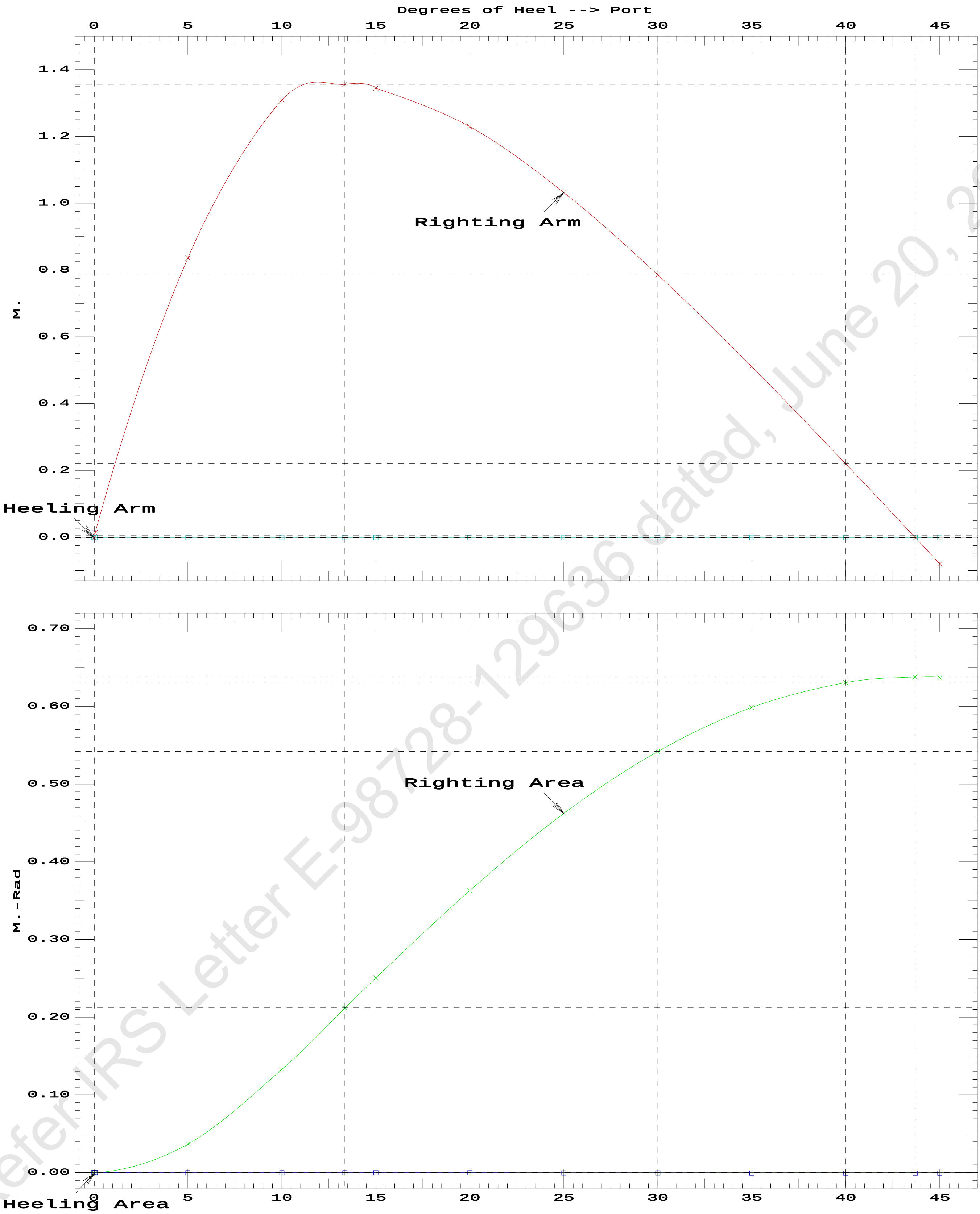
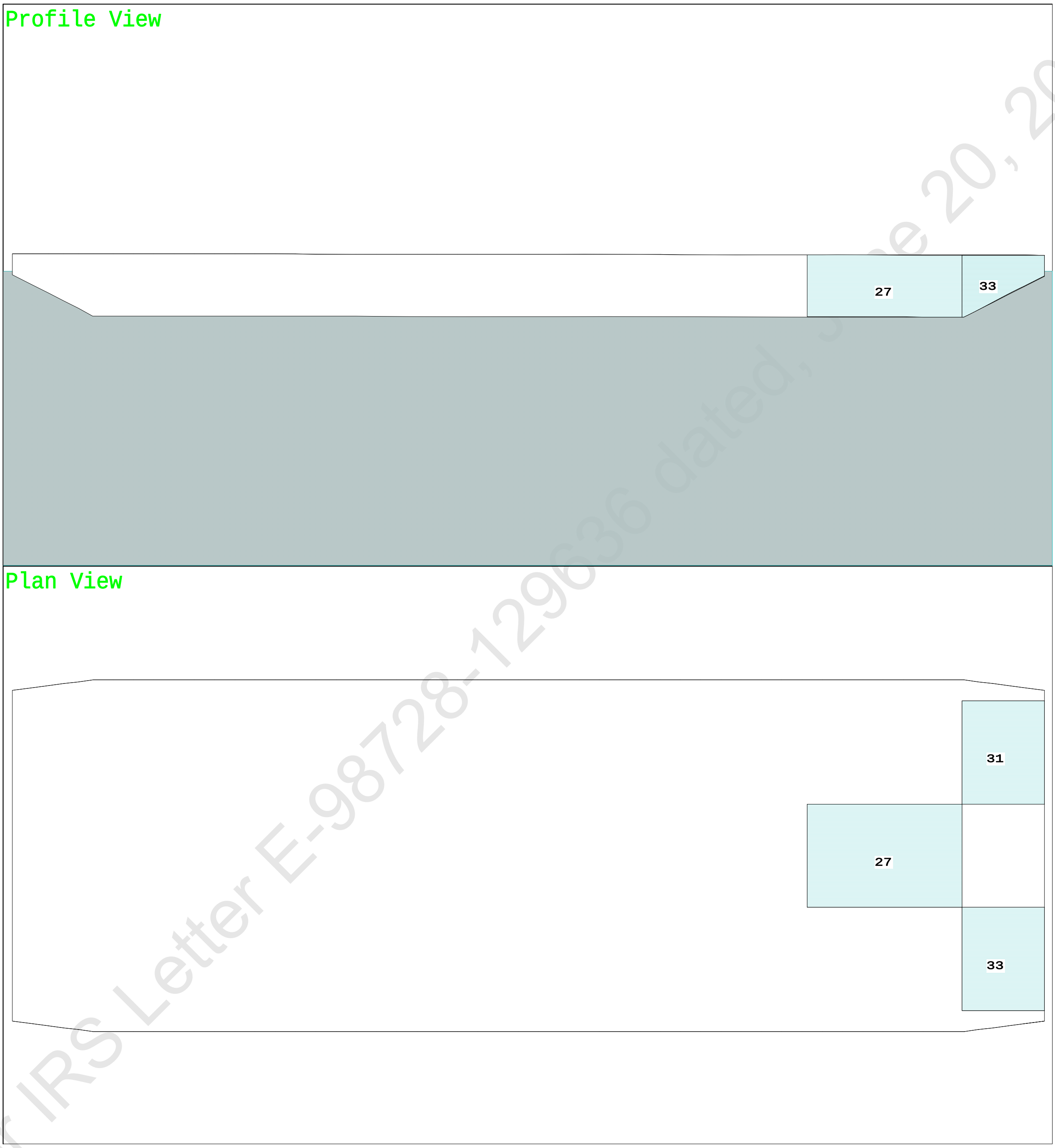
GHS 16.68C **PONTOON**



27-01-20 14:18:51 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Stbd

Condition Graphic - Draft: 2.248 @ 50.000f, 2.176 @ 0.000 Heel: zero



Tanks 27 06_WBTK.C 31 07_WBTK.P 33 07_WBTK.S

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27-01-20 14:18:51    Cybermarine Technologies PTE. Ltd.
GHS 16.68C                         PONTOON
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Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo

Crane Lifting 51.30T @ 12m Radius Stbd

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                 WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 2.248 @ 50.00f, 2.212 @ 25.00f, 2.176 @ 0.00
                     Trim: Fwd 0.072/50.000,  Heel: zero
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                          389.39  25.000f  0.000   3.000
Crane                               150.00   7.500f  0.000   5.200
Crane Load                           51.30   7.500f  0.000  33.218
Deck Cargo                          895.54  25.000f  0.000   5.000
  Total Fixed--------->           1,486.23  22.630f  0.000   5.470
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.C      1.000    1.025       113.01  42.250f  0.000   1.500    78.48*
07_WBTK.P      1.000    1.025        40.19  47.659f  4.971p  1.911    41.85*
07_WBTK.S      1.000    1.025        40.19  47.659f  4.971s  1.911    41.85*
  Total Tanks--------->             193.38  44.498f  0.000   1.671   162.18
  Total Weight-------->           1,679.61  25.147f  0.000   5.033
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025     1,679.60  25.153f  0.000   1.163
------------------------------------------------------------------------------
               Righting Arms:                      0.000   0.000
Part--------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          843.7  25.008f  0.000    105.64  12.147
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        8.65           34.19    101.77   8.277
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                          SUMMARY OF LOADING
       188.7 Cu.M.  (11%) SALT WATER          0.0 Cu.M.   (0%) VOID

      150.00 MT  Crane                      51.30 MT  Crane Load
      895.54 MT  Deck Cargo

DRAFT AT FP (M) =    2.248
DRAFT AT MS (M) =    2.212
DRAFT AT AP (M) =    2.176
DRAFT AT FWD MARK (M) =    2.242
DRAFT AT AFT MARK (M) =    2.182
                            HYDROSTATIC PROPERTIES
                 Trim: Fwd 0.072/50.000, No Heel, VCG = 5.033

    LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
   Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
   2.212     1,679.60  25.153f  1.163     8.65  25.008f   34.19  101.77   8.277
  Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                             Trim is per 50.00m.
  Draft is from USK.                               Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 162.2 m.-MT
```

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GHS 16.68C **PONTOON**

Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Stbd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 2.248 @ 50.00f, 2.212 @ 25.00f, 2.176 @ 0.00
Trim: Fwd 0.072/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.405	1.477	0.05140	3.08
CRANE.C	34.4	3.018	4.090	0.05140	7.23
DECKCARGO.C	98.0	2.796	3.869	0.05140	19.49
Total wind heeling moment to starboard------------->					29.79

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.147f TCG = 0.000 VCG = 5.033
Free Surface Adjustment: 0.097
Adjusted CG: LCG = 25.147f TCG = 0.000 VCG = 5.129

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
2.164	0.08f	0.00	1,679.6	0.000	-0.018	0.0000	2.625
2.164	0.08f	0.12s	1,679.6	0.000	0.000	-0.0000	2.505
2.161	0.08f	2.62s	1,679.6	0.000	0.365	0.0080	0.000
2.151	0.09f	5.12s	1,679.6	0.000	0.723	0.0317	-2.495
2.188	0.13f	10.12s	1,679.6	0.000	1.032	0.1134	-7.495
2.197	0.13f	10.51s	1,679.6	0.000	1.033	0.1205	-7.886
2.329	0.16f	15.12s	1,679.6	0.000	0.944	0.2022	-12.495
2.512	0.20f	20.12s	1,679.6	0.000	0.730	0.2761	-17.495
2.704	0.24f	25.12s	1,679.6	0.000	0.454	0.3282	-22.495
2.876	0.30f	30.12s	1,679.6	0.000	0.135	0.3542	-27.495
2.939	0.32f	32.13s	1,679.6	0.000	0.000	0.3566	-29.503
3.026	0.35f	35.12s	1,679.6	0.000	-0.206	0.3513	-32.495
3.152	0.40f	40.12s	1,679.6	0.000	-0.557	0.3181	-37.495
3.254	0.44f	45.12s	1,679.6	0.000	-0.911	0.2540	-42.495
3.331	0.49f	50.12s	1,679.6	0.000	-1.262	0.1592	-47.495
3.384	0.52f	55.12s	1,679.6	0.000	-1.607	0.0340	-52.495
3.411	0.56f	60.12s	1,679.6	0.000	-1.941	-0.1209	-57.495

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 162.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 29.79 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.1205	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	32.01	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.50	P

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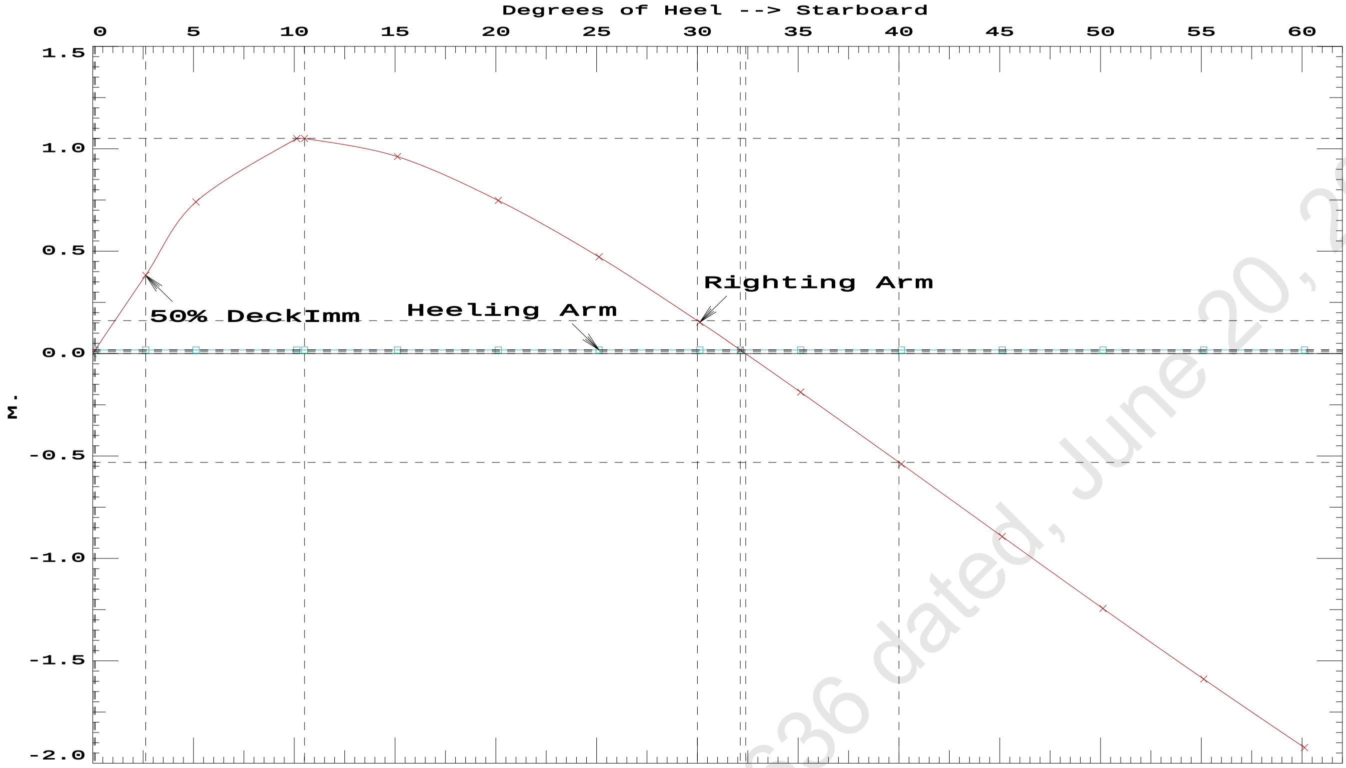
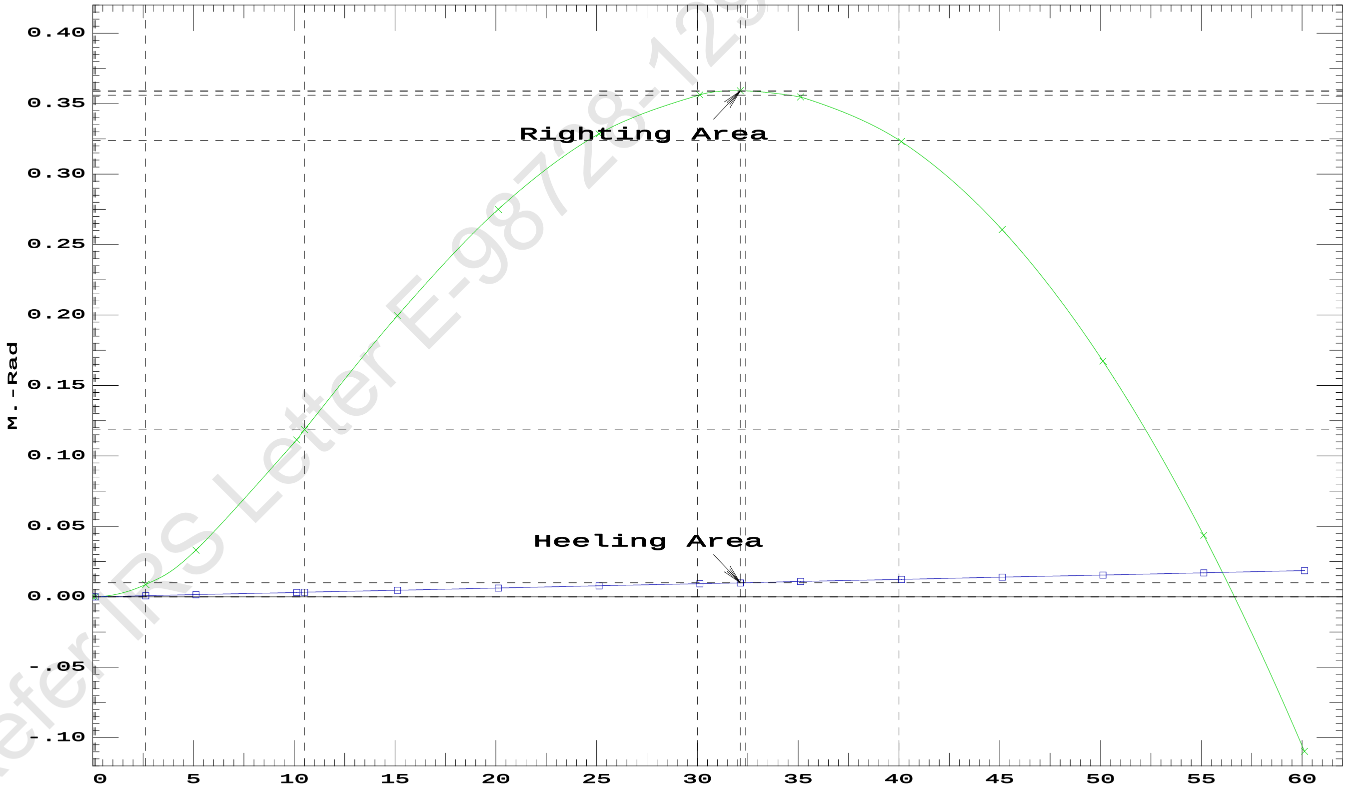
GHS 16.68C PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Stbd





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GHS 16.68C

PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.147f TCG = 0.000 VCG = 5.033
Free Surface Adjustment: 0.097
Adjusted CG: LCG = 25.147f TCG = 0.004p VCG = 5.129

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.161	0.08f	2.49s	1,679.6	0.000	0.000	0.0000
2.148	0.10f	7.49s	1,679.6	0.000	0.615	0.0268
2.205	0.13f	10.84s	1,679.6	0.000	0.688	0.0663
2.247	0.14f	12.49s	1,679.6	0.000	0.670	0.0858
2.413	0.18f	17.49s	1,679.6	0.000	0.508	0.1380
2.604	0.22f	22.49s	1,679.6	0.000	0.260	0.1722
2.767	0.26f	26.88s	1,679.6	0.000	0.000	0.1824
2.788	0.27f	27.49s	1,679.6	0.000	-0.038	0.1822
2.950	0.32f	32.49s	1,679.6	0.000	-0.370	0.1647
3.089	0.37f	37.49s	1,679.6	0.000	-0.717	0.1174
3.203	0.42f	42.49s	1,679.6	0.000	-1.070	0.0394
3.294	0.46f	47.49s	1,679.6	0.000	-1.424	-0.0694
3.359	0.50f	52.49s	1,679.6	0.000	-1.772	-0.2089
3.400	0.54f	57.49s	1,679.6	0.000	-2.113	-0.3785
3.416	0.57f	62.49s	1,679.6	0.000	-2.442	-0.5773

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 162.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 615.60

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	2.49 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	187.7 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		2.169 P

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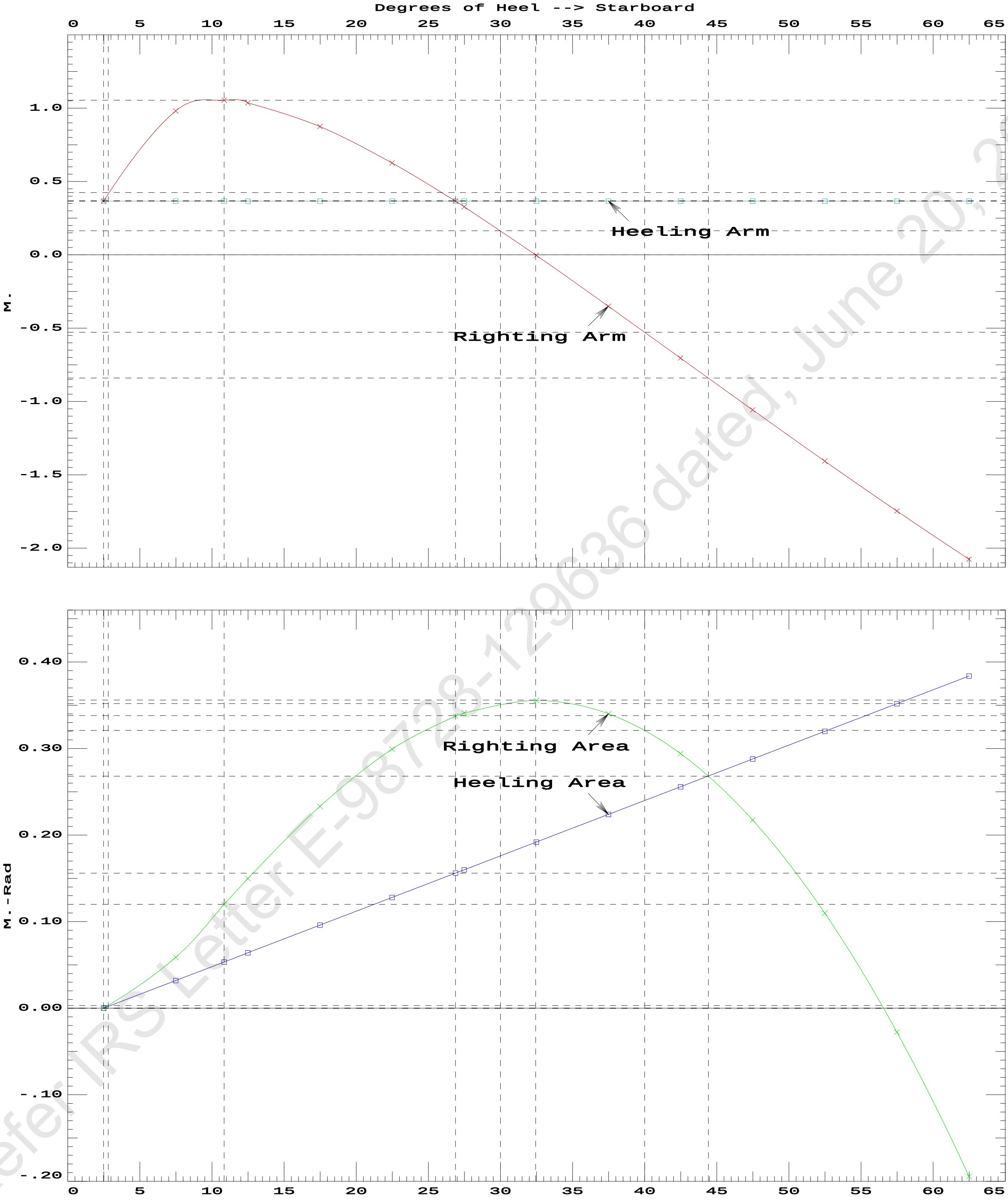
GHS 16.68C PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo

Crane Lifting 51.30T @ 12m Radius Stbd



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GHS 16.68C **PONTOON**

Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Stbd
Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.248 @ 50.00f, 2.211 @ 25.00f, 2.175 @ 0.00
Trim: Fwd 0.073/50.000, Heel: Stbd 2.49 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			389.39	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	7.500f	0.000	33.218	
Deck Cargo			895.54	25.000f	0.000	5.000	
Total Fixed--------->			1,486.23	22.630f	0.000	5.470	
06_WBTK.C	1.000	1.025	113.01	42.250f	0.000	1.500	78.48*
07_WBTK.P	1.000	1.025	40.19	47.659f	4.971p	1.911	41.85*
07_WBTK.S	1.000	1.025	40.19	47.659f	4.971s	1.911	41.85*
Total Tanks--------->			193.38	44.498f	0.000	1.671	162.18
Total Weight-------->			1,679.61	25.147f	0.000	5.033	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,679.61	25.153f	0.534s	1.175

Righting Arms:	0.000	0.367s
External Arms:	0.000	0.367s
Residual Righting Arms:	0.000	0.000s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	841.9	25.032f	0.137s	104.79	12.111
		MT/cm 8.63				

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 2.49

```
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GHS 16.68C                          PONTOON

                          After hook load drop
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 2.319 @ 50.00f, 2.153 @ 25.00f, 1.986 @ 0.00
                  Trim: Fwd 0.333/50.000,  Heel: 0.00 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           389.39  25.000f  0.000   3.000
Crane                                150.00   7.500f  0.000   5.200
Deck Cargo                           895.54  25.000f  0.000   5.000
  Total Fixed--------->            1,434.93  23.171f  0.000   4.478
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.C      1.000    1.025        113.01  42.250f  0.000   1.500     78.48*
07_WBTK.P      1.000    1.025         40.19  47.659f  4.971p  1.911     41.85*
07_WBTK.S      1.000    1.025         40.19  47.659f  4.971s  1.911     41.85*
  Total Tanks--------->              193.38  44.498f  0.000   1.671    162.18
  Total Weight-------->            1,628.31  25.703f  0.000   4.145
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,628.31  25.724f  0.002s  1.134
------------------------------------------------------------------------------
               Righting Arms:                 0.000   0.002s
               External Arms:                 0.000   0.001
      Residual Righting Arms:                 0.000   0.001s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         836.1  25.247f  0.030s   106.05  12.449
                                      MT/cm
                                       8.57
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 5928.37; t3: 30.00; t: 15.00
limmarg: 2694.99; t3: 15.00; t: 7.50
limmarg: 787.60; t3: 7.50; t: 3.75
limmarg: 128.13; t3: 3.75; t: 1.88
limmarg: -43.07; t3: 1.87; t: 0.94
limmarg: 28.30; t3: 2.81; t: 0.47
limmarg: -10.96; t3: 2.34; t: 0.23
limmarg: 7.36; t3: 2.57; t: 0.12
limmarg: -2.41; t3: 2.45; t: 0.06
limmarg: 2.42; t3: 2.51; t: 0.03
limmarg: -0.01; t3: 2.48; t: 0.01

              Theta3 is 2.48 degrees, as demonstrated below.
```

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GHS 16.68C **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 23.171f TCG = 0.000 VCG = 4.478

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.969	0.39f	2.49p	1,628.3	0.000	-0.419	0.0000
1.974	0.38f	0.00	1,628.3	0.000	0.001	-0.0091
1.969	0.39f	2.48s	1,628.3	0.000	0.416	-0.0000
1.957	0.39f	5.00s	1,628.3	0.000	0.835	0.0275
1.921	0.55f	10.00s	1,628.3	0.000	1.308	0.1232
1.954	0.66f	13.35s	1,628.3	0.000	1.355	0.2021
1.981	0.72f	15.00s	1,628.4	0.000	1.344	0.2410
2.086	0.89f	20.00s	1,628.3	0.000	1.228	0.3550
2.181	1.09f	25.00s	1,628.4	0.000	1.031	0.4542
2.250	1.32f	30.00s	1,628.3	0.000	0.785	0.5338

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1434.93 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Res. Area Ratio from abs -2.5 deg to abs 2.5	>	1.000	1.000 P

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GHS 16.68C **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 23.171f TCG = 0.000 VCG = 4.478

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.969	0.39f	2.49p	1,628.3	0.000	-0.419	0.0000
1.974	0.38f	0.00	1,628.3	0.000	0.001	-0.0091
1.969	0.39f	2.48s	1,628.3	0.000	0.416	-0.0000
1.957	0.39f	5.00s	1,628.3	0.000	0.835	0.0275
1.921	0.55f	10.00s	1,628.3	0.000	1.308	0.1232
1.954	0.66f	13.35s	1,628.3	0.000	1.355	0.2021
1.981	0.72f	15.00s	1,628.4	0.000	1.344	0.2410
2.086	0.89f	20.00s	1,628.3	0.000	1.228	0.3550
2.181	1.09f	25.00s	1,628.4	0.000	1.031	0.4542
2.250	1.32f	30.00s	1,628.3	0.000	0.785	0.5338
2.298	1.54f	35.00s	1,628.3	0.000	0.510	0.5904
2.327	1.76f	40.00s	1,628.3	0.000	0.219	0.6224
2.336	1.92f	43.66s	1,628.3	0.000	0.000	0.6294
2.336	1.97f	45.00s	1,628.3	0.000	-0.081	0.6284

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1434.93 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Res. Area Ratio from abs -2.5 deg to RAzero	>	1.000		70.499 P
(2) Angle from abs 2.5 deg to RAzero	>	20.00	deg	41.18 P

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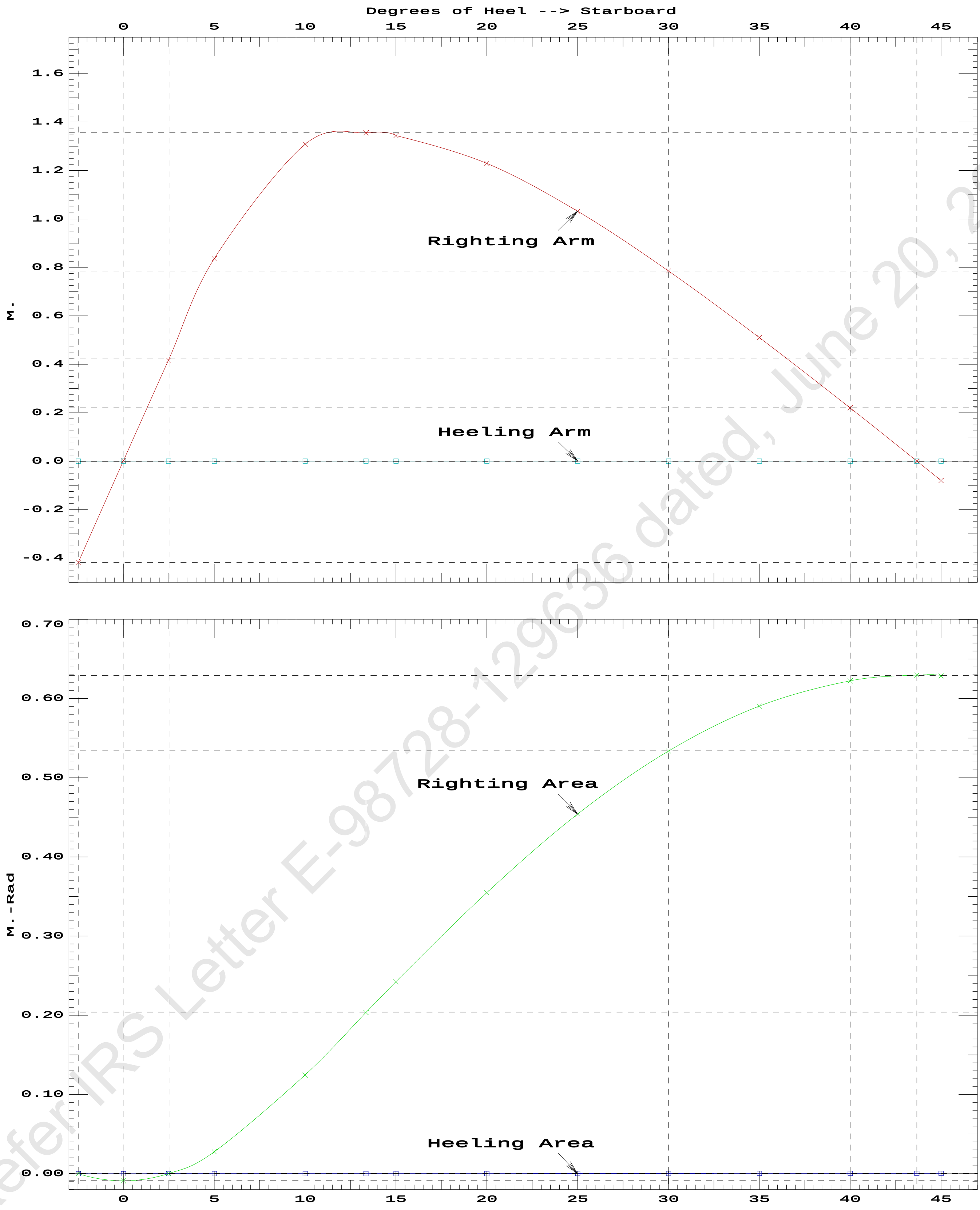
GHS 16.68C **PONTOON**



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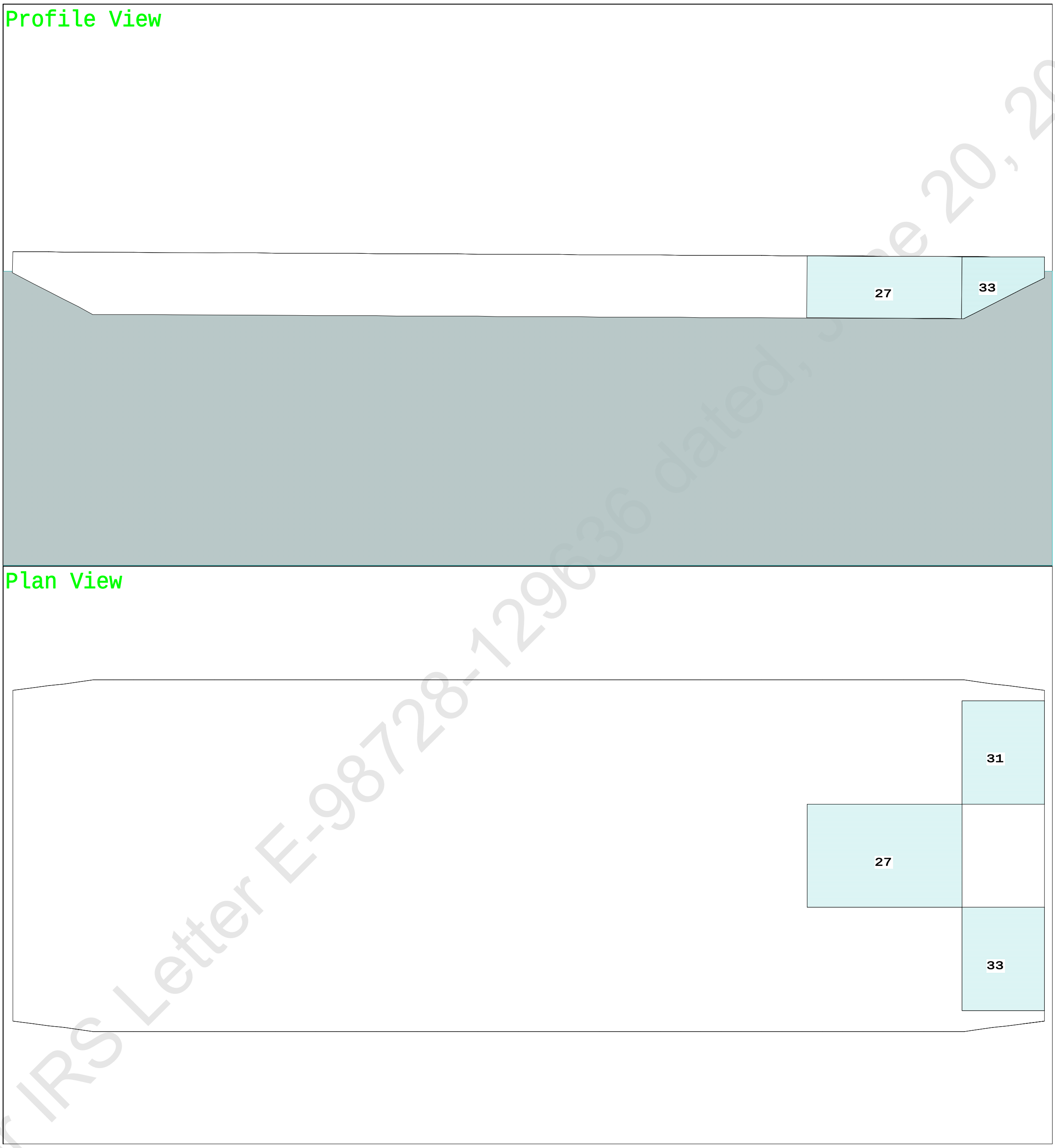
GHS 16.68C PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Fwd

Condition Graphic - Draft: 2.338 @ 50.000f, 2.086 @ 0.000 Heel: zero



Tanks 27 06_WBTK.C 31 07_WBTK.P 33 07_WBTK.S

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GHS 16.68C

PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.338 @ 50.00f, 2.212 @ 25.00f, 2.086 @ 0.00
Trim: Fwd 0.252/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			389.39	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	19.500f	0.000	33.218	
Deck Cargo			895.54	25.000f	0.000	5.000	
Total Fixed--------->			1,486.23	23.044f	0.000	5.470	
06_WBTK.C	1.000	1.025	113.01	42.250f	0.000	1.500	78.48*
07_WBTK.P	1.000	1.025	40.19	47.659f	4.971p	1.911	41.85*
07_WBTK.S	1.000	1.025	40.19	47.659f	4.971s	1.911	41.85*
Total Tanks--------->			193.38	44.498f	0.000	1.671	162.18
Total Weight-------->			1,679.61	25.514f	0.000	5.033	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,679.61	25.533f	0.000	1.164	

Righting Arms:				0.000	0.000	
Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	843.7	25.029f	0.000	105.64	12.148
		MT/cm		m.-MT/cm	GML	GMT
		8.65		34.19	101.77	8.279

Distances in METERS.--------------------------------------Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

188.7 Cu.M. (11%) SALT WATER 0.0 Cu.M. (0%) VOID

150.00 MT Crane 51.30 MT Crane Load
895.54 MT Deck Cargo

DRAFT AT FP (M) = 2.338
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.086
DRAFT AT FWD MARK (M) = 2.318
DRAFT AT AFT MARK (M) = 2.106

HYDROSTATIC PROPERTIES
Trim: Fwd 0.252/50.000, No Heel, VCG = 5.033

Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.212	1,679.61	25.533f	1.164	8.65	25.029f	34.19	101.77	8.279

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 162.2 m.-MT

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GHS 16.68C **PONTOON**

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Fwd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 2.338 @ 50.00f, 2.212 @ 25.00f, 2.086 @ 0.00
Trim: Fwd 0.252/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.408	1.482	0.05140	3.08
CRANE.C	34.4	3.081	4.154	0.05140	7.34
DECKCARGO.C	98.0	2.787	3.861	0.05140	19.45
Total wind heeling moment to starboard------------->					29.87

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.514f TCG = 0.000 VCG = 5.033
Free Surface Adjustment: 0.097
Adjusted CG: LCG = 25.514f TCG = 0.000 VCG = 5.129

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
2.073	0.29f	0.00	1,679.6	0.000	-0.018	0.0000	2.365
2.074	0.29f	0.12s	1,679.6	0.000	0.000	-0.0000	2.245
2.070	0.29f	2.36s	1,679.6	0.000	0.326	0.0064	0.000
2.058	0.30f	5.12s	1,679.6	0.000	0.722	0.0316	-2.755
2.054	0.44f	10.12s	1,679.6	0.000	1.025	0.1128	-7.755
2.062	0.45f	10.66s	1,679.6	0.000	1.026	0.1224	-8.291
2.154	0.57f	15.12s	1,679.8	0.000	0.937	0.2009	-12.755
2.296	0.70f	20.12s	1,679.6	0.000	0.723	0.2749	-17.755
2.439	0.85f	25.12s	1,679.7	0.000	0.445	0.3264	-22.755
2.556	1.03f	30.12s	1,679.6	0.000	0.127	0.3516	-27.755
2.594	1.10f	32.00s	1,679.6	0.000	0.000	0.3537	-29.639
2.650	1.21f	35.12s	1,679.6	0.000	-0.214	0.3479	-32.755
2.723	1.38f	40.12s	1,679.6	0.000	-0.565	0.3140	-37.755
2.774	1.54f	45.12s	1,679.6	0.000	-0.918	0.2493	-42.755
2.804	1.69f	50.12s	1,679.6	0.000	-1.269	0.1539	-47.755
2.815	1.82f	55.12s	1,679.6	0.000	-1.612	0.0281	-52.755
2.808	1.93f	60.12s	1,679.6	0.000	-1.946	-0.1272	-57.755

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 162.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 29.87 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.1224	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	31.88	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.24	P

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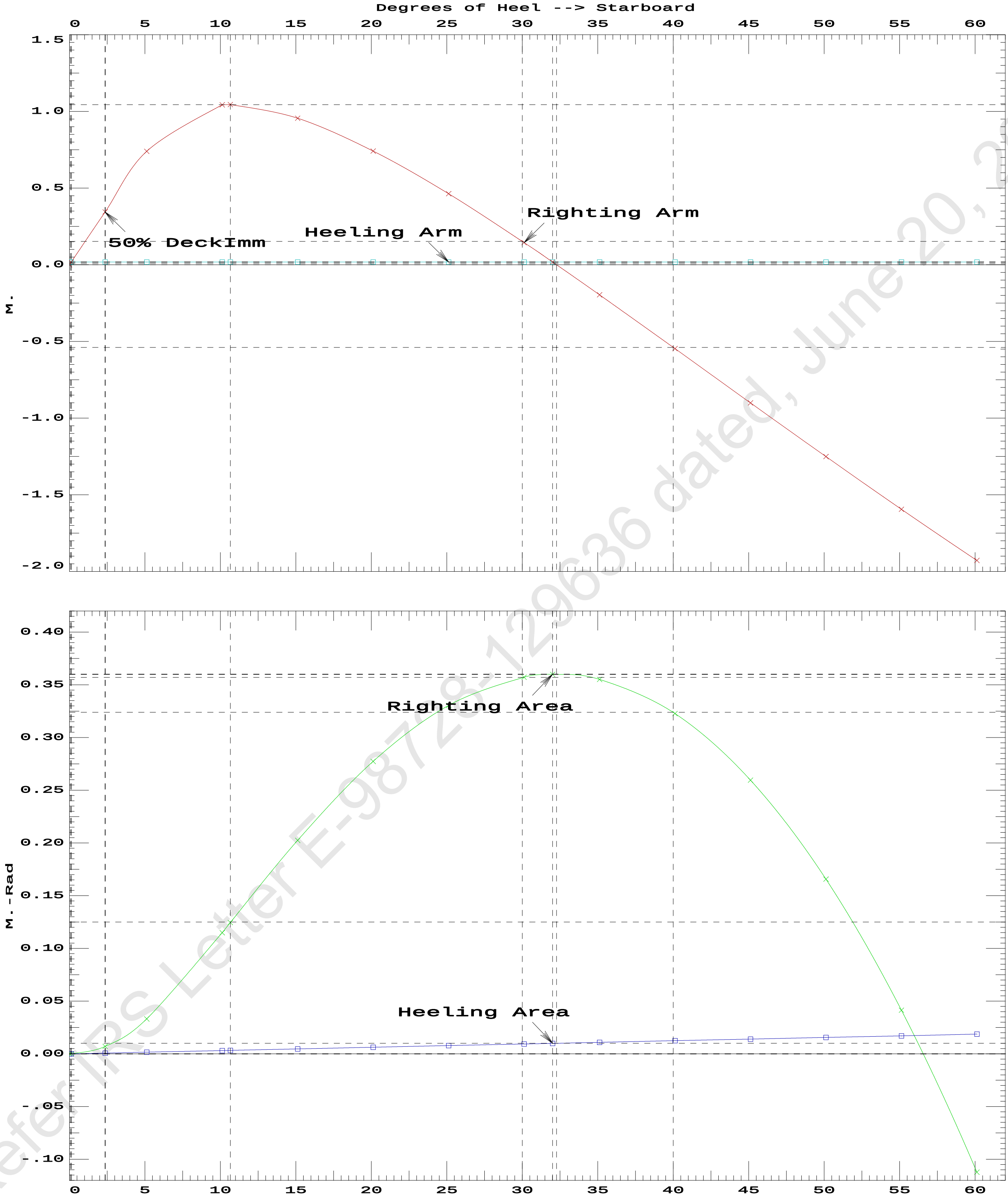
GHS 16.68C PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Fwd



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PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo

Crane Lifting 51.30T @ 12m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.514f TCG = 0.000 VCG = 5.033
Free Surface Adjustment: 0.097
Adjusted CG: LCG = 25.514f TCG = 0.000 VCG = 5.129

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.073	0.29f	0.00s	1,679.6	0.000	0.000	0.0000
2.059	0.30f	5.00s	1,679.6	0.000	0.723	0.0315
2.053	0.43f	10.00s	1,679.6	0.000	1.042	0.1137
2.060	0.45f	10.54s	1,679.6	0.000	1.043	0.1234
2.151	0.57f	15.00s	1,679.8	0.000	0.958	0.2034
2.292	0.70f	20.00s	1,679.6	0.000	0.746	0.2794
2.436	0.85f	25.00s	1,679.7	0.000	0.469	0.3329
2.554	1.02f	30.00s	1,679.6	0.000	0.151	0.3602
2.599	1.10f	32.25s	1,679.6	0.000	0.000	0.3632
2.648	1.20f	35.00s	1,679.6	0.000	-0.189	0.3587
2.721	1.37f	40.00s	1,679.6	0.000	-0.540	0.3270
2.773	1.54f	45.00s	1,679.6	0.000	-0.893	0.2645
2.804	1.69f	50.00s	1,679.6	0.000	-1.244	0.1713
2.815	1.82f	55.00s	1,679.6	0.000	-1.587	0.0477
2.809	1.93f	60.00s	1,679.6	0.000	-1.921	-0.1055

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 162.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM------------STABILITY CRITERIA FOR CRANE OPERAT		----Min/Max		--------Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	175265	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		1084.89	P

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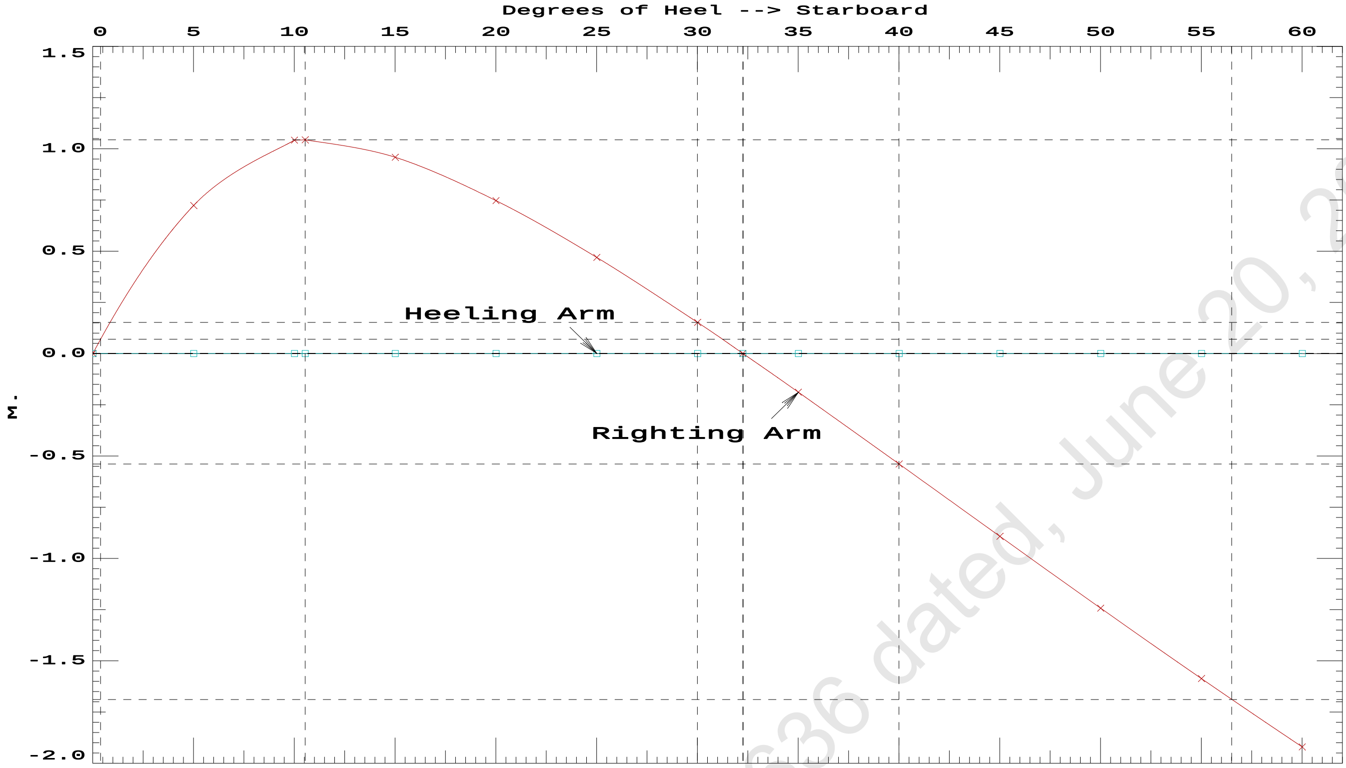
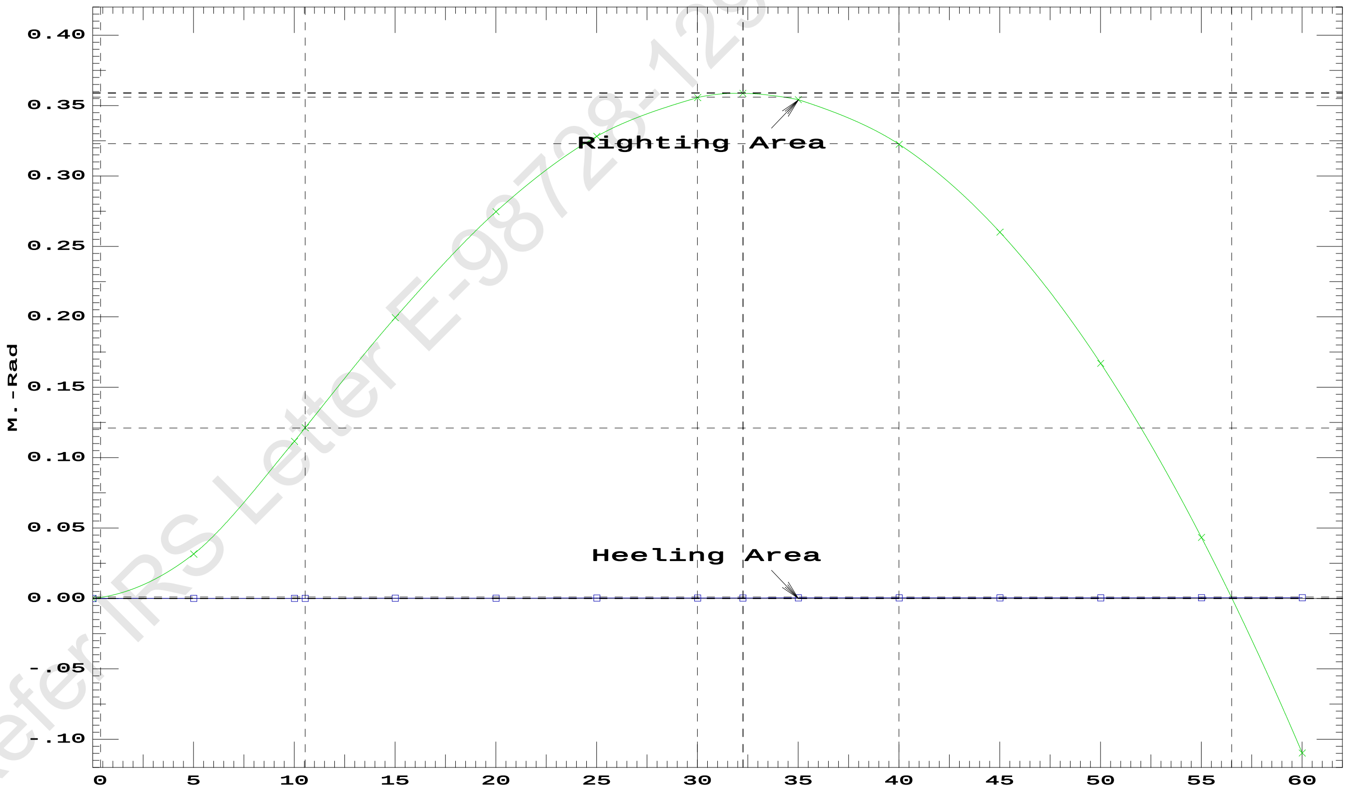
GHS 16.68C PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo

Crane Lifting 51.30T @ 12m Radius Fwd





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GHS 16.68C   **PONTOON**

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 51.30T @ 12m Radius Fwd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.338 @ 50.00f, 2.212 @ 25.00f, 2.085 @ 0.00
Trim: Fwd 0.253/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			389.39	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	19.500f	0.000	33.218	
Deck Cargo			895.54	25.000f	0.000	5.000	
Total Fixed--------->			1,486.23	23.044f	0.000	5.470	
06_WBTK.C	1.000	1.025	113.01	42.250f	0.000	1.500	78.48*
07_WBTK.P	1.000	1.025	40.19	47.659f	4.971p	1.911	41.85*
07_WBTK.S	1.000	1.025	40.19	47.659f	4.971s	1.911	41.85*
Total Tanks--------->			193.38	44.498f	0.000	1.671	162.18
Total Weight-------->			1,679.61	25.514f	0.000	5.033	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,679.61	25.534f	0.001	1.164

Righting Arms:	0.001	0.001s
External Arms:	0.000	0.001
Residual Righting Arms:	0.001	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	843.7	25.029f	0.000	105.64	12.148
		MT/cm		**m.-MT/cm**	**GML**	**GMT**
		8.65		34.19	101.77	8.279

Distances in METERS.---Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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GHS 16.68C                             PONTOON

                              After hook load drop
                     WEIGHT and DISPLACEMENT and WATERPLANE STATUS
                USK draft: 2.319 @ 50.00f, 2.153 @ 25.00f, 1.986 @ 0.00
                       Trim: Fwd 0.333/50.000,  Heel: 0.00 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           389.39  25.000f  0.000   3.000
Crane                                150.00   7.500f  0.000   5.200
Deck Cargo                           895.54  25.000f  0.000   5.000
  Total Fixed--------->            1,434.93  23.171f  0.000   4.478
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.C      1.000    1.025        113.01  42.250f  0.000   1.500     78.48*
07_WBTK.P      1.000    1.025         40.19  47.659f  4.971p  1.911     41.85*
07_WBTK.S      1.000    1.025         40.19  47.659f  4.971s  1.911     41.85*
  Total Tanks--------->              193.38  44.498f  0.000   1.671    162.18
  Total Weight-------->            1,628.31  25.703f  0.000   4.145
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,628.30  25.724f  0.002s  1.134
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.002s
               External Arms:                  0.000   0.001
      Residual Righting Arms:                  0.000   0.001s
Part--------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          836.1  25.247f  0.030s   106.05  12.449
                                       MT/cm
                                        8.57
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

               Theta3 is 0.07 degrees, as demonstrated below.
```

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GHS 16.68C **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 23.171f TCG = 0.000 VCG = 4.478

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.974	0.38f	0.00	1,628.3	0.000	-0.001	0.0000
1.974	0.38f	0.07p	1,628.4	0.000	0.014	0.0000
1.957	0.39f	5.00p	1,628.3	0.000	0.836	0.0366
1.921	0.55f	10.00p	1,628.3	0.000	1.309	0.1354
1.954	0.66f	13.35p	1,628.3	0.000	1.356	0.2144
1.981	0.72f	15.00p	1,628.4	0.000	1.345	0.2533
2.086	0.89f	20.00p	1,628.3	0.000	1.230	0.3674
2.181	1.09f	25.00p	1,628.4	0.000	1.033	0.4667
2.250	1.32f	30.00p	1,628.3	0.000	0.786	0.5464

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1434.93 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 0.1	>	1.000	LARGE

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GHS 16.68C **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 23.171f TCG = 0.000 VCG = 4.478

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.974	0.38f	0.00	1,628.3	0.000	-0.001	0.0000
1.974	0.38f	0.07p	1,628.4	0.000	0.014	0.0000
1.957	0.39f	5.00p	1,628.3	0.000	0.836	0.0366
1.921	0.55f	10.00p	1,628.3	0.000	1.309	0.1354
1.954	0.66f	13.35p	1,628.3	0.000	1.356	0.2144
1.981	0.72f	15.00p	1,628.4	0.000	1.345	0.2533
2.086	0.89f	20.00p	1,628.3	0.000	1.230	0.3674
2.181	1.09f	25.00p	1,628.4	0.000	1.033	0.4667
2.250	1.32f	30.00p	1,628.3	0.000	0.786	0.5464
2.298	1.54f	35.00p	1,628.3	0.000	0.511	0.6032
2.327	1.76f	40.00p	1,628.3	0.000	0.220	0.6353
2.336	1.92f	43.68p	1,628.3	0.000	0.000	0.6424
2.336	1.97f	45.00p	1,628.3	0.000	-0.079	0.6415

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1434.93 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	43.61 P

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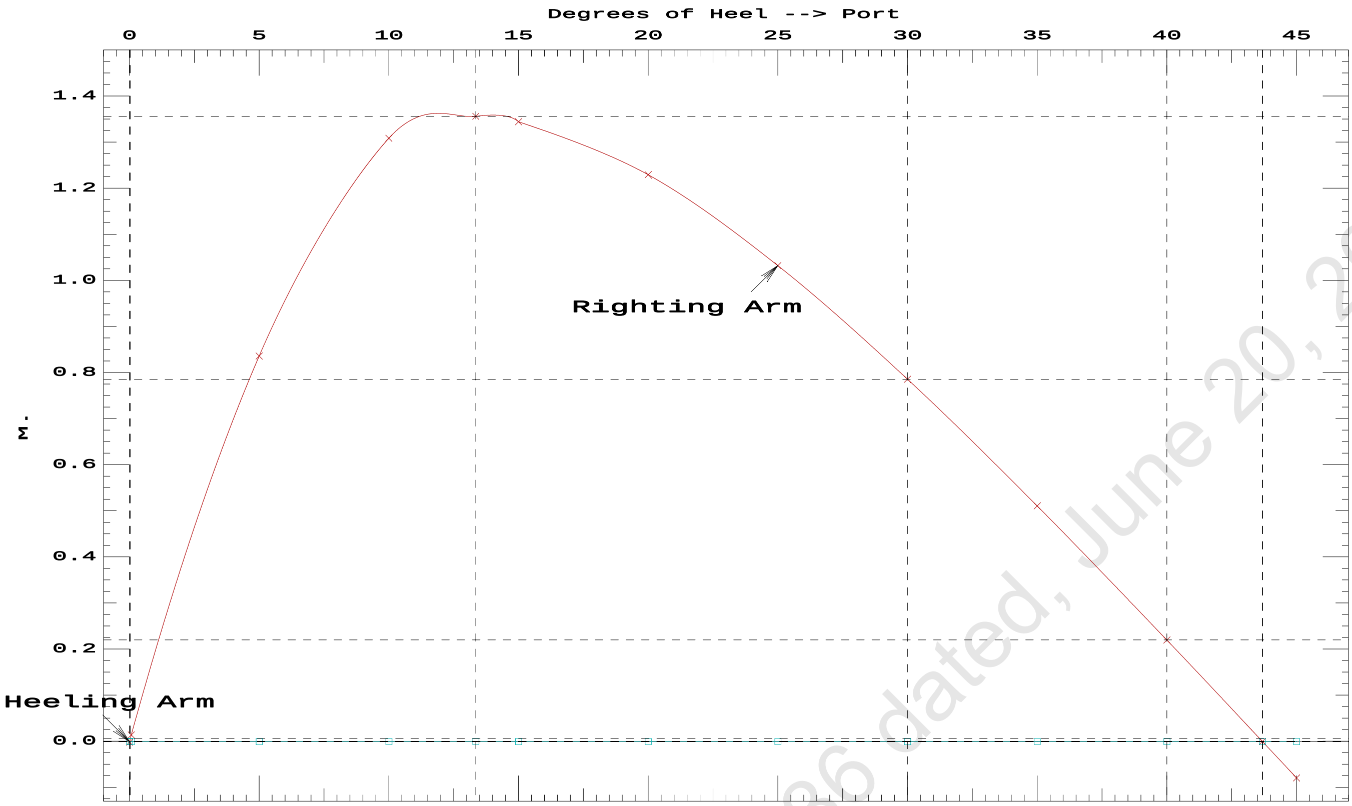
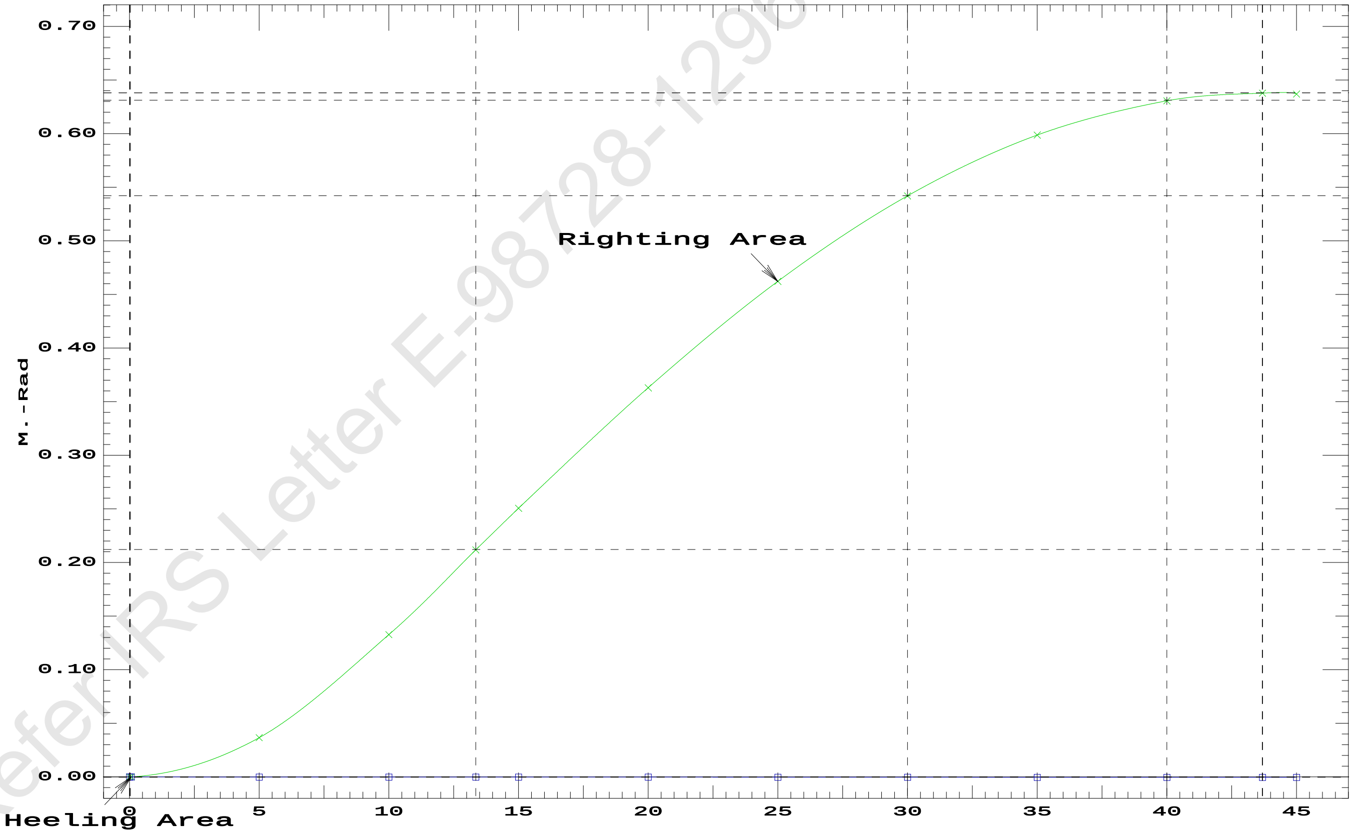
GHS 16.68C **PONTOON**





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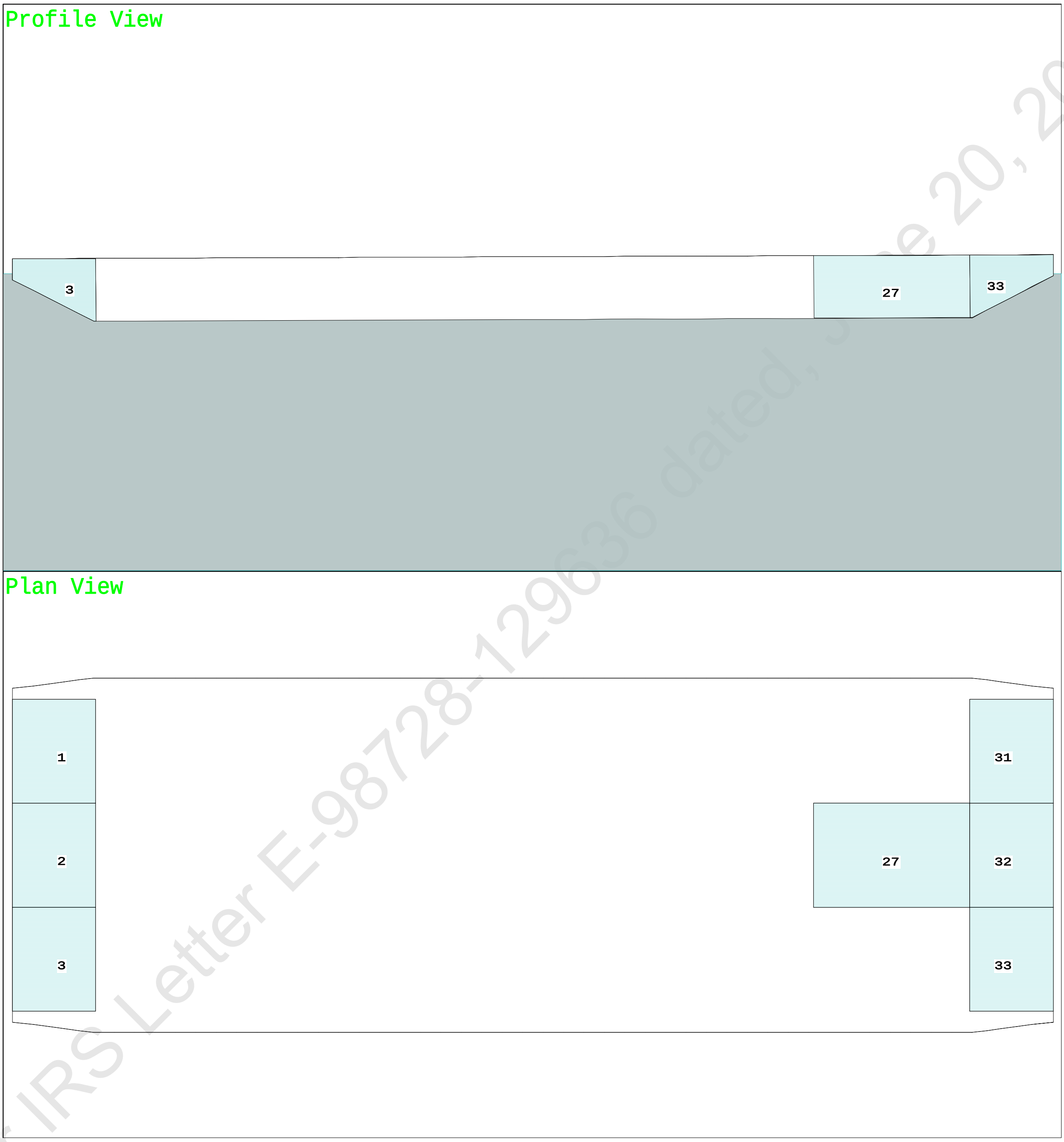
GHS 16.68C PONTOON

Condition 5 : Lightship + Crane + Ballast + Deck Cargo
Towing Condition with Max. draft 2.212m

Condition Graphic - Draft: 2.114 @ 50.000f, 2.310 @ 0.000 Heel: zero



Tanks

	2 01_WBTK.C	27 06_WBTK.C	32 07_WBTK.C
1 01_WBTK.P	3 01_WBTK.S	31 07_WBTK.P	33 07_WBTK.S

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27-01-20 14:18:57    Cybermarine Technologies PTE. Ltd.
GHS 16.68C                          PONTOON
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Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Towing Condition with Max. draft 2.212m

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                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 2.114 @ 50.00f, 2.212 @ 25.00f, 2.310 @ 0.00
                     Trim: Aft 0.195/50.000,  Heel: zero
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           389.39  25.000f  0.000   3.000
Crane                                150.00   7.500f  0.000   5.200
Deck Cargo                           785.12  25.000f  0.000   5.000
  Total Fixed--------->            1,324.51  23.018f  0.000   4.435
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
01_WBTK.P      1.000    1.025         40.19   2.341f  4.971p  1.911     41.85*
01_WBTK.C      1.000    1.025         40.67   2.337f  0.000   1.901     41.85*
01_WBTK.S      1.000    1.025         40.19   2.341f  4.971s  1.911     41.85*
06_WBTK.C      1.000    1.025        113.01  42.250f  0.000   1.500     78.48*
07_WBTK.P      1.000    1.025         40.19  47.659f  4.971p  1.911     41.85*
07_WBTK.C      1.000    1.025         40.67  47.663f  0.000   1.901     41.85*
07_WBTK.S      1.000    1.025         40.19  47.659f  4.971s  1.911     41.85*
  Total Tanks--------->              355.10  30.490f  0.000   1.778    329.60
  Total Weight-------->            1,679.61  24.598f  0.000   3.873
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,679.61  24.587f  0.000   1.164
------------------------------------------------------------------------------
               Righting Arms:                 0.000   0.000
Part--------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         843.7  24.977f  0.000    105.64  12.047
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.65           34.58    102.93   9.338
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                          SUMMARY OF LOADING
       346.4 Cu.M.  (20%) SALT WATER           0.0 Cu.M.   (0%) VOID

      150.00 MT  Crane                       785.12 MT  Deck Cargo

DRAFT AT FP (M) =     2.114
DRAFT AT MS (M) =     2.212
DRAFT AT AP (M) =     2.310
DRAFT AT FWD MARK (M) =    2.130
DRAFT AT AFT MARK (M) =    2.294
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GHS 16.68C

PONTOON

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Towing Condition with Max. draft 2.212m

HYDROSTATIC PROPERTIES
Trim: Aft 0.195/50.000, No Heel, VCG = 3.873

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.212	1,679.61	24.587f	1.164	8.65	24.977f	34.58	102.93	9.338

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 329.6 m.-MT

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 2.114 @ 50.00f, 2.212 @ 25.00f, 2.310 @ 0.00
Trim: Aft 0.195/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.407	1.480	0.05140	3.08
CRANE.C	34.4	2.924	3.997	0.05140	7.06
DECKCARGO.C	98.0	2.810	3.883	0.05140	19.56
Total wind heeling moment to starboard------------->					29.70

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.598f TCG = 0.000 VCG = 3.873
Free Surface Adjustment: 0.196
Adjusted CG: LCG = 24.598f TCG = 0.000 VCG = 4.069

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% Area	DeckImm Angle
2.298	0.22a	0.00	1,679.6	0.000	-0.017	0.0000	2.447
2.298	0.22a	0.11s	1,679.6	0.000	0.000	-0.0000	2.342
2.296	0.23a	2.45s	1,679.6	0.000	0.384	0.0078	0.000
2.288	0.23a	5.11s	1,679.6	0.000	0.815	0.0357	-2.658
2.389	0.33a	10.11s	1,679.6	0.000	1.214	0.1293	-7.658
2.487	0.39a	12.69s	1,679.6	0.000	1.240	0.1853	-10.248
2.593	0.44a	15.11s	1,679.6	0.000	1.218	0.2370	-12.658
2.834	0.54a	20.11s	1,679.6	0.000	1.092	0.3393	-17.658
3.092	0.65a	25.11s	1,679.6	0.000	0.900	0.4267	-22.658
3.344	0.78a	30.11s	1,679.6	0.000	0.663	0.4953	-27.658
3.573	0.91a	35.11s	1,679.6	0.000	0.400	0.5419	-32.658
3.777	1.04a	40.11s	1,679.6	0.000	0.123	0.5648	-37.658
3.856	1.09a	42.27s	1,679.6	0.000	0.000	0.5671	-39.820
3.952	1.16a	45.11s	1,679.6	0.000	-0.163	0.5631	-42.658
4.097	1.27a	50.11s	1,679.6	0.000	-0.451	0.5364	-47.658
4.211	1.37a	55.11s	1,679.6	0.000	-0.739	0.4844	-52.658
4.291	1.46a	60.11s	1,679.6	0.000	-1.024	0.4075	-57.658

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 329.6 m.-MT was applied to artificially modify the CG.

27-01-20 14:18:57 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

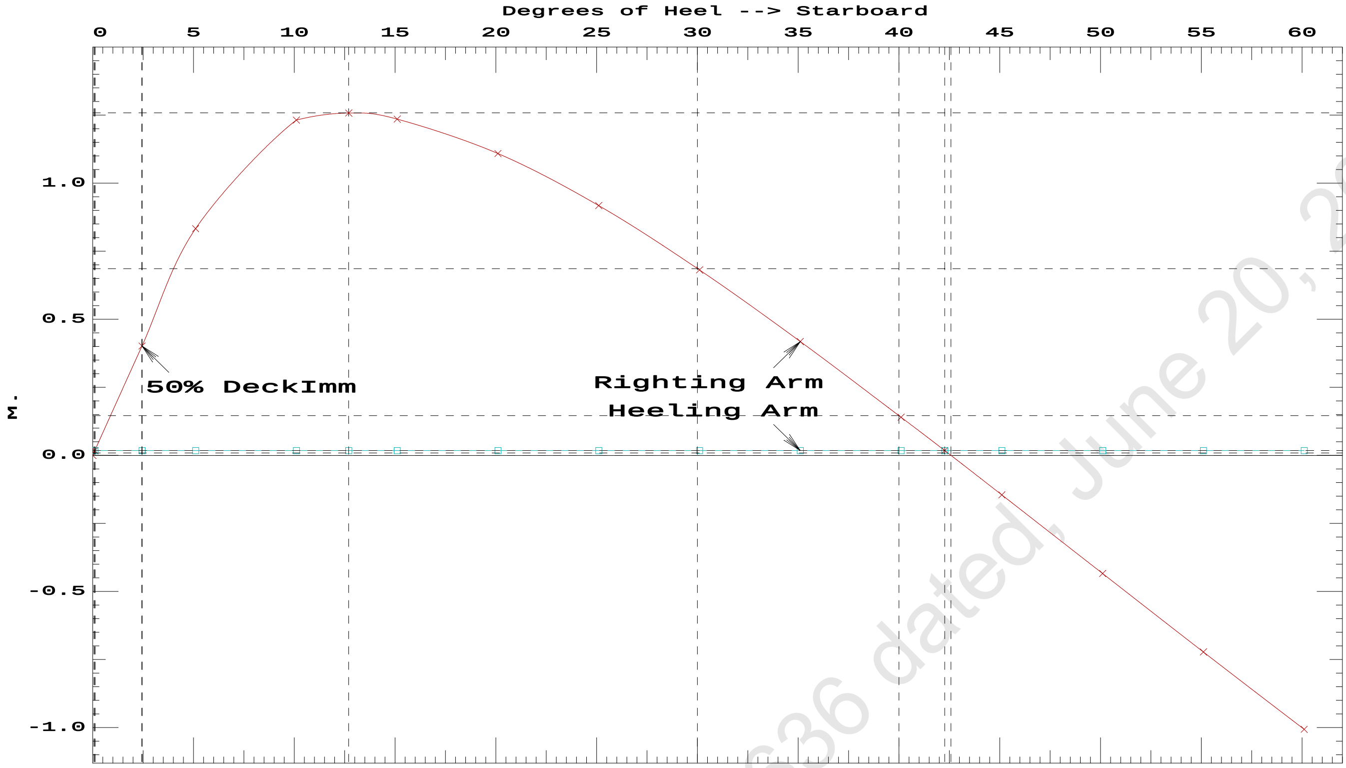
Condition 5 : Lightship + Crane + Ballast + Deck Cargo
Towing Condition with Max. draft 2.212m

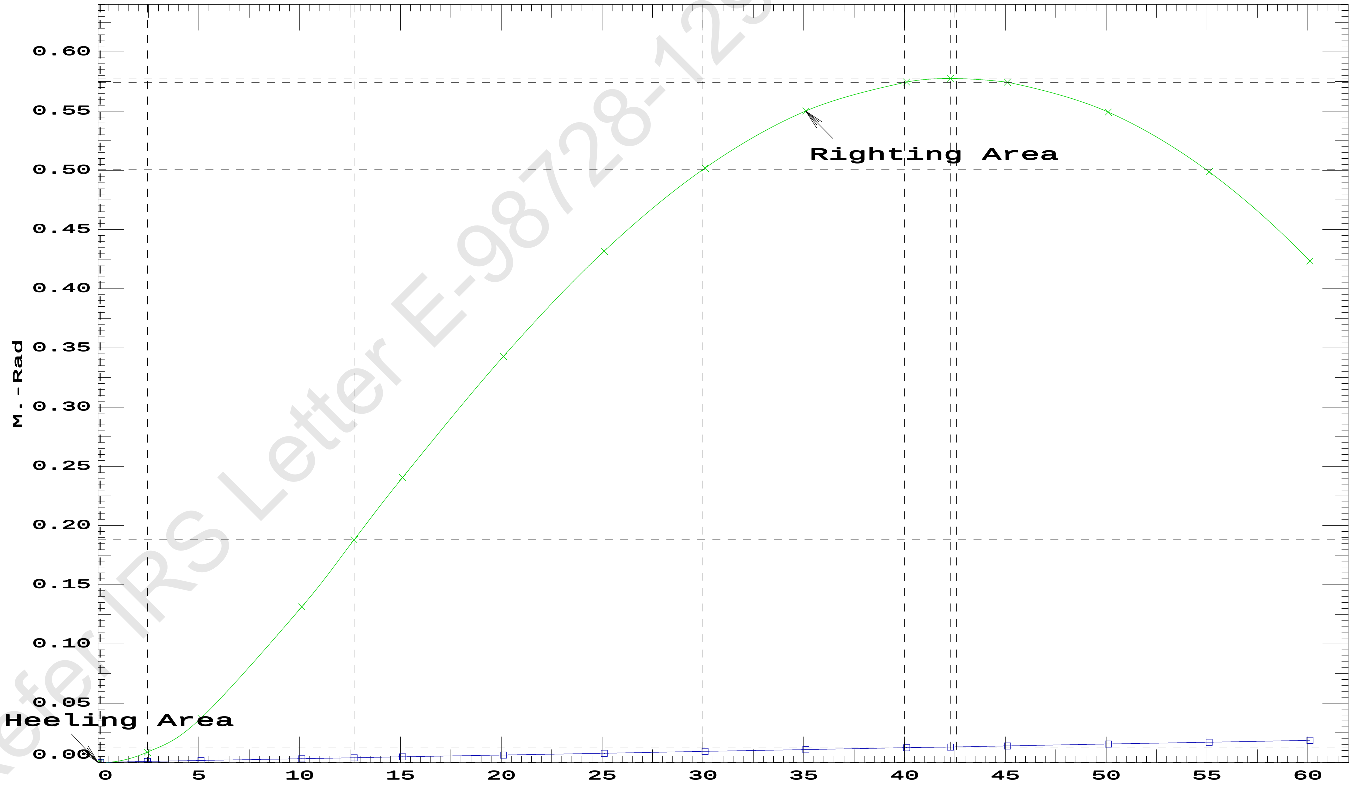
```
  Note:  The Residual Righting Arms shown above are in excess of the
         wind heeling arms derived from these moments (in m.-MT):
                   Stbd heeling moment = 29.70 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA          >   0.0800 m.-Rad 0.1853 P
(2) Angle from Equilibrium to RAzero or Flood      >    20.00  deg    42.16 P
(3) Angle from Equilibrium to 50% Dk Imm Angle     >     0.00  deg     2.34 P
-----------------------------------------------------------------------------
```

27-01-20 14:18:57 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONTOON

Condition 5 : Lightship + Crane + Ballast + Deck Cargo
Towing Condition with Max. draft 2.212m



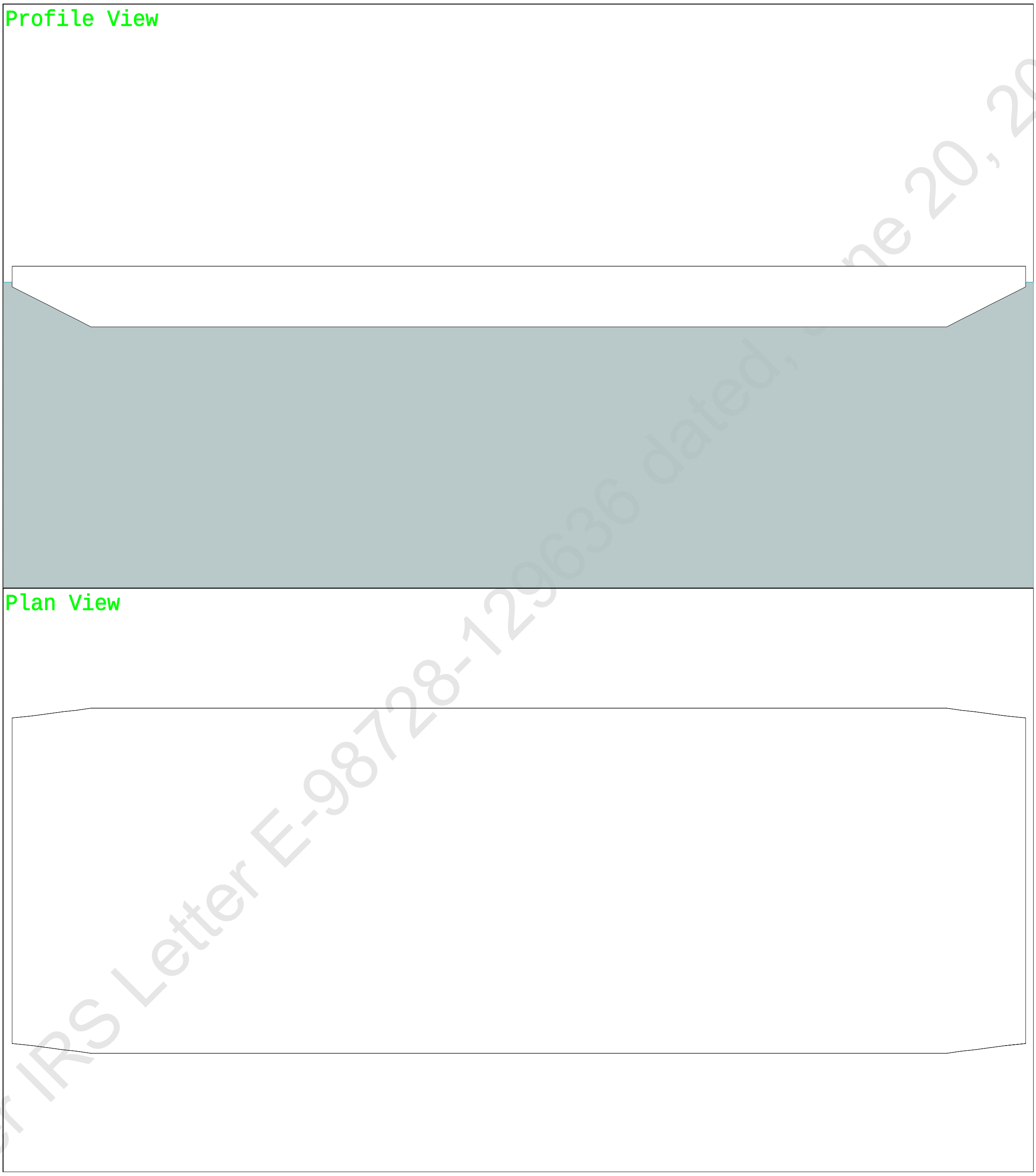


27-01-20 14:18:59 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

Condition 6 : Lightship + Max.Deck Cargo

Max. deck cargo loading condition

Condition Graphic - Draft: 2.212 Trim: zero Heel: zero



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27-01-20 14:18:59    Cybermarine Technologies PTE. Ltd.
GHS 16.68C                        PONTOON
```

Condition 6 : Lightship + Max.Deck Cargo

Max. deck cargo loading condition

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00
                          Trim: zero,  Heel: zero
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG-------FSM
LIGHT SHIP                          389.39  25.000f  0.000   3.000
Deck Cargo                        1,290.22  25.000f  0.000   5.000
  Total Weight-------->           1,679.61  25.000f  0.000   4.536
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG
  Total Tanks--------->               0.00                                0.00
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                     1.025    1,679.61  25.000f  0.000   1.163
-----------------------------------------------------------------------------
               Righting Arms:                  0.000   0.000
Part--------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         843.7  25.000f  0.000   105.63  12.243
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.65            34.35   102.26   8.870
Distances in METERS.-----------------------------------------Moments in m.-MT.

                               SUMMARY OF LOADING
         0.0 Cu.M.   (0%) SALT WATER           0.0 Cu.M.   (0%) VOID

    1,290.22 MT  Deck Cargo

DRAFT AT FP (M) =    2.212
DRAFT AT MS (M) =    2.212
DRAFT AT AP (M) =    2.212
DRAFT AT FWD MARK (M) =    2.212
DRAFT AT AFT MARK (M) =    2.212
```

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GHS 16.68C

PONTOON

Condition 6 : Lightship + Max.Deck Cargo

Max. deck cargo loading condition

HYDROSTATIC PROPERTIES
No Trim, No Heel, VCG = 4.536

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.212	1,679.61	25.000f	1.163	8.65	25.000f	34.35	102.26	8.870

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK.

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00
Trim: zero, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.405	1.477	0.05140	3.07
DECKCARGO.C	117.6	2.800	3.872	0.05140	23.41
Total wind heeling moment to starboard------------->					26.48

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
LCG = 25.000f TCG = 0.000 VCG = 4.536

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
2.200	0.00	0.00	1,679.6	0.000	-0.016	0.0000	2.732
2.200	0.00	0.10s	1,679.6	0.000	0.000	-0.0000	2.631
2.197	0.00	2.73s	1,679.6	0.000	0.410	0.0094	0.000
2.188	0.00	5.10s	1,679.6	0.000	0.775	0.0339	-2.369
2.242	0.00	10.10s	1,679.6	0.000	1.138	0.1226	-7.369
2.284	0.00	11.64s	1,679.7	0.000	1.149	0.1533	-8.907
2.400	0.00	15.10s	1,679.6	0.000	1.102	0.2222	-12.369
2.598	0.00	20.10s	1,679.5	0.000	0.937	0.3127	-17.369
2.810	0.00	25.10s	1,679.6	0.000	0.709	0.3850	-22.369
3.005	0.00	30.10s	1,679.6	0.000	0.436	0.4353	-27.369
3.177	0.00	35.10s	1,679.6	0.000	0.139	0.4606	-32.369
3.246	0.00	37.36s	1,679.6	0.000	0.000	0.4633	-34.626
3.324	0.00	40.10s	1,679.6	0.000	-0.171	0.4592	-37.369
3.447	0.00	45.10s	1,679.6	0.000	-0.487	0.4306	-42.369
3.543	0.00	50.10s	1,679.6	0.000	-0.803	0.3743	-47.369
3.612	0.00	55.10s	1,679.6	0.000	-1.117	0.2905	-52.369
3.654	0.00	60.10s	1,679.6	0.000	-1.423	0.1796	-57.369

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 26.48 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.1533	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	37.26	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.63	P

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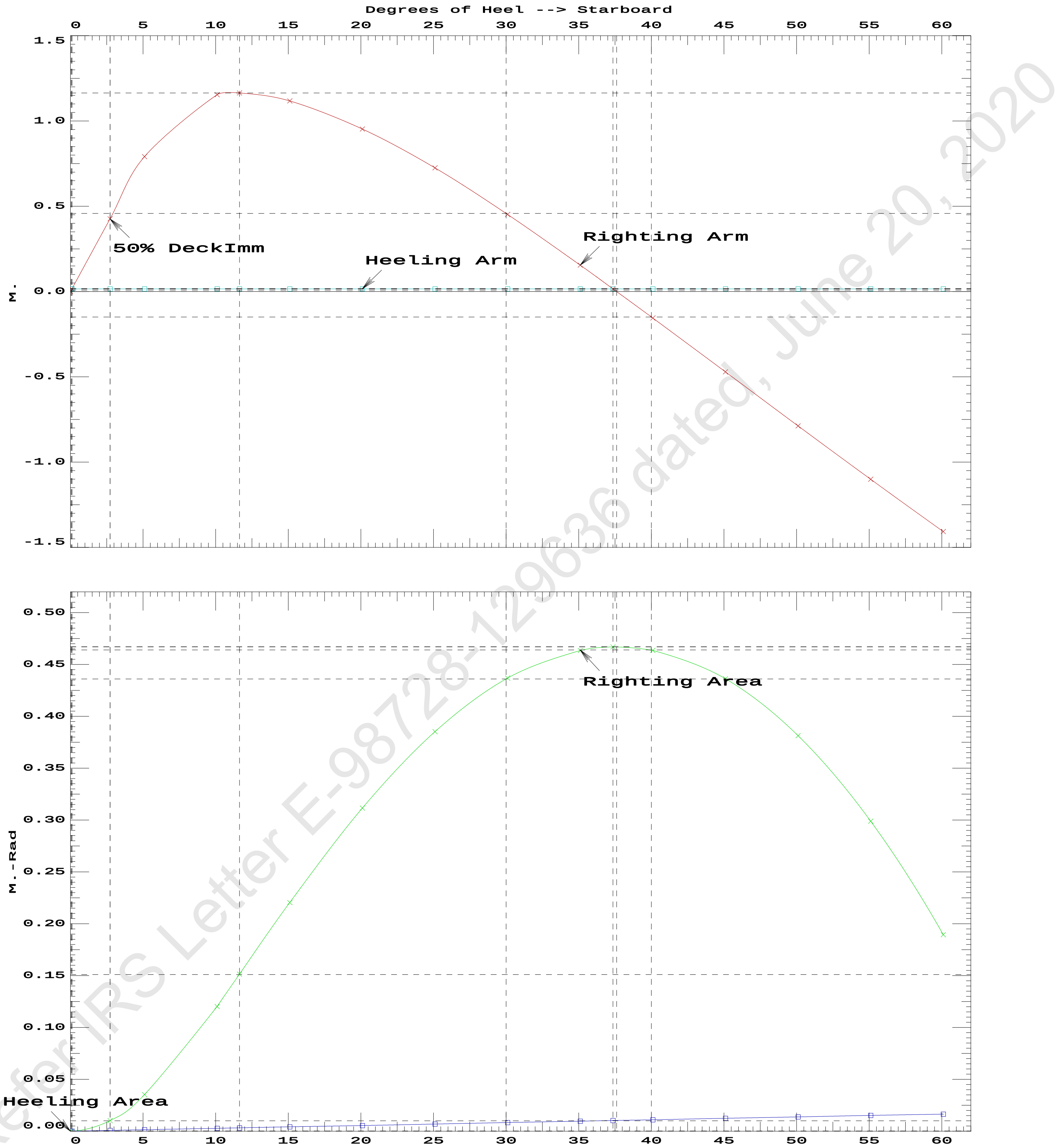
GHS 16.68C

PONTOON

Condition 6 : Lightship + Max.Deck Cargo

Max. deck cargo loading condition



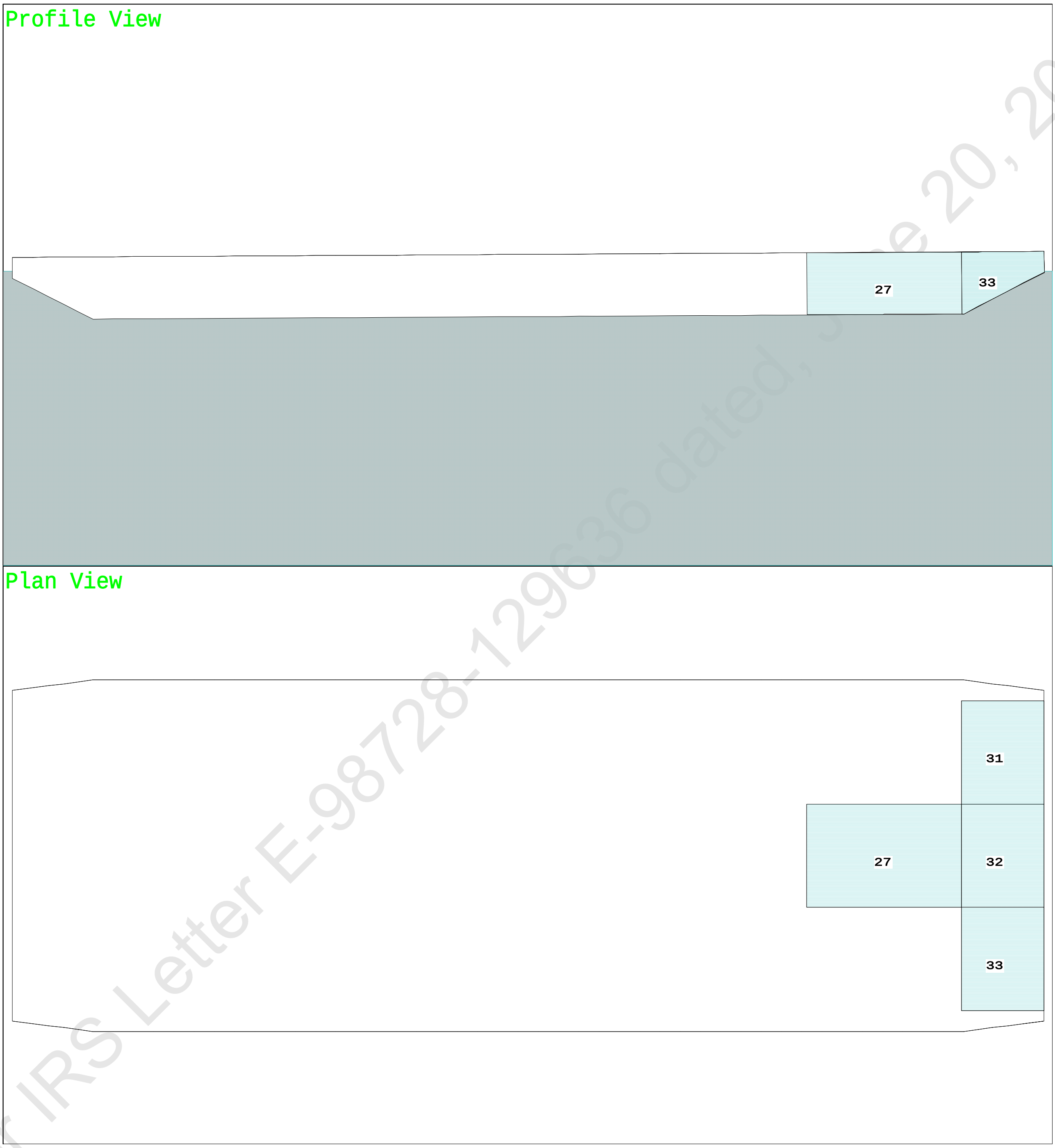
27-01-20 14:19:00 Cybermarine Technologies PTE. Ltd.
GHS 16.68C

PONTOON

Condition 7 : Lightship + Load at Aft + Ballast + Deck Cargo

Crane Lifting 36.60T @ 18m Radius Aft

Condition Graphic - Draft: 2.067 @ 50.000f, 2.356 @ 0.000 Heel: zero



Tanks 27 06_WBTK.C 31 07_WBTK.P 32 07_WBTK.C 33 07_WBTK.S

27-01-20 14:19:00 Cybermarine Technologies PTE. Ltd.
GHS 16.68C

PONTOON

Condition 7 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.067 @ 50.00f, 2.212 @ 25.00f, 2.356 @ 0.00
Trim: Aft 0.289/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			389.39	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	10.500a	0.000	76.977	
Deck Cargo			769.57	25.000f	0.000	5.000	
Total Fixed--------->			1,445.56	21.075f	0.000	6.318	
06_WBTK.C	1.000	1.025	113.01	42.250f	0.000	1.500	78.48*
07_WBTK.P	1.000	1.025	40.19	47.659f	4.971p	1.911	41.85*
07_WBTK.C	1.000	1.025	40.67	47.663f	0.000	1.901	41.85*
07_WBTK.S	1.000	1.025	40.19	47.659f	4.971s	1.911	41.85*
Total Tanks--------->			234.05	45.048f	0.000	1.711	204.04
Total Weight-------->			1,679.61	24.415f	0.000	5.676	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,679.61	24.389f	0.000	1.165	

Part	SpGr	WPA	LCF	TCF	BML	BMT
Righting Arms:			0.000	0.000		
Total Waterplane---->	1.025	843.7	24.967f	0.000	105.63	12.122
		MT/cm		m.-MT/cm	GML	GMT
		8.65		33.97	101.12	7.610

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

228.3 Cu.M. (13%) SALT WATER   0.0 Cu.M. (0%) VOID

250.00 MT Crane   36.60 MT Crane Load
769.57 MT Deck Cargo

DRAFT AT FP (M) = 2.067
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.356
DRAFT AT FWD MARK (M) = 2.090
DRAFT AT AFT MARK (M) = 2.333

HYDROSTATIC PROPERTIES
Trim: Aft 0.289/50.000, No Heel, VCG = 5.676

Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.212	1,679.61	24.389f	1.165	8.65	24.967f	33.97	101.12	7.610

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK.   Formal Free Surface included.

Note: GMT includes the formal free surface moment 204.0 m.-MT

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GHS 16.68C **PONTOON**

Condition 7 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Aft

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 2.067 @ 50.00f, 2.212 @ 25.00f, 2.356 @ 0.00
Trim: Aft 0.289/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.409	1.483	0.05140	3.09
CRANE.C	34.4	2.891	3.965	0.05140	7.01
DECKCARGO.C	98.0	2.815	3.888	0.05140	19.59
Total wind heeling moment to starboard------------->					29.68

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.415f TCG = 0.000 VCG = 5.676
Free Surface Adjustment: 0.121
Adjusted CG: LCG = 24.416f TCG = 0.000 VCG = 5.798

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
2.344	0.33a	0.00	1,679.6	0.000	-0.017	0.0000	2.311
2.344	0.33a	0.13s	1,679.6	0.000	0.000	-0.0000	2.183
2.343	0.33a	2.31s	1,679.6	0.000	0.292	0.0055	0.000
2.336	0.34a	5.13s	1,679.6	0.000	0.663	0.0290	-2.817
2.449	0.49a	9.76s	1,679.6	0.000	0.907	0.0966	-7.452
2.463	0.50a	10.13s	1,679.6	0.000	0.906	0.1024	-7.817
2.692	0.66a	15.13s	1,679.6	0.000	0.761	0.1765	-12.817
2.957	0.81a	20.13s	1,679.7	0.000	0.491	0.2320	-17.817
3.241	0.99a	25.13s	1,679.6	0.000	0.158	0.2608	-22.817
3.368	1.08a	27.31s	1,679.6	0.000	0.000	0.2639	-25.000
3.526	1.19a	30.13s	1,679.6	0.000	-0.212	0.2587	-27.817
3.788	1.40a	35.13s	1,679.6	0.000	-0.602	0.2233	-32.817
4.024	1.61a	40.13s	1,679.6	0.000	-0.999	0.1535	-37.817
4.229	1.80a	45.13s	1,679.6	0.000	-1.395	0.0490	-42.817
4.401	1.97a	50.13s	1,679.6	0.000	-1.785	-0.0898	-47.817
4.536	2.12a	55.13s	1,679.6	0.000	-2.163	-0.2621	-52.817
4.632	2.25a	60.13s	1,679.6	0.000	-2.527	-0.4668	-57.817

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 204.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 29.68 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.0966	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	27.18	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.18	P

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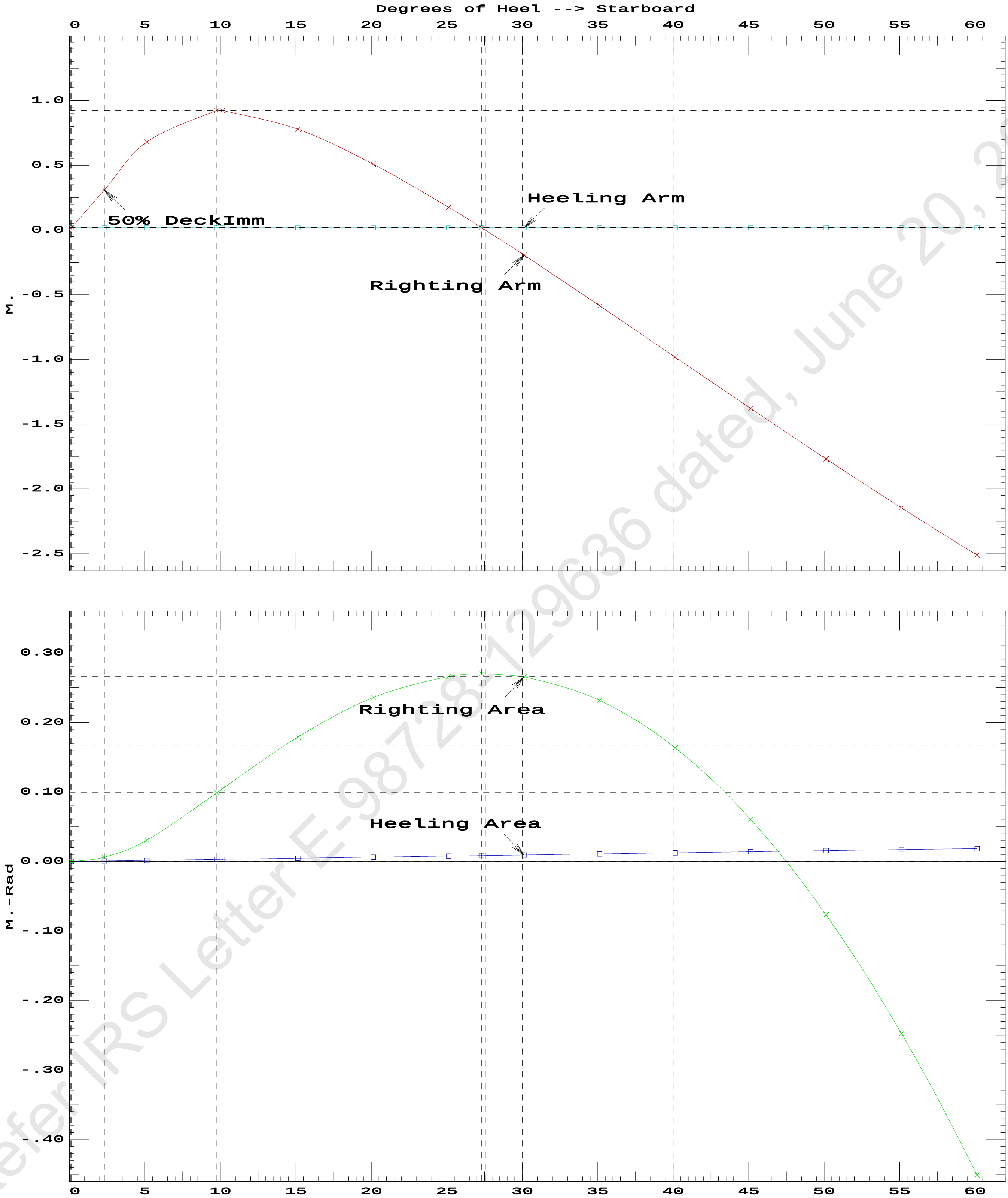
GHS 16.68C PONTOON

Condition 7 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Aft



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GHS 16.68C

PONTOON

Condition 7 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.415f TCG = 0.000 VCG = 5.676
Free Surface Adjustment: 0.121
Adjusted CG: LCG = 24.416f TCG = 0.000 VCG = 5.798

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.344	0.33a	0.00s	1,679.6	0.000	0.000	0.0000
2.336	0.34a	5.00s	1,679.6	0.000	0.664	0.0290
2.448	0.49a	9.73s	1,679.6	0.000	0.924	0.0989
2.458	0.50a	10.00s	1,679.6	0.000	0.923	0.1033
2.685	0.65a	15.00s	1,679.6	0.000	0.784	0.1792
2.950	0.81a	20.00s	1,679.8	0.000	0.515	0.2368
3.234	0.98a	25.00s	1,679.6	0.000	0.184	0.2677
3.381	1.09a	27.54s	1,679.6	0.000	0.000	0.2718
3.519	1.19a	30.00s	1,679.6	0.000	-0.186	0.2678
3.782	1.40a	35.00s	1,679.6	0.000	-0.576	0.2348
4.018	1.60a	40.00s	1,679.6	0.000	-0.972	0.1673
4.225	1.79a	45.00s	1,679.6	0.000	-1.368	0.0652
4.397	1.97a	50.00s	1,679.6	0.000	-1.758	-0.0713
4.533	2.12a	55.00s	1,679.6	0.000	-2.137	-0.2413
4.630	2.25a	60.00s	1,679.6	0.000	-2.501	-0.4438

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 204.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	155161 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		951.105 P

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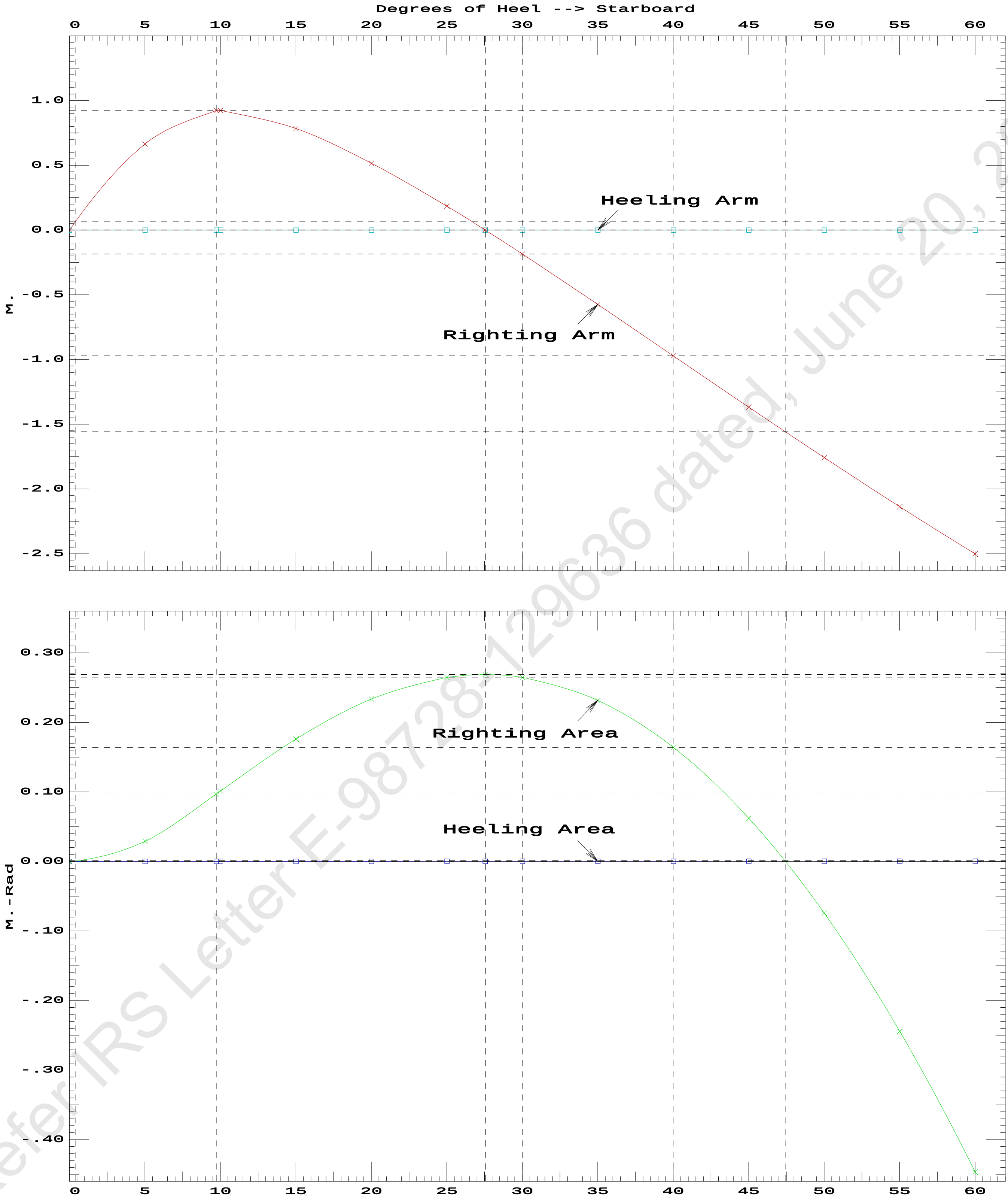
GHS 16.68C PONTOON

Condition 7 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Aft



27-01-20 14:19:02 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

Condition 7 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Aft
Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.067 @ 50.00f, 2.212 @ 25.00f, 2.356 @ 0.00
Trim: Aft 0.289/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			389.39	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	10.500a	0.000	76.977	
Deck Cargo			769.57	25.000f	0.000	5.000	
Total Fixed--------->			1,445.56	21.075f	0.000	6.318	
06_WBTK.C	1.000	1.025	113.01	42.250f	0.000	1.500	78.48*
07_WBTK.P	1.000	1.025	40.19	47.659f	4.971p	1.911	41.85*
07_WBTK.C	1.000	1.025	40.67	47.663f	0.000	1.901	41.85*
07_WBTK.S	1.000	1.025	40.19	47.659f	4.971s	1.911	41.85*
Total Tanks--------->			234.05	45.048f	0.000	1.711	204.04
Total Weight-------->			1,679.61	24.415f	0.000	5.676	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,679.61	24.389f	0.001s	1.165	

Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	843.7	24.967f	0.000	105.63	12.122
		MT/cm		m.-MT/cm	GML	GMT
		8.65		33.97	101.12	7.610

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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27-01-20 14:19:02    Cybermarine Technologies PTE. Ltd.
GHS 16.68C                          PONTOON

                          After hook load drop
               WEIGHT and DISPLACEMENT and WATERPLANE STATUS
          USK draft: 2.216 @ 50.00f, 2.170 @ 25.00f, 2.124 @ 0.00
                 Trim: Fwd 0.092/50.000,  Heel: 0.00 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                            389.39  25.000f  0.000   3.000
Crane                                 250.00   7.500f  0.000   5.200
Deck Cargo                            769.57  25.000f  0.000   5.000
  Total Fixed--------->             1,408.96  21.895f  0.000   4.483
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.C      1.000    1.025         113.01  42.250f  0.000   1.500     78.48*
07_WBTK.P      1.000    1.025          40.19  47.659f  4.971p  1.911     41.85*
07_WBTK.C      1.000    1.025          40.67  47.663f  0.000   1.901     41.85*
07_WBTK.S      1.000    1.025          40.19  47.659f  4.971s  1.911     41.85*
  Total Tanks--------->               234.05  45.048f  0.000   1.711   204.04
  Total Weight-------->             1,643.01  25.193f  0.000   4.088
                                    Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       1,643.01  25.199f  0.001   1.141
-----------------------------------------------------------------------------
               Righting Arms:                   0.000   0.001s
               External Arms:                   0.000   0.001
      Residual Righting Arms:                   0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          843.3  25.011f  0.000    107.86  12.379
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        8.64            34.47    104.91   9.431
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.
```

```
limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01
```

Theta3 is 0.07 degrees, as demonstrated below.

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GHS 16.68C **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 21.895f TCG = 0.000 VCG = 4.483

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.111	0.11f	0.00s	1,643.0	0.000	0.000	0.0000
2.112	0.11f	0.07s	1,643.0	0.000	0.011	0.0000
2.099	0.11f	5.00s	1,643.0	0.000	0.834	0.0364
2.114	0.15f	10.00s	1,643.1	0.000	1.301	0.1351
2.185	0.18f	13.22s	1,643.0	0.000	1.343	0.2103
2.234	0.20f	15.00s	1,643.0	0.000	1.330	0.2519
2.399	0.25f	20.00s	1,643.0	0.000	1.216	0.3647
2.568	0.30f	25.00s	1,643.0	0.000	1.026	0.4631
2.717	0.37f	30.00s	1,643.0	0.000	0.785	0.5425

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1408.96 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 0.1	>	1.000	LARGE

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GHS 16.68C **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 21.895f TCG = 0.000 VCG = 4.483

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.111	0.11f	0.00s	1,643.0	0.000	0.000	0.0000
2.112	0.11f	0.07s	1,643.0	0.000	0.011	0.0000
2.099	0.11f	5.00s	1,643.0	0.000	0.834	0.0364
2.114	0.15f	10.00s	1,643.1	0.000	1.301	0.1351
2.185	0.18f	13.22s	1,643.0	0.000	1.343	0.2103
2.234	0.20f	15.00s	1,643.0	0.000	1.330	0.2519
2.399	0.25f	20.00s	1,643.0	0.000	1.216	0.3647
2.568	0.30f	25.00s	1,643.0	0.000	1.026	0.4631
2.717	0.37f	30.00s	1,643.0	0.000	0.785	0.5425
2.844	0.43f	35.00s	1,643.0	0.000	0.516	0.5995
2.949	0.49f	40.00s	1,643.0	0.000	0.231	0.6323
3.015	0.53f	43.95s	1,643.0	0.000	0.000	0.6403
3.030	0.54f	45.00s	1,643.0	0.000	-0.062	0.6397

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1408.96 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	43.88 P

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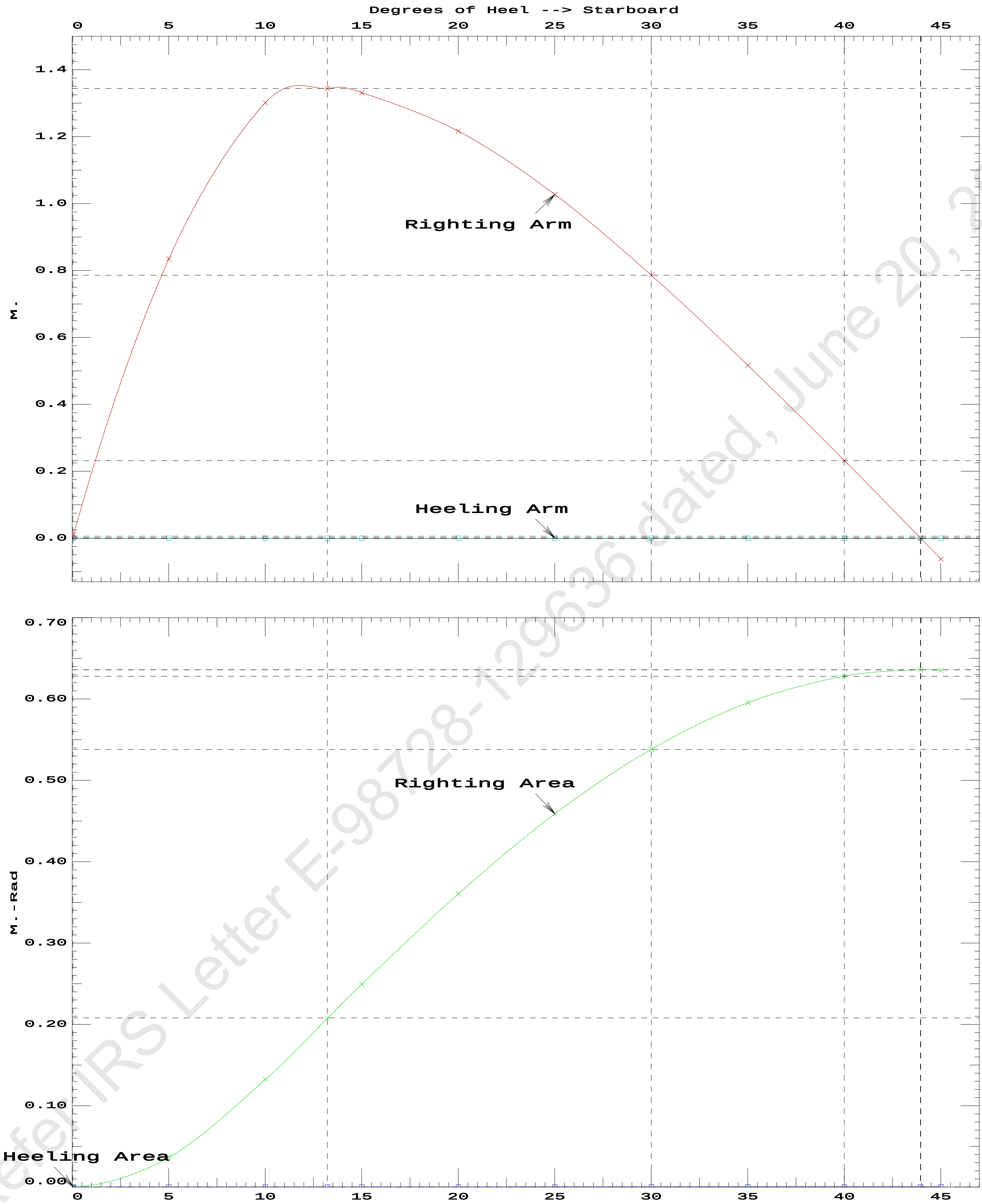
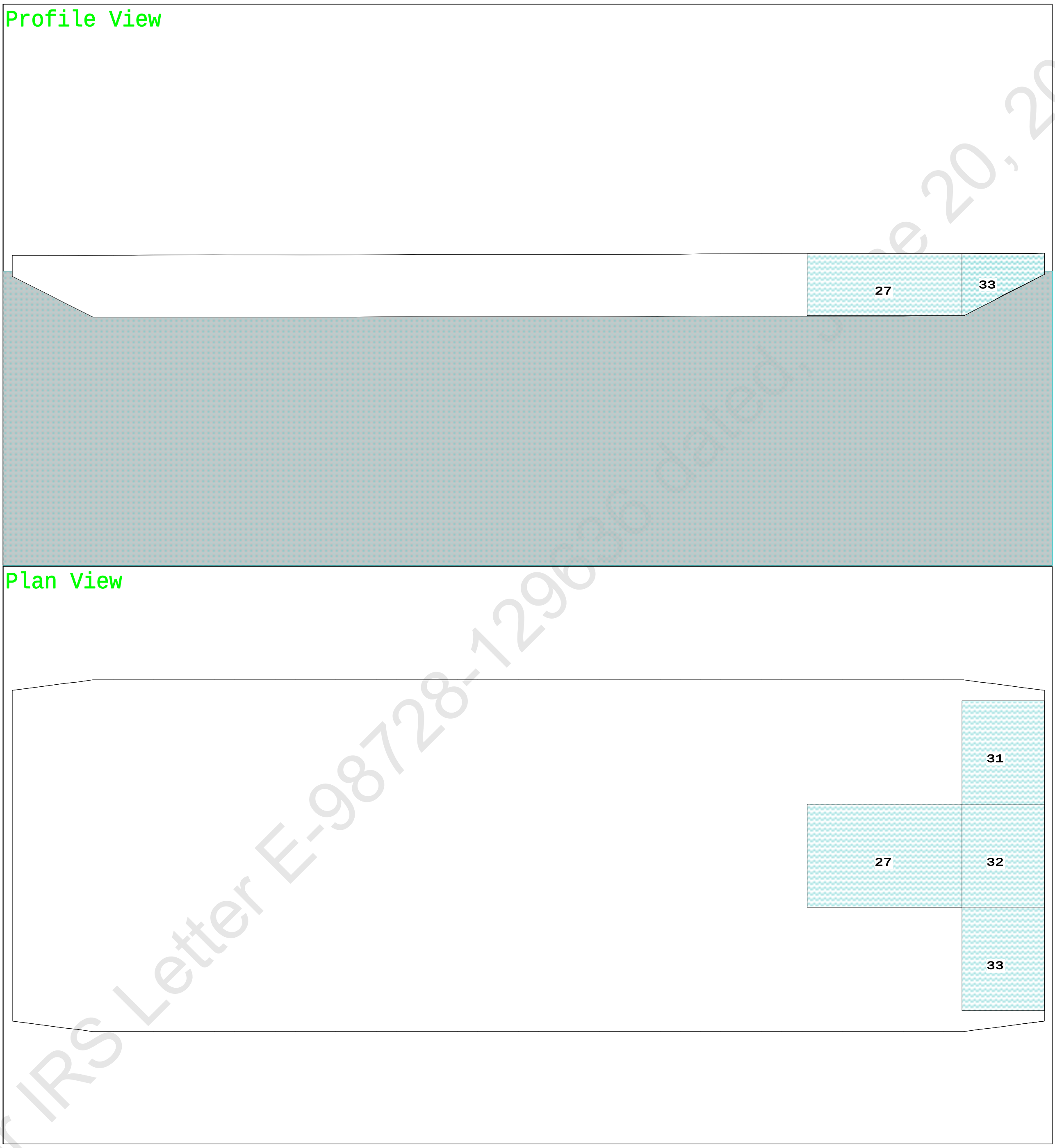
GHS 16.68C **PONTOON**



27-01-20 14:19:04 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

Condition 8 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Stbd

Condition Graphic - Draft: 2.164 @ 50.000f, 2.260 @ 0.000 Heel: zero



Tanks 27 06_WBTK.C 31 07_WBTK.P 32 07_WBTK.C 33 07_WBTK.S

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GHS 16.68C **PONTOON**

Condition 8 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.164 @ 50.00f, 2.212 @ 25.00f, 2.260 @ 0.00
Trim: Aft 0.095/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			389.39	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	7.500f	0.000	76.977	
Deck Cargo			769.57	25.000f	0.000	5.000	
Total Fixed--------->			1,445.56	21.530f	0.000	6.318	
06_WBTK.C	1.000	1.025	113.01	42.250f	0.000	1.500	78.48*
07_WBTK.P	1.000	1.025	40.19	47.659f	4.971p	1.911	41.85*
07_WBTK.C	1.000	1.025	40.67	47.663f	0.000	1.901	41.85*
07_WBTK.S	1.000	1.025	40.19	47.659f	4.971s	1.911	41.85*
Total Tanks--------->			234.05	45.048f	0.000	1.711	204.04
Total Weight-------->			1,679.61	24.808f	0.000	5.676	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,679.61	24.799f	0.000	1.163	

Part	SpGr	WPA	LCF	TCF	BML	BMT
Righting Arms:			0.000	0.000		
Total Waterplane---->	1.025	843.7	24.989f	0.000	105.63	12.122
		MT/cm		m.-MT/cm	GML	GMT
		8.65		33.97	101.12	7.609

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

228.3 Cu.M. (13%) SALT WATER 0.0 Cu.M. (0%) VOID

250.00 MT Crane 36.60 MT Crane Load
769.57 MT Deck Cargo

DRAFT AT FP (M) = 2.164
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.260
DRAFT AT FWD MARK (M) = 2.172
DRAFT AT AFT MARK (M) = 2.252

HYDROSTATIC PROPERTIES
Trim: Aft 0.095/50.000, No Heel, VCG = 5.676

Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.212	1,679.61	24.799f	1.163	8.65	24.989f	33.97	101.12	7.609

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 204.0 m.-MT

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GHS 16.68C **PONTOON**

Condition 8 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Stbd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 2.164 @ 50.00f, 2.212 @ 25.00f, 2.260 @ 0.00
Trim: Aft 0.095/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.405	1.478	0.05140	3.08
CRANE.C	34.4	2.959	4.031	0.05140	7.12
DECKCARGO.C	98.0	2.805	3.877	0.05140	19.53
Total wind heeling moment to starboard------------->					29.73

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.808f TCG = 0.000 VCG = 5.676
Free Surface Adjustment: 0.121
Adjusted CG: LCG = 24.808f TCG = 0.000 VCG = 5.798

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
2.248	0.11a	0.00	1,679.6	0.000	-0.017	0.0000	2.593
2.248	0.11a	0.13s	1,679.6	0.000	0.000	-0.0000	2.463
2.245	0.11a	2.59s	1,679.6	0.000	0.330	0.0071	0.000
2.236	0.11a	5.13s	1,679.6	0.000	0.665	0.0291	-2.537
2.300	0.16a	9.62s	1,679.6	0.000	0.916	0.0951	-7.027
2.315	0.16a	10.13s	1,679.6	0.000	0.914	0.1032	-7.537
2.496	0.22a	15.13s	1,679.6	0.000	0.769	0.1780	-12.537
2.716	0.27a	20.13s	1,679.6	0.000	0.499	0.2343	-17.537
2.952	0.32a	25.13s	1,679.6	0.000	0.169	0.2638	-22.537
3.060	0.36a	27.46s	1,679.6	0.000	0.000	0.2673	-24.866
3.177	0.39a	30.13s	1,679.6	0.000	-0.201	0.2626	-27.537
3.379	0.46a	35.13s	1,679.6	0.000	-0.592	0.2282	-32.537
3.556	0.53a	40.13s	1,679.6	0.000	-0.989	0.1593	-37.537
3.705	0.59a	45.13s	1,679.6	0.000	-1.386	0.0557	-42.537
3.827	0.65a	50.13s	1,679.6	0.000	-1.776	-0.0823	-47.537
3.918	0.70a	55.13s	1,679.6	0.000	-2.156	-0.2540	-52.537
3.977	0.74a	60.13s	1,679.6	0.000	-2.521	-0.4581	-57.537

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 204.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 29.73 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.0951	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	27.33	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.46	P

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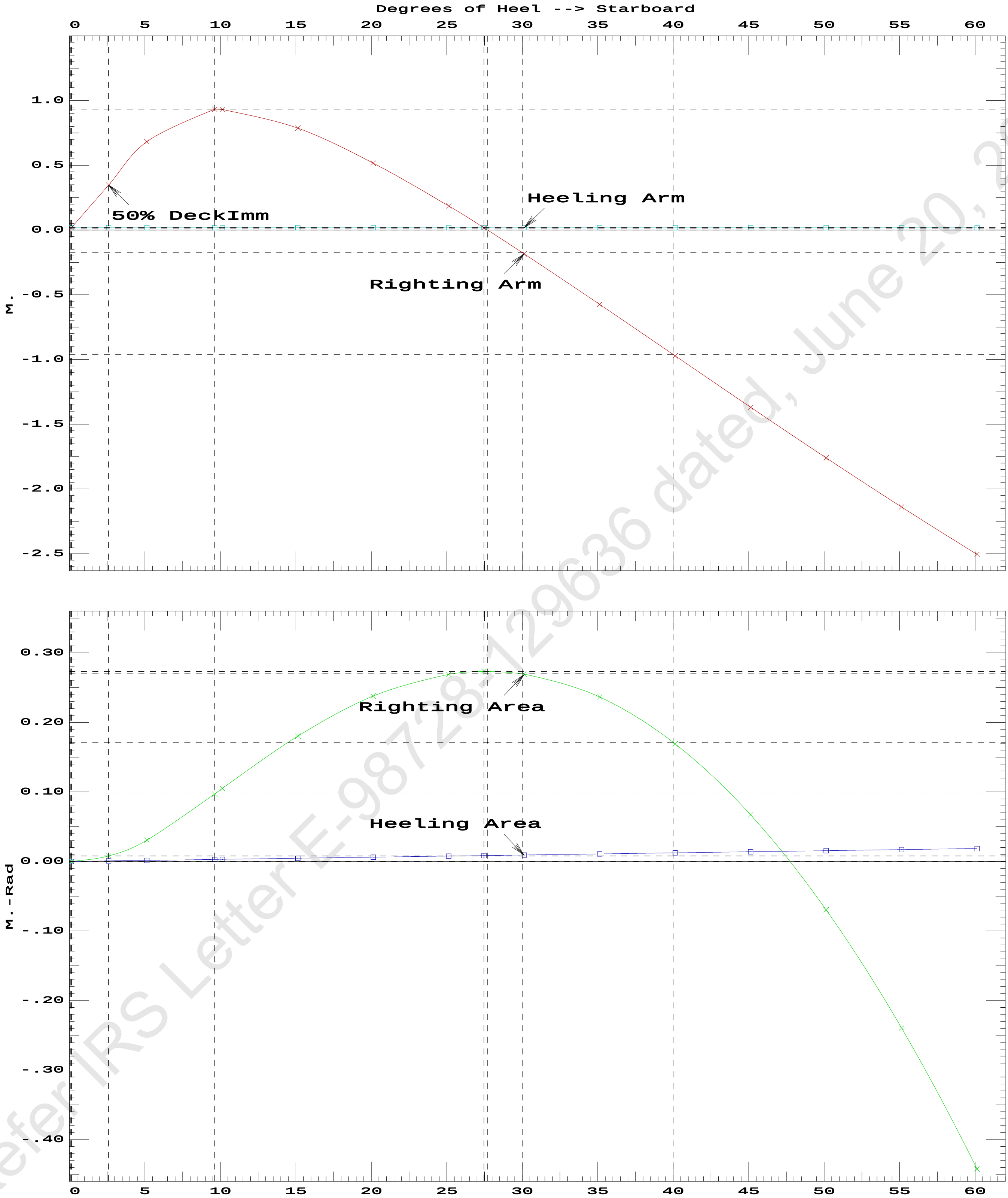
GHS 16.68C PONTOON

Condition 8 : Lightship + Load at Stbd + Ballast + Deck Cargo

Crane Lifting 36.60T @ 18m Radius Stbd



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GHS 16.68C

PONTOON

Condition 8 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.808f TCG = 0.000 VCG = 5.676
Free Surface Adjustment: 0.121
Adjusted CG: LCG = 24.808f TCG = 0.006p VCG = 5.798

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.244	0.11a	2.88s	1,679.6	0.000	0.000	0.0000
2.259	0.14a	7.88s	1,679.6	0.000	0.519	0.0226
2.307	0.16a	9.85s	1,679.6	0.000	0.547	0.0412
2.407	0.19a	12.88s	1,679.6	0.000	0.488	0.0694
2.614	0.24a	17.88s	1,679.6	0.000	0.261	0.1029
2.812	0.29a	22.18s	1,679.6	0.000	0.000	0.1131
2.845	0.29a	22.88s	1,679.5	0.000	-0.046	0.1128
3.079	0.36a	27.88s	1,679.6	0.000	-0.401	0.0937
3.291	0.43a	32.88s	1,679.6	0.000	-0.785	0.0422
3.479	0.50a	37.88s	1,679.6	0.000	-1.180	-0.0434
3.642	0.56a	42.88s	1,679.6	0.000	-1.578	-0.1638
3.776	0.62a	47.88s	1,679.6	0.000	-1.972	-0.3187
3.881	0.68a	52.88s	1,679.6	0.000	-2.358	-0.5077
3.954	0.72a	57.88s	1,679.6	0.000	-2.731	-0.7298
3.996	0.76a	62.88s	1,679.6	0.000	-3.087	-0.9838

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 204.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 658.80

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	2.88 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	139.4 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		1.856 P

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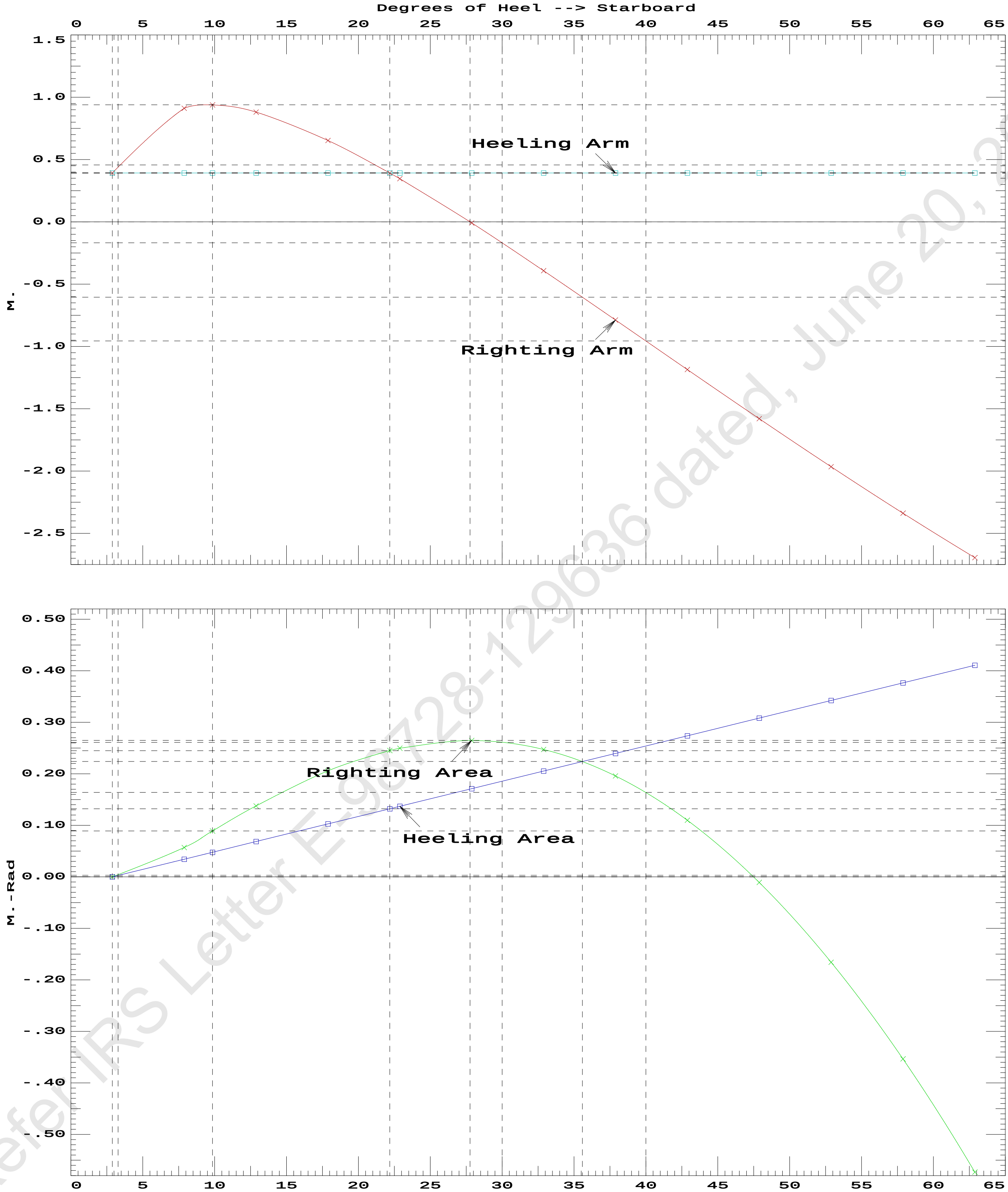
GHS 16.68C

PONTOON

Condition 8 : Lightship + Load at Stbd + Ballast + Deck Cargo

Crane Lifting 36.60T @ 18m Radius Stbd



27-01-20 14:19:08 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

Condition 8 : Lightship + Load at Stbd + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Stbd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.163 @ 50.00f, 2.211 @ 25.00f, 2.259 @ 0.00
Trim: Aft 0.096/50.000, Heel: Stbd 2.88 deg.

```
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                            389.39  25.000f  0.000   3.000
Crane                                 250.00   7.500f  0.000   5.200
Crane Load                             36.60   7.500f  0.000  76.977
Deck Cargo                            769.57  25.000f  0.000   5.000
  Total Fixed--------->             1,445.56  21.530f  0.000   6.318
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.C      1.000    1.025         113.01  42.250f  0.000   1.500     78.48*
07_WBTK.P      1.000    1.025          40.19  47.659f  4.971p  1.911     41.85*
07_WBTK.C      1.000    1.025          40.67  47.663f  0.000   1.901     41.85*
07_WBTK.S      1.000    1.025          40.19  47.659f  4.971s  1.911     41.85*
  Total Tanks--------->               234.05  45.048f  0.000   1.711    204.04
  Total Weight-------->             1,679.61  24.808f  0.000   5.676
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       1,679.62  24.799f  0.619s  1.179
------------------------------------------------------------------------------
               Righting Arms:                   0.000   0.392s
               External Arms:                   0.000   0.392s
      Residual Righting Arms:                   0.000   0.000s
Part--------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          841.3  24.963f  0.161s   104.53  12.079
                                       MT/cm
                                        8.62
Distances in METERS.----------------------------------------Moments in m.-MT.
```

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 2.89

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GHS 16.68C **PONTOON**

After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.216 @ 50.00f, 2.170 @ 25.00f, 2.124 @ 0.00
Trim: Fwd 0.092/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			389.39	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Deck Cargo			769.57	25.000f	0.000	5.000	
Total Fixed--------->			1,408.96	21.895f	0.000	4.483	
06_WBTK.C	1.000	1.025	113.01	42.250f	0.000	1.500	78.48*
07_WBTK.P	1.000	1.025	40.19	47.659f	4.971p	1.911	41.85*
07_WBTK.C	1.000	1.025	40.67	47.663f	0.000	1.901	41.85*
07_WBTK.S	1.000	1.025	40.19	47.659f	4.971s	1.911	41.85*
Total Tanks--------->			234.05	45.048f	0.000	1.711	204.04
Total Weight-------->			1,643.01	25.193f	0.000	4.088	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,643.01	25.199f	0.001	1.141	

Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	843.3	25.011f	0.000	107.86	12.379
		MT/cm		m.-MT/cm	GML	GMT
		8.64		34.47	104.91	9.431

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 4342.06; t3: 30.00; t: 15.00
limmarg: 1962.36; t3: 15.00; t: 7.50
limmarg: 556.12; t3: 7.50; t: 3.75
limmarg: 67.89; t3: 3.75; t: 1.88
limmarg: -57.89; t3: 1.88; t: 0.94
limmarg: -5.39; t3: 2.81; t: 0.47
limmarg: 28.68; t3: 3.28; t: 0.23
limmarg: 11.03; t3: 3.05; t: 0.12
limmarg: 2.69; t3: 2.93; t: 0.06
limmarg: -1.39; t3: 2.87; t: 0.03
limmarg: 0.60; t3: 2.90; t: 0.01

Theta3 is 2.90 degrees, as demonstrated below.

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GHS 16.68C **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 21.895f TCG = 0.000 VCG = 4.483

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.107	0.11f	2.88p	1,643.0	0.000	-0.485	0.0000
2.112	0.11f	0.00s	1,643.0	0.000	0.000	-0.0122
2.107	0.11f	2.90s	1,643.0	0.000	0.486	0.0001
2.098	0.11f	5.00s	1,643.0	0.000	0.834	0.0243
2.114	0.15f	10.00s	1,643.1	0.000	1.301	0.1195
2.185	0.18f	13.22s	1,643.0	0.000	1.343	0.1948
2.234	0.20f	15.00s	1,643.0	0.000	1.330	0.2363
2.399	0.25f	20.00s	1,643.0	0.000	1.216	0.3492
2.568	0.30f	25.00s	1,643.0	0.000	1.026	0.4475
2.717	0.37f	30.00s	1,643.0	0.000	0.785	0.5270

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1408.96 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Res. Area Ratio from abs -2.9 deg to abs 2.9	>	1.000	1.006 P

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GHS 16.68C **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 21.895f TCG = 0.000 VCG = 4.483

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.107	0.11f	2.88p	1,643.0	0.000	-0.485	0.0000
2.112	0.11f	0.00s	1,643.0	0.000	0.000	-0.0122
2.107	0.11f	2.90s	1,643.0	0.000	0.486	0.0001
2.098	0.11f	5.00s	1,643.0	0.000	0.834	0.0243
2.114	0.15f	10.00s	1,643.1	0.000	1.301	0.1195
2.185	0.18f	13.22s	1,643.0	0.000	1.343	0.1948
2.234	0.20f	15.00s	1,643.0	0.000	1.330	0.2363
2.399	0.25f	20.00s	1,643.0	0.000	1.216	0.3492
2.568	0.30f	25.00s	1,643.0	0.000	1.026	0.4475
2.717	0.37f	30.00s	1,643.0	0.000	0.785	0.5270
2.844	0.43f	35.00s	1,643.0	0.000	0.516	0.5840
2.949	0.49f	40.00s	1,643.0	0.000	0.231	0.6167
3.015	0.53f	43.95s	1,643.0	0.000	0.000	0.6247
3.030	0.54f	45.00s	1,643.0	0.000	-0.062	0.6241

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1408.96 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max	Attained
(1) Res. Area Ratio from abs -2.9 deg to RAzero	>	1.000	52.149 P
(2) Angle from abs 2.9 deg to RAzero	>	20.00 deg	41.05 P

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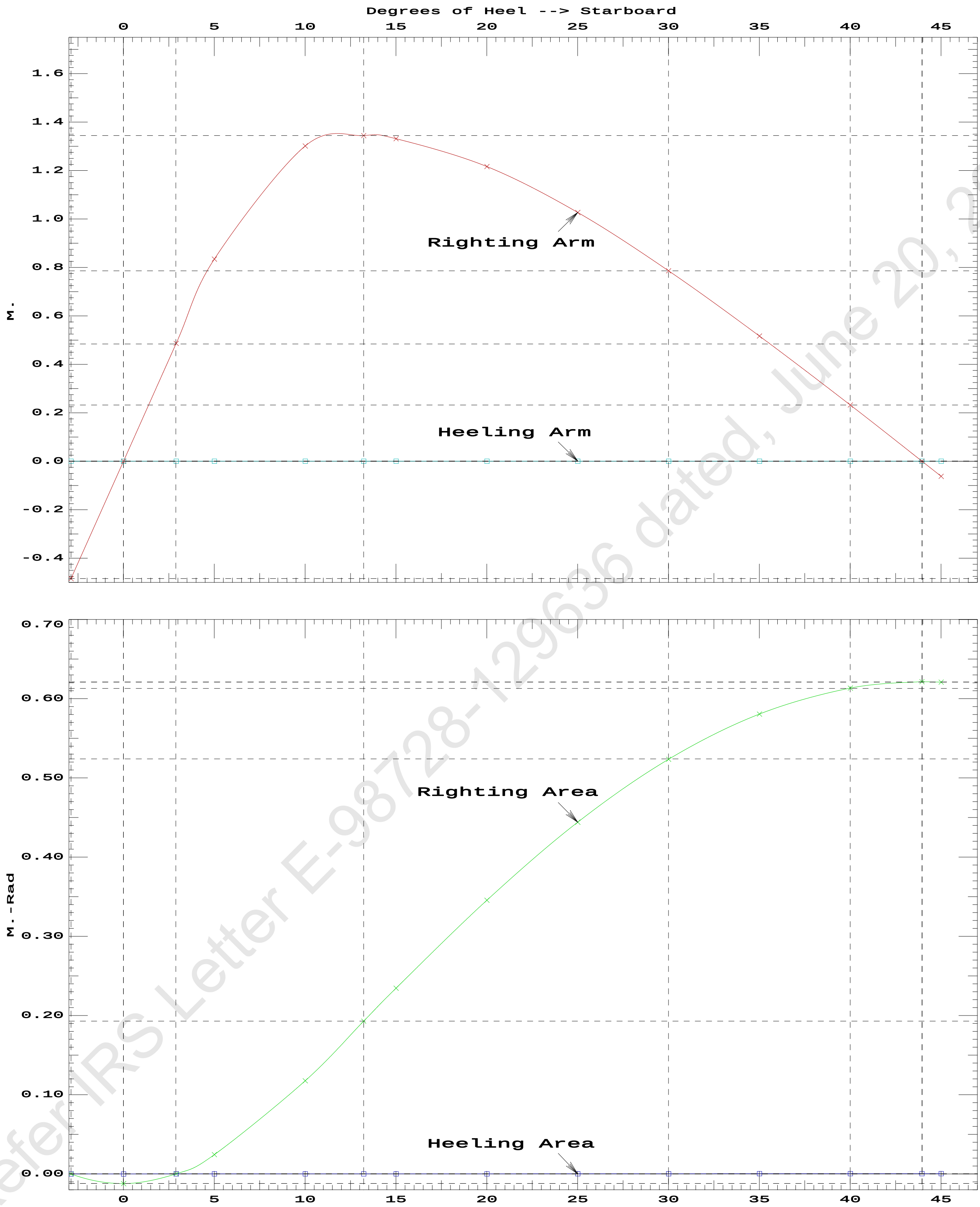
GHS 16.68C PONTOON



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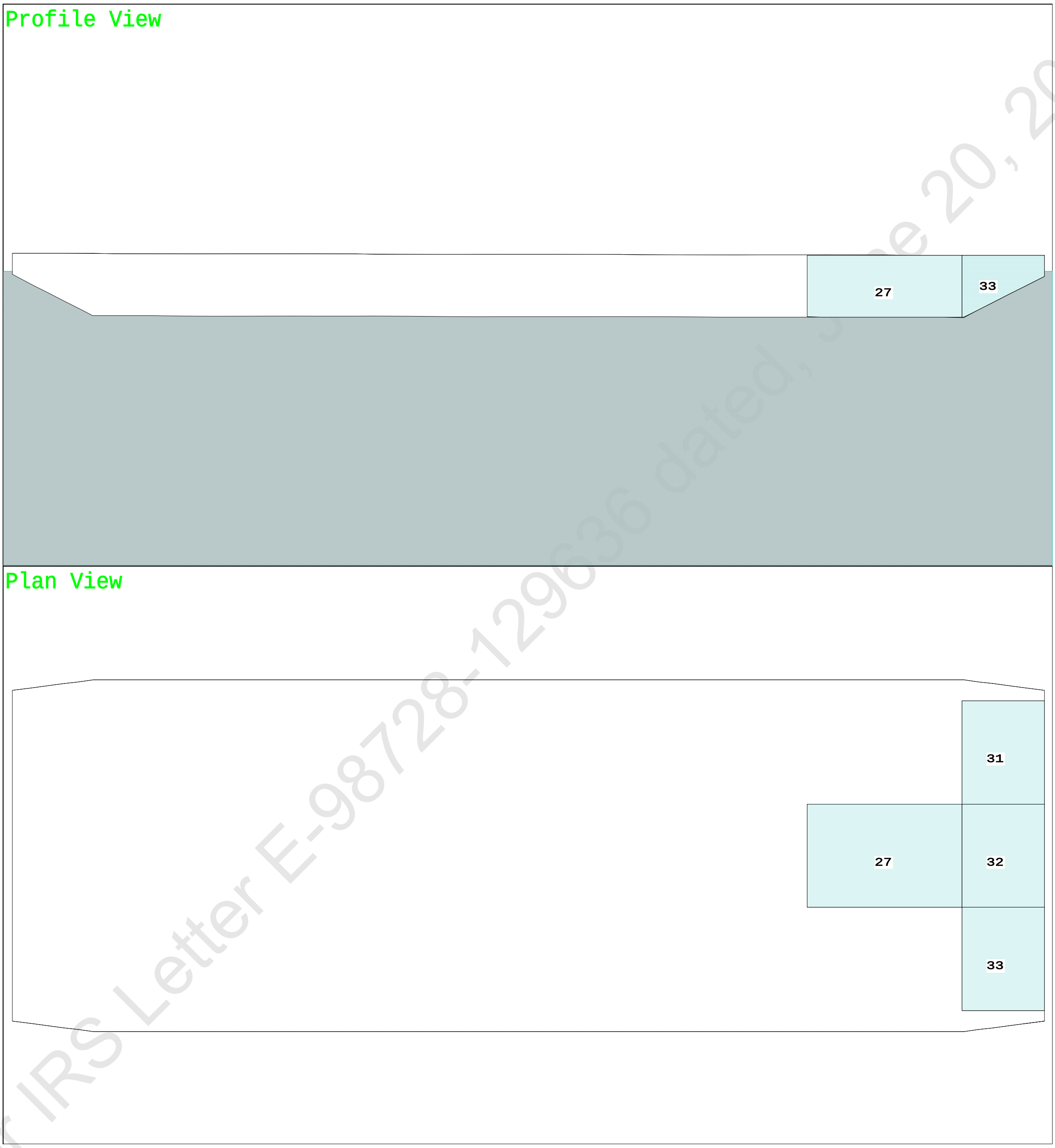
PONTOON

Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Fwd

Condition Graphic - Draft: 2.261 @ 50.000f, 2.163 @ 0.000 Heel: zero



Tanks				
	27 06_WBTK.C	31 07_WBTK.P	32 07_WBTK.C	33 07_WBTK.S

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GHS 16.68C                          PONTOON
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Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo

Crane Lifting 36.60T @ 18m Radius Fwd

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 2.261 @ 50.00f, 2.212 @ 25.00f, 2.163 @ 0.00
                    Trim: Fwd 0.099/50.000,  Heel: zero
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           389.39  25.000f  0.000   3.000
Crane                                250.00   7.500f  0.000   5.200
Crane Load                            36.60  25.500f  0.000  76.977
Deck Cargo                           769.57  25.000f  0.000   5.000
  Total Fixed--------->            1,445.56  21.986f  0.000   6.318
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.C      1.000    1.025        113.01  42.250f  0.000   1.500    78.48*
07_WBTK.P      1.000    1.025         40.19  47.659f  4.971p  1.911    41.85*
07_WBTK.C      1.000    1.025         40.67  47.663f  0.000   1.901    41.85*
07_WBTK.S      1.000    1.025         40.19  47.659f  4.971s  1.911    41.85*
  Total Tanks--------->              234.05  45.048f  0.000   1.711   204.04
  Total Weight-------->            1,679.61  25.200f  0.000   5.676
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,679.61  25.209f  0.000   1.163
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         843.7  25.011f  0.000    105.63  12.122
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.65             33.97    101.12   7.609
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                               SUMMARY OF LOADING
       228.3 Cu.M.  (13%) SALT WATER           0.0 Cu.M.   (0%) VOID

      250.00 MT  Crane                        36.60 MT  Crane Load
      769.57 MT  Deck Cargo

DRAFT AT FP (M) =    2.261
DRAFT AT MS (M) =    2.212
DRAFT AT AP (M) =    2.163
DRAFT AT FWD MARK (M) =    2.253
DRAFT AT AFT MARK (M) =    2.170
                            HYDROSTATIC PROPERTIES
                 Trim: Fwd 0.099/50.000, No Heel, VCG = 5.676

    LCF   Displacement   Buoyancy-Ctr.  Weight/          Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  2.212     1,679.61  25.209f  1.163     8.65  25.011f   33.97  101.12   7.609
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                              Trim is per 50.00m.
 Draft is from USK.                                 Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 204.0 m.-MT
```

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GHS 16.68C **PONTOON**

Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Fwd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 2.261 @ 50.00f, 2.212 @ 25.00f, 2.163 @ 0.00
Trim: Fwd 0.099/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.406	1.478	0.05140	3.08
CRANE.C	34.4	3.027	4.099	0.05140	7.24
DECKCARGO.C	98.0	2.795	3.867	0.05140	19.48
Total wind heeling moment to starboard------------->					29.80

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.200f TCG = 0.000 VCG = 5.676
Free Surface Adjustment: 0.121
Adjusted CG: LCG = 25.200f TCG = 0.000 VCG = 5.798

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
2.151	0.11f	0.00	1,679.6	0.000	-0.017	0.0000	2.586
2.150	0.11f	0.13s	1,679.6	0.000	0.000	-0.0000	2.456
2.147	0.11f	2.59s	1,679.6	0.000	0.329	0.0070	0.000
2.137	0.12f	5.13s	1,679.6	0.000	0.665	0.0291	-2.544
2.161	0.17f	9.75s	1,679.6	0.000	0.915	0.0971	-7.160
2.168	0.17f	10.13s	1,679.6	0.000	0.914	0.1032	-7.544
2.303	0.22f	15.13s	1,679.6	0.000	0.769	0.1780	-12.544
2.479	0.28f	20.13s	1,679.6	0.000	0.499	0.2342	-17.544
2.664	0.34f	25.13s	1,679.6	0.000	0.169	0.2638	-22.544
2.743	0.37f	27.46s	1,679.6	0.000	0.000	0.2672	-24.870
2.827	0.41f	30.13s	1,679.6	0.000	-0.202	0.2626	-27.544
2.968	0.48f	35.13s	1,679.6	0.000	-0.592	0.2281	-32.544
3.085	0.55f	40.13s	1,679.6	0.000	-0.989	0.1592	-37.544
3.179	0.61f	45.13s	1,679.6	0.000	-1.386	0.0556	-42.544
3.249	0.67f	50.13s	1,679.6	0.000	-1.776	-0.0824	-47.544
3.295	0.73f	55.13s	1,679.6	0.000	-2.156	-0.2541	-52.544
3.318	0.77f	60.13s	1,679.6	0.000	-2.521	-0.4583	-57.544

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 204.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 29.80 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.0971	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	27.33	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.46	P

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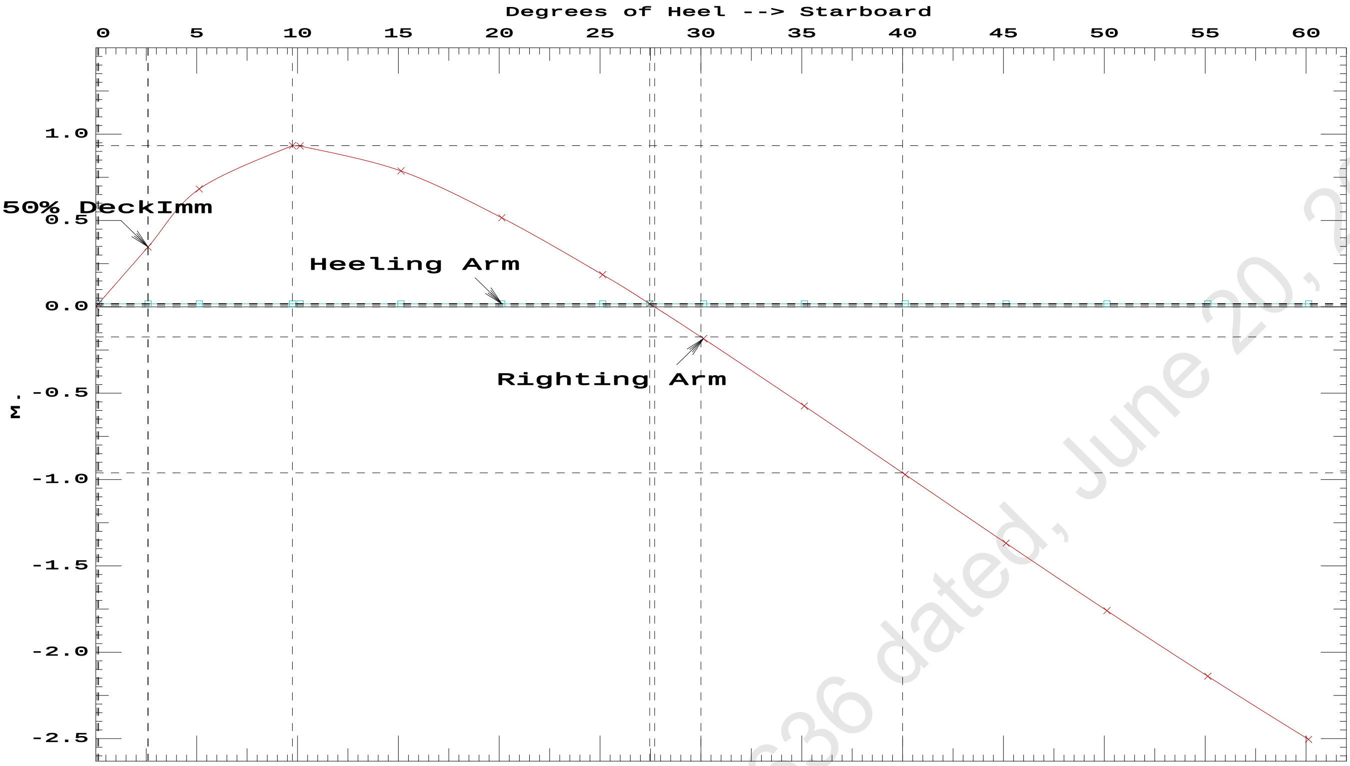
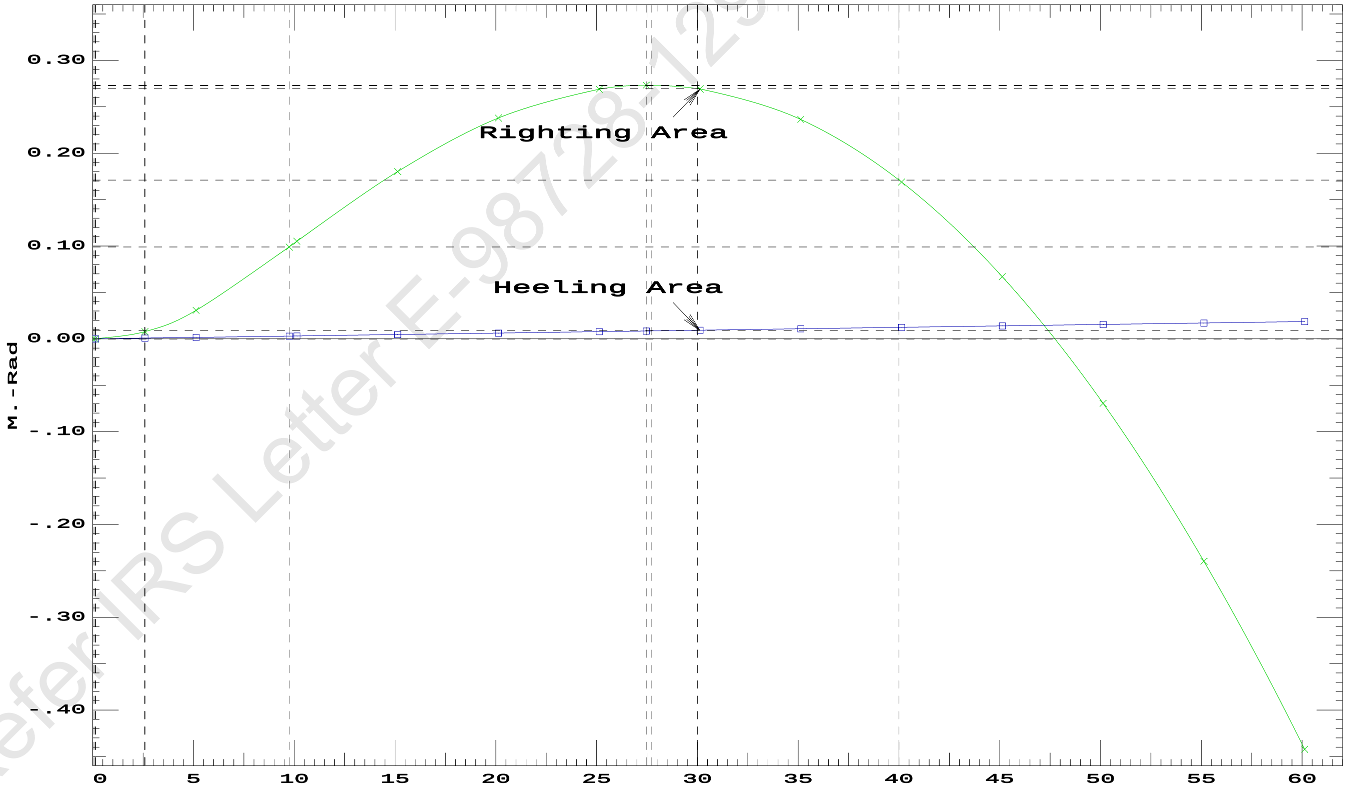
Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Fwd





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GHS 16.68C

PONTOON

Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo

Crane Lifting 36.60T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.200f TCG = 0.000   VCG = 5.676
Free Surface Adjustment: 0.121
Adjusted CG: LCG = 25.200f TCG = 0.000   VCG = 5.798

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.150	0.11f	0.00s	1,679.6	0.000	0.000	0.0000
2.138	0.12f	5.00s	1,679.7	0.000	0.665	0.0290
2.160	0.17f	9.71s	1,679.6	0.000	0.932	0.0992
2.166	0.17f	10.00s	1,679.6	0.000	0.932	0.1040
2.299	0.22f	15.00s	1,679.6	0.000	0.792	0.1806
2.475	0.28f	20.00s	1,679.6	0.000	0.523	0.2389
2.660	0.33f	25.00s	1,679.6	0.000	0.194	0.2707
2.750	0.37f	27.68s	1,679.6	0.000	0.000	0.2753
2.824	0.41f	30.00s	1,679.6	0.000	-0.175	0.2717
2.965	0.48f	35.00s	1,679.6	0.000	-0.565	0.2396
3.083	0.55f	40.00s	1,679.6	0.000	-0.962	0.1731
3.177	0.61f	45.00s	1,679.6	0.000	-1.359	0.0718
3.248	0.67f	50.00s	1,679.6	0.000	-1.750	-0.0639
3.295	0.72f	55.00s	1,679.6	0.000	-2.130	-0.2332
3.318	0.77f	60.00s	1,679.6	0.000	-2.496	-0.4352

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 204.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	156608 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		958.045 P

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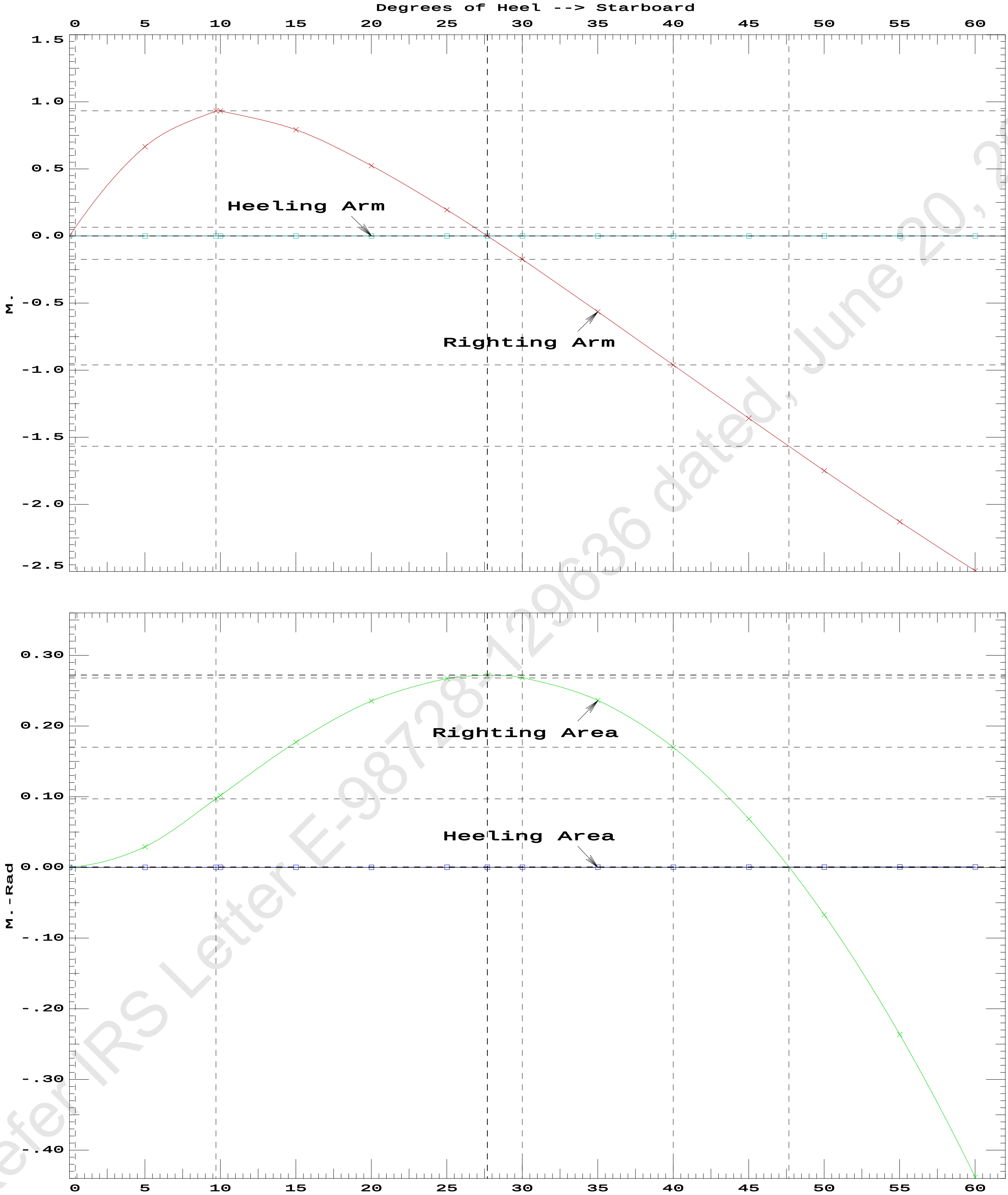
Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Fwd



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GHS 16.68C **PONTOON**

Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 36.60T @ 18m Radius Fwd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.261 @ 50.00f, 2.212 @ 25.00f, 2.163 @ 0.00
Trim: Fwd 0.099/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			389.39	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	25.500f	0.000	76.977	
Deck Cargo			769.57	25.000f	0.000	5.000	
Total Fixed--------->			1,445.56	21.986f	0.000	6.318	
06_WBTK.C	1.000	1.025	113.01	42.250f	0.000	1.500	78.48*
07_WBTK.P	1.000	1.025	40.19	47.659f	4.971p	1.911	41.85*
07_WBTK.C	1.000	1.025	40.67	47.663f	0.000	1.901	41.85*
07_WBTK.S	1.000	1.025	40.19	47.659f	4.971s	1.911	41.85*
Total Tanks--------->			234.05	45.048f	0.000	1.711	204.04
Total Weight-------->			1,679.61	25.200f	0.000	5.676	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,679.61	25.209f	0.001s	1.163	

	LCG	TCG
Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	843.7	25.011f	0.000	105.63	12.122
		MT/cm		m.-MT/cm	GML	GMT
		8.65		33.97	101.12	7.609

Distances in METERS.---Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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GHS 16.68C                       PONTOON

                          After hook load drop
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 2.216 @ 50.00f, 2.170 @ 25.00f, 2.124 @ 0.00
                   Trim: Fwd 0.092/50.000,  Heel: 0.00 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           389.39  25.000f  0.000   3.000
Crane                                250.00   7.500f  0.000   5.200
Deck Cargo                           769.57  25.000f  0.000   5.000
  Total Fixed--------->            1,408.96  21.895f  0.000   4.483
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.C      1.000    1.025        113.01  42.250f  0.000   1.500     78.48*
07_WBTK.P      1.000    1.025         40.19  47.659f  4.971p  1.911     41.85*
07_WBTK.C      1.000    1.025         40.67  47.663f  0.000   1.901     41.85*
07_WBTK.S      1.000    1.025         40.19  47.659f  4.971s  1.911     41.85*
  Total Tanks--------->              234.05  45.048f  0.000   1.711   204.04
  Total Weight-------->            1,643.01  25.193f  0.000   4.088
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,643.01  25.198f  0.001   1.141
-----------------------------------------------------------------------------
               Righting Arms:                 0.000   0.001s
               External Arms:                 0.000   0.001
      Residual Righting Arms:                 0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         843.3  25.011f  0.000    107.86  12.379
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.64             34.47    104.91   9.431
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

                Theta3 is 0.07 degrees, as demonstrated below.
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GHS 16.68C **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 21.895f TCG = 0.000 VCG = 4.483

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.111	0.11f	0.00s	1,643.0	0.000	0.000	0.0000
2.112	0.11f	0.07s	1,643.0	0.000	0.011	0.0000
2.099	0.11f	5.00s	1,643.0	0.000	0.834	0.0364
2.114	0.15f	10.00s	1,643.1	0.000	1.301	0.1351
2.185	0.18f	13.22s	1,643.0	0.000	1.343	0.2103
2.234	0.20f	15.00s	1,643.0	0.000	1.330	0.2519
2.399	0.25f	20.00s	1,643.0	0.000	1.216	0.3647
2.568	0.30f	25.00s	1,643.0	0.000	1.026	0.4631
2.717	0.37f	30.00s	1,643.0	0.000	0.785	0.5425

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1408.96 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 0.1	>	1.000	LARGE

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GHS 16.68C **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 21.895f TCG = 0.000 VCG = 4.483

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.111	0.11f	0.00s	1,643.0	0.000	0.000	0.0000
2.112	0.11f	0.07s	1,643.0	0.000	0.011	0.0000
2.099	0.11f	5.00s	1,643.0	0.000	0.834	0.0364
2.114	0.15f	10.00s	1,643.1	0.000	1.301	0.1351
2.185	0.18f	13.22s	1,643.0	0.000	1.343	0.2103
2.234	0.20f	15.00s	1,643.0	0.000	1.330	0.2519
2.399	0.25f	20.00s	1,643.0	0.000	1.216	0.3647
2.568	0.30f	25.00s	1,643.0	0.000	1.026	0.4631
2.717	0.37f	30.00s	1,643.0	0.000	0.785	0.5425
2.844	0.43f	35.00s	1,643.0	0.000	0.516	0.5995
2.949	0.49f	40.00s	1,643.0	0.000	0.231	0.6323
3.015	0.53f	43.95s	1,643.0	0.000	0.000	0.6403
3.030	0.54f	45.00s	1,643.0	0.000	-0.062	0.6397

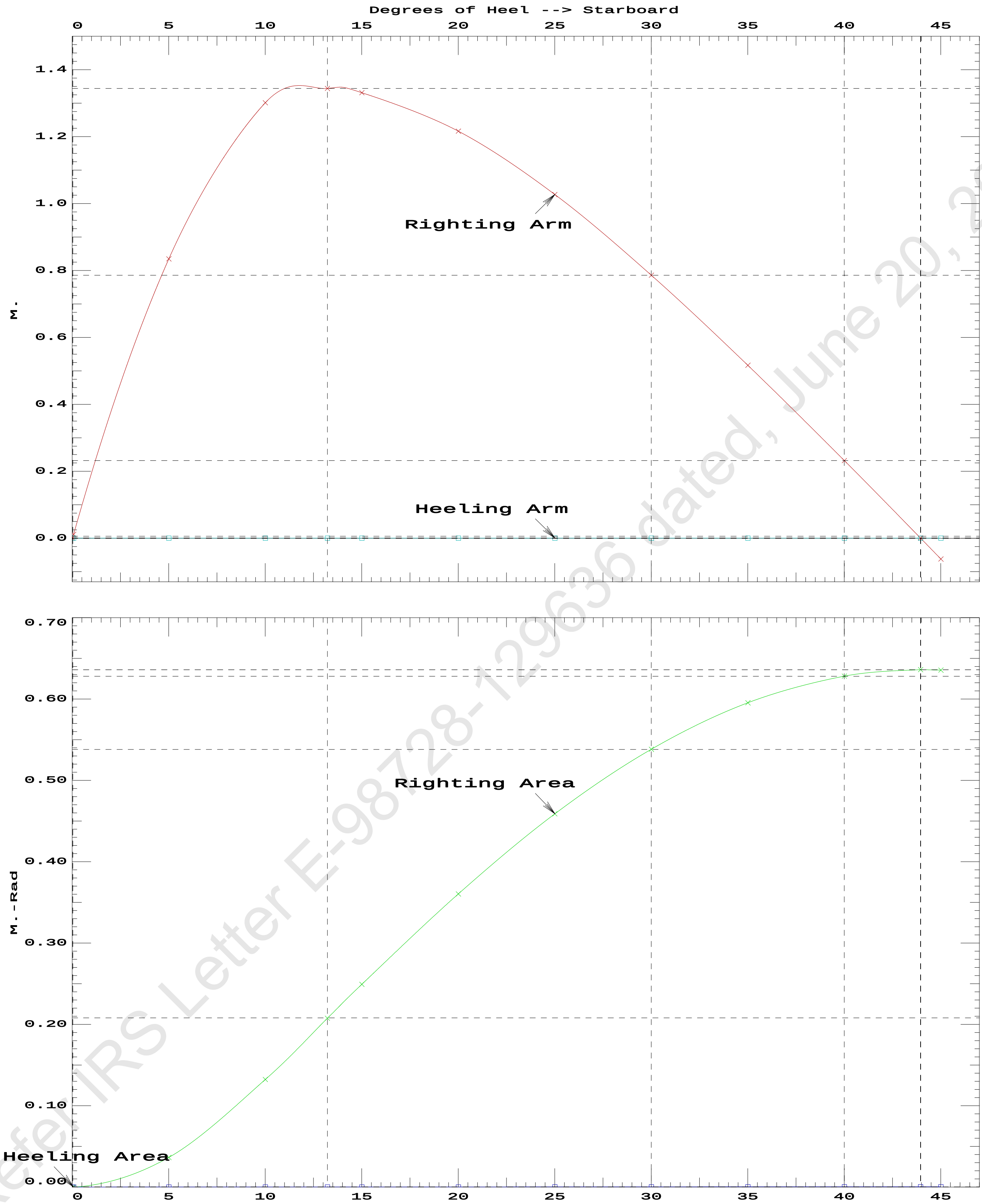
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1408.96 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	43.88 P

27-01-20 14:19:11 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONTOON



LIGHTSHIP PARTICULARS

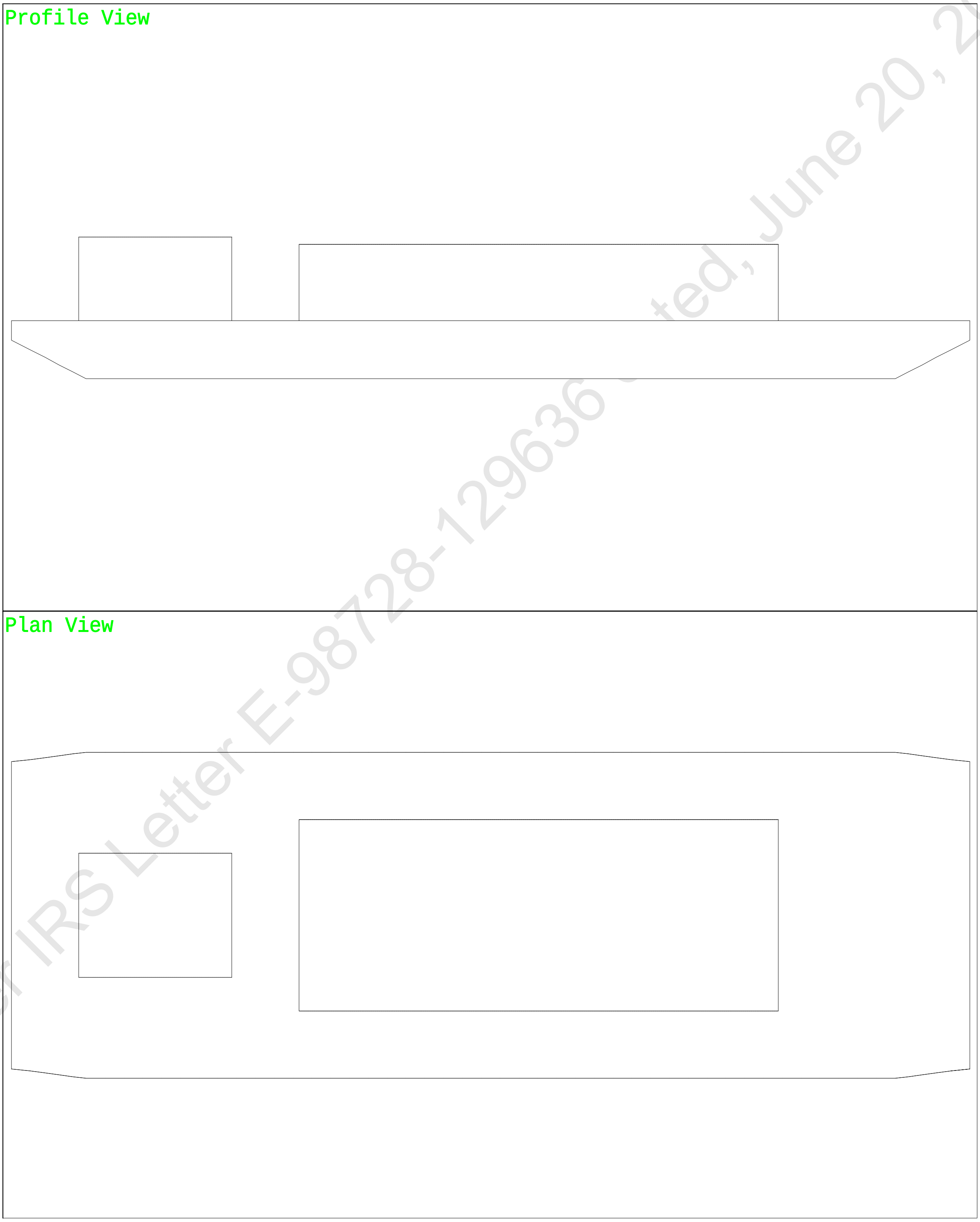
Lightship Weight	=	389.39	T
VCG	=	3.000	M above Baseline
LCG	=	25.000	M fwd of AP (Fr.0)
TCG	=	0.000	M off centerline

(Reference: Lightship particulars are taken from the approved draft survey report.)

03-01-20 15:19:17 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

WINDAGE AREA CALCULATION AT DIFFERENT DRAFTS



03-01-20 15:19:17 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

LATERAL PLANE STATUS
USK draft: 0.600 @ 50.00f, 0.600 @ 25.00f, 0.600 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	26.1	25.000f	-0.297	117.2	25.000f	1.236
CRANE.C				34.4	7.500f	4.604
DECKCARGO.C				98.0	27.500f	4.412
Total Lateral Plane->	26.1	25.000f	-0.297	249.6	23.571f	2.947

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.700 @ 50.00f, 0.700 @ 25.00f, 0.700 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	30.5	25.000f	-0.346	112.8	25.000f	1.183
CRANE.C				34.4	7.500f	4.504
DECKCARGO.C				98.0	27.500f	4.312
Total Lateral Plane->	30.5	25.000f	-0.346	245.2	23.545f	2.899

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.800 @ 50.00f, 0.800 @ 25.00f, 0.800 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	35.1	25.000f	-0.394	108.2	25.000f	1.130
CRANE.C				34.4	7.500f	4.404
DECKCARGO.C				98.0	27.500f	4.212
Total Lateral Plane->	35.1	25.000f	-0.394	240.6	23.518f	2.853

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.900 @ 50.00f, 0.900 @ 25.00f, 0.900 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	39.6	25.000f	-0.443	103.7	25.000f	1.078
CRANE.C				34.4	7.500f	4.304
DECKCARGO.C				98.0	27.500f	4.112
Total Lateral Plane->	39.6	25.000f	-0.443	236.0	23.489f	2.808

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.000 @ 50.00f, 1.000 @ 25.00f, 1.000 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	44.2	25.000f	-0.492	99.1	25.000f	1.025
CRANE.C				34.4	7.500f	4.204
DECKCARGO.C				98.0	27.500f	4.012
Total Lateral Plane->	44.2	25.000f	-0.492	231.5	23.459f	2.762

Distances in METERS.

```
03-01-20 15:19:17    Cybermarine Technologies PTE. Ltd.
GHS 16.68C                         PONTOON

                          LATERAL PLANE STATUS
           USK draft: 1.100 @ 50.00f, 1.100 @ 25.00f, 1.100 @ 0.00
                        Trim: zero,  Heel: zero
Part------------------------------LPA------LCP-----HCP-------LPA------LCP-----HCP
HULL                             48.8  25.000f  -0.541      94.5  25.000f   0.972
CRANE.C                                                     34.4   7.500f   4.104
DECKCARGO.C                                                 98.0  27.500f   3.912
  Total Lateral Plane->          48.8  25.000f  -0.541     226.9  23.428f   2.717
Distances in METERS.--------------------------------------------------------------

                          LATERAL PLANE STATUS
           USK draft: 1.200 @ 50.00f, 1.200 @ 25.00f, 1.200 @ 0.00
                        Trim: zero,  Heel: zero
Part------------------------------LPA------LCP-----HCP-------LPA------LCP-----HCP
HULL                             53.5  25.000f  -0.589      89.8  25.000f   0.921
CRANE.C                                                     34.4   7.500f   4.004
DECKCARGO.C                                                 98.0  27.500f   3.812
  Total Lateral Plane->          53.5  25.000f  -0.589     222.2  23.395f   2.673
Distances in METERS.--------------------------------------------------------------

                          LATERAL PLANE STATUS
           USK draft: 1.300 @ 50.00f, 1.300 @ 25.00f, 1.300 @ 0.00
                        Trim: zero,  Heel: zero
Part------------------------------LPA------LCP-----HCP-------LPA------LCP-----HCP
HULL                             58.2  25.000f  -0.637      85.1  25.000f   0.869
CRANE.C                                                     34.4   7.500f   3.904
DECKCARGO.C                                                 98.0  27.500f   3.712
  Total Lateral Plane->          58.2  25.000f  -0.637     217.4  23.360f   2.630
Distances in METERS.--------------------------------------------------------------

                          LATERAL PLANE STATUS
           USK draft: 1.400 @ 50.00f, 1.400 @ 25.00f, 1.400 @ 0.00
                        Trim: zero,  Heel: zero
Part------------------------------LPA------LCP-----HCP-------LPA------LCP-----HCP
HULL                             62.9  25.000f  -0.685      80.3  25.000f   0.817
CRANE.C                                                     34.4   7.500f   3.804
DECKCARGO.C                                                 98.0  27.500f   3.612
  Total Lateral Plane->          62.9  25.000f  -0.685     212.7  23.324f   2.588
Distances in METERS.--------------------------------------------------------------

                          LATERAL PLANE STATUS
           USK draft: 1.500 @ 50.00f, 1.500 @ 25.00f, 1.500 @ 0.00
                        Trim: zero,  Heel: zero
Part------------------------------LPA------LCP-----HCP-------LPA------LCP-----HCP
HULL                             67.7  25.000f  -0.733      75.6  25.000f   0.765
CRANE.C                                                     34.4   7.500f   3.704
DECKCARGO.C                                                 98.0  27.500f   3.512
  Total Lateral Plane->          67.7  25.000f  -0.733     208.0  23.285f   2.545
Distances in METERS.--------------------------------------------------------------
```

03-01-20 15:19:17 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

LATERAL PLANE STATUS
USK draft: 1.600 @ 50.00f, 1.600 @ 25.00f, 1.600 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	72.6	25.000f	-0.780	70.7	25.000f	0.714
CRANE.C				34.4	7.500f	3.604
DECKCARGO.C				98.0	27.500f	3.412
Total Lateral Plane->	72.6	25.000f	-0.780	203.1	23.244f	2.505

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.700 @ 50.00f, 1.700 @ 25.00f, 1.700 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	77.5	25.000f	-0.828	65.8	25.000f	0.664
CRANE.C				34.4	7.500f	3.504
DECKCARGO.C				98.0	27.500f	3.312
Total Lateral Plane->	77.5	25.000f	-0.828	198.2	23.200f	2.466

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.800 @ 50.00f, 1.800 @ 25.00f, 1.800 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	82.4	25.000f	-0.876	60.9	25.000f	0.613
CRANE.C				34.4	7.500f	3.404
DECKCARGO.C				98.0	27.500f	3.212
Total Lateral Plane->	82.4	25.000f	-0.876	193.3	23.155f	2.427

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.900 @ 50.00f, 1.900 @ 25.00f, 1.900 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	87.3	25.000f	-0.924	56.0	25.000f	0.562
CRANE.C				34.4	7.500f	3.304
DECKCARGO.C				98.0	27.500f	3.112
Total Lateral Plane->	87.3	25.000f	-0.924	188.4	23.107f	2.389

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 2.000 @ 50.00f, 2.000 @ 25.00f, 2.000 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	92.2	25.000f	-0.972	51.1	25.000f	0.511
CRANE.C				34.4	7.500f	3.204
DECKCARGO.C				98.0	27.500f	3.012
Total Lateral Plane->	92.2	25.000f	-0.972	183.5	23.056f	2.352

Distances in METERS.

03-01-20 15:19:17 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON**

LATERAL PLANE STATUS
USK draft: 2.100 @ 50.00f, 2.100 @ 25.00f, 2.100 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	97.2	25.000f	-1.019	46.1	25.000f	0.461
CRANE.C				34.4	7.500f	3.104
DECKCARGO.C				98.0	27.500f	2.912
Total Lateral Plane->	97.2	25.000f	-1.019	178.5	23.002f	2.316

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	102.8	25.000f	-1.072	40.5	25.000f	0.405
CRANE.C				34.4	7.500f	2.992
DECKCARGO.C				98.0	27.500f	2.800
Total Lateral Plane->	102.8	25.000f	-1.072	172.9	22.937f	2.277

Distances in METERS.

IRCLASS
Indian Register of Shipping
CIN: U61100MH1975NPL018244
52 A, Adi Shankaracharya Marg, Opp. Powai Lake,
Powai, Mumbai - 400 072, India
Phone : +91 22 30519400 / +91 22 7119 9400
Fax : +91 22 2570 3611
E-mail : ho@irclass.org ★ Website : www.irclass.org

Our ref: E-98298-129011 27-January-2020

Company : Cybermarine Knowledge System

Dear Sir/Madam,

Your document (detailed below) has been reviewed.

Project details	
Project number	NC002438
Project name	Chisti Marine Engineering Works - CHISTI/010
Project Manager	Dhruvesh Singh
Project In-Charge	Veerendra P

Document Details	
Plan No.	CM20-V-102-0205
Plan Title	Draft Survey Report
Rev.No.	0
Upload.No.	1
Submitted Date	20-January-2020
Review Status	Approved
Parent Plan Title	

Notes

1. The review status Approved indicates:

   The following lightship particulars are approved:
   Lightship Displacement - 389.39 T
   Lightship L.C.G - 25.000 m from AP (Fr.0)
   Lightship V.C.G - 3.000 m above BL
   Lightship T.C.G - 0.000 m off-centerline

   Trim and Stability Booklet prepared based on the approved Lightship particulars should be submitted for approval.

   Copy of the approved draft survey report should form part of the booklet.

Thanking you,
Yours Sincerely,
For INDIAN REGISTER OF SHIPPING

Ramiz Ismail
Surveyor

This is an electronically published document and does not require signature.

DRAFT SURVEY REPORT
OF
KB-23
(YARD NO. CHISTI/010)

APPROVED
DATE: 27 Jan 2020
INDIAN REGISTER OF SHIPPING
MUMBAI

SEE LETTER E-98298-129011

LIGHTSHIP PARTICULARS:

Lightship Displacement	-	389.39 T
Lightship L.C.G	-	25.000 m from AP (Fr.0)
Lightship V.C.G	-	3.000 m above BL
Lightship T.C.G	-	0.000 m off-centerline

Rev.	Description	Date	Prepared	Review

Issued for : **Approval**

Client: **K.B. SHIPPING & CO. (JAMNAGAR)** Builder: **CHISTI MARINE ENGINEERING WORKS**	Project: **PONTOON (YARD NO. CHISTI/010)**

Doc #				Document Title:	Prepared	**OM**
CM20-V-102-0205				**DRAFT SURVEY REPORT**	Checked	**MG**
Scale	Size	Sheet	Date		Review 1	**SN**
NTS	A4	1 of 15	20-01-2020		Review 2	

Cybermarine
www.cybermarine.net info@cybermarine.net

10, Bukit Batok Crescent,
#12-01, The Spire Building,
Singapore-658079.

501, Technocity IT Premises Co-op. Soc. Ltd,
Plot No. X-5/3, Mahape,
Maharashtra- 400710, India.

22503, Katy Freeway #7,
Katy, Texas,
USA 77450.

GENERAL PARTICULARS

Name of the Vessel	KB-23
Client	K.B.SHIPPING & CO., JAMNAGAR
Builder	CHISTI MARINE ENGINEERING WORKS
Yard No.	CHISTI/010
Type of Vessel	PONTOON
Class Notation	卐SU-PONTOON, “INDIAN COASTAL SERVICE” DESCRIPTIVE NOTE “FOR CRANE OPERATION” “DECK STRENGTHENED FOR {10T/M^2}

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	17.000 M
Depth (Mld.)	3.000 M
Draft (Mld.)	2.200 M

References

AP	at Fr.0
FP	at Fr.100
Midship	at Fr.50

Frame spacing

Throughout	500 mm

DRAFT SURVEY REPORT : KB-23 (Yard No. CHISTI/010)

Location of Test	Sikka,Jamnagar
Date	20-Jan-20
Weather	Calm
Wind	Light
Specific Gravity	1.025
Restraining Line	Loose
Witnessed by	Mr.Amit Garg (KB Shipping Rep.)
	Mr.Chiranjeev Malviya (IRS Surveyor)

Drafts :

(w.r.t keel line)

Location		Draft Port (m)	Draft Stbd (m)	Mean Draft (m)
Fwd	(Fr.92)	0.600	0.600	0.600
Mid	(Fr.50)	0.600	0.600	0.600
Aft	(Fr.8)	0.600	0.600	0.600

Hydrostatics

Correction for Drafts

To correct for observed drafts at marks and obtain Drafts at perps.

LBP	-	50.000	m

Observed Drafts (wrt keel line)

Fwd	-	0.600	m
Aft	-	0.600	m

Correction to Drafts (wrt Keel line)

Location	At Fwd (Fr.92)	At Aft (Fr.8)	
Draft at Marks	0.600	0.600	m
Trim at Marks		0.000	m
Distance of Marks from PP	4.000	4.000	m
Distance between Marks		42.000	m
Draft correction factor		0.000	
Correction at PP	0.000	0.000	
Corrected Draft at PP	0.600	0.600	
Trim		0.000	m
Mean Draft		0.600	m
From Hydrostatics against above Mean Draft and Trim			
Displacement at density(T/M^3)=1.025 T/m^3		389.39	T
LCB		25.000	m
KMt		34.428	m
LCG		25.000	m
TCG		0.000	m

NOTE

1. Hydrostatic Particulars computed by Computer Program at Sp.Gr. 1.025
2. LCB, LCG are defined from AP, Fwd of AP is +ve and Aft of AP is -ve
3. Trim by Fwd is -ve and Trim by Aft is +ve.
4. KB is measured from Base Line.
5. Draft is measured from Keel Line

```
20-01-20 17:31:02    Cybermarine Technologies PTE. Ltd.
GHS 16.68C                     PONTOON

                               HYDROSTATIC PROPERTIES
                       No Trim, No Heel, Fixed VCG = 0.000

 Draft@  Displacement   Buoyancy-Ctr.  Weight/         Moment/
 25.000f---Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----KML-----KMT
  0.600       389.39  25.000f  0.295     6.93  25.000f   22.82  293.08  34.428
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                             Trim is per 50.00m.
 Draft is from USK.
```

Weights to go off Board

No	Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
-	-	0.00	0.000	0.00	0.000	0.00	0.000	0.00
	TOTAL	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Weights to go on Board

No	Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
-	-	0.000	0.000	0.00	0.000	0.00	0.000	0.00
	TOTAL	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Lightship Particulars

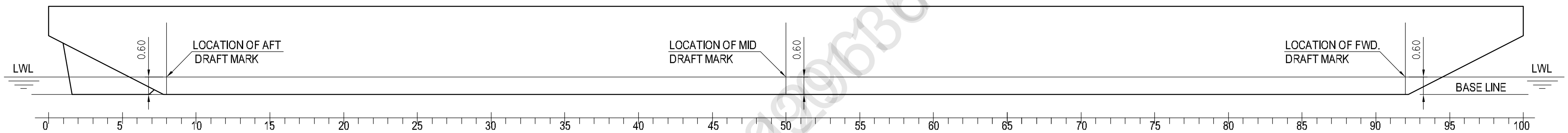
Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
Particulars as Draft Survey	389.39	25.000	9734.750	3.000	1168.170	0.000	0.000
Weights to go Off board	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Weights to go on board	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Lightship Particulars	389.39	25.000	9734.75	3.000	1168.17	0.000	0.00

Lightship Particulars

Displacement	**389.39**	**T**
LCG	**25.000**	**m From AP**
TCG	**0.000**	**m Off-Centreline**
VCG	**3.000**	**m above BL**

NOTE
1. Hydrostatic Particulars computed by Computer Program at Sp.Gr. 1.025
2. LCB, LCG are defined from AP, Fwd of AP is +ve and Aft of AP is -ve
3. Trim by Fwd is -ve and Trim by Aft is +ve.
4. KB is measured from Base Line.
5. Draft is measured from Keel Line

DRAFT READING



FRAME SPACING 500mm THROUGHOUT

PROFILE

LIGHT-SHIP SURVEY PROCEDURE AND REPORTING FORMAT
Part I

Pre requisites for Conducting Light-Ship Survey
(To be provided /confirmed by Builder/Owner/Operator/Consultant prior commencement of Light-Ship Survey)

Name Of Builder/owner rep conducting trial	CHISI MARINE ENGINEERING WORKS
Hull Number/Vessel name	CHISTI-010/ K.B 23
Type of Vessel (Cargo/passenger)	PONTOON
Location of Vessel	SIKKA, JAMNAGAR
Date of Proposed Light-Ship Survey	20.01.2020
Draft of Vessel and trim	0.600 Meter, Trim- zero
Depth of Water	4.0 Meter
Anticipated tide on date of survey	5.0M, 20/JAN/2020
Weather prediction for proposed date	CALM
Availability of following reports from builder/Owner/Owners rep	
Copy of Approved Inclining experiment report	☐ YES N/A ☐ NO
Details of Periodical Draft Survey report/any Draft Survey report conducted after the approval of Final TSB.	☐ YES N/A ☐ NO
Declaration of weights to go on	☐ YES N/A ☐ NO
Declaration of weights to go off -tank contents - Loose items	☐ YES N/A ☐ NO

Facilities/Arrangement for conduct of Light-Ship Survey	
Availability of boat to take draft reading	☐ YES ☐ NO YES
Availability of draft mark plan/approved document like Prelim TSB/Final TSB for confirming location of draft marks	☐ YES ☐ NO YES
Safety Requirements- lighting, gas freeing, accessibility of areas requiring inspection as part of Survey	☐ YES ☐ NO YES
Availability of sounding tapes, hydrometer etc	☐ YES ☐ NO YES
Closing/isolating cross connected tanks	☐ YES ☐ NO N/A
Removal of gangway during draft reading/experiment	☐ YES ☐ NO YES
Permanent ballast used ? ; if any and its details	☐ YES ☐ NO N/A
If vessel has any list, reason for the list is established.	☐ YES ☐ NO NO
A trial is conducted to see whether the vessel's draft corresponding to the loading condition is matching with the approved hydrostatics.	☐ YES ☐ NO N/A

NOTE : The responsibility of the IRS Surveyor is to verify that the survey is conducted according to accepted procedures and that all basic measurements and data are correctly taken and recorded. The responsibility for making preparation, conducting lightship survey and preparing report rest with owner/operator.

Confirm Readiness for test

AMIT GARG   Amit Garg

Name & Signature of Shipyard /Owners rep

Sheet for recording Weights to Go-on, Go-off, relocated and Tank Soundings

Weights To Go-on						
Sl No.	**Weight Description**	**Weight (t)**	**L.C.G (m)**	**V.C.G (m)**	**T.C.G (m)**	**Surveyor Verification**
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----

Weights To Go-off						
Sl No.	**Weight Description**	**Weight (t)**	**L.C.G (m)**	**V.C.G (m)**	**T.C.G (m)**	**Surveyor Verification**
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
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						☐
						☐
						☐
						☐
						☐

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----

Tank Soundings				
Sl No.	**Tank Name**	**Sounding (cm /m)**	**Density** T/m3	**Surveyor Endorsement**
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐
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				☐
				☐
				☐
				☐

Note: Tank name should be mentioned as per the approved TSB / Capacity Plan.

Weights To Be Relocated									
Sl No.	Weight Description	Weight (t)	L.C.G (m)		V.C.G (m)		T.C.G (m)		Surveyor Verification
			Initial	Final	Initial	Final	Initial	Final	
									☐
									☐
									☐
									☐
									☐
									☐
									☐
									☐
									☐
									☐
									☐
									☐

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----

LIGHTSHIP SURVEY PROCEDURE AND REPORTING FORMAT
Part II

REPORTING FORM FOR LIGHTSHIP SURVEY

Attending Surveyor should witness Light-Ship Survey only if the Part-I submitted by Owner/Builder/Operator/Consultant is found satisfactory

General Particulars

Name of Vessel	K.B 23
Builder/ Owner	CHISTI MARINE ENGINEERING WORKS
Yard Number/ IR Number	CHISTI-010
Type of Vessel — Passenger	☐
Non Passenger	☐ PONTOON
Date(dd/mm/yyyy);Local Time of inspection/test	20 TH JAN 2020, 1100-1200 HRS
Wind / Weather condition	1 KNOTS, CALM
Location of Test	SIKKA, JAMNAGAR
Depth of water where vessel is moored, m	4.5 METER
Method of reading Draft(By Boat-Y/N)	YES

Location of Draft Marks	Frame Number	Draft Reading (Port side),m	Draft Reading (STBD Side),m
Aft	8	0.600 METER	0.600 METER
Midship	50	0.600 METER	0.600 METER
Forward	92	0.600 METER	0.600 METER
Note : *Recommended to attach Photographs of draft; as record*			

Density of Water

Results of 3 samples from Forward, Aft and Mid of the vessel		
Sample 1: (ford)	1.025	**t/m3**
Sample 2: (mid)	1.025	**t/m3**
Sample 3: (aft)	1.025	**t/m3**

Verification of weights to go on / go off /re-location	*SEE PART 1*
Note: *See Part 1 (the sheet for recording the above data) , verify and endorse.*	

Special Comments on Light Ship Survey, if any.
Draft Survey carried out satisfactorily.

Attending surveyor to forward both forms Part 1 and Part 2 together with completed attachments 1 and 2 to HO.

Name & designation of Attending Surveyor: CHIRANJEEV MALVIYA

SURVEYOR

20/01/2020

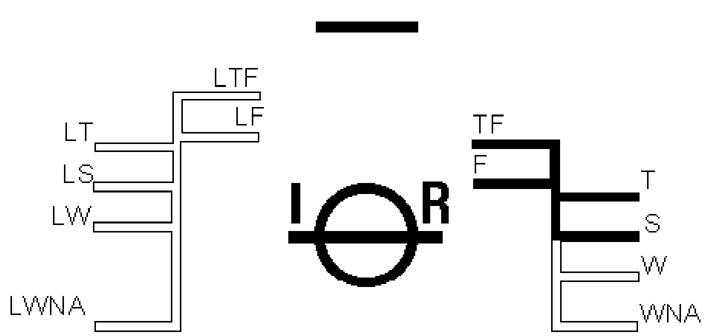
MEMORANDUM OF FREEBOARDS
09 January 2020

	Type B freeboard		
Chisti Marine Engineering Works - 010	Freeboard Length	:	48.00 m
K.B. - 23	Summer WL Extreme Draught - load line amidships	:	2.212 m

The upper edge of the deckline from which these freeboards are measured is 0 mm below top of * ~~Wood~~ / *steel Deck *at side / *~~continued to side~~. (*Delete words not applicable).

Freeboards from Deck Line			Seasonal, Fresh Water, or Timber allowances	
Tropical Fresh Water	715	**mm.**	95	**mm. above summer**
Fresh Water	761	**mm.**	49	**mm. above summer**
Tropical	764	**mm.**	46	**mm. above summer**
Summer	810	**mm.**	**Upper edge of line through centre**	
Winter	Not	**mm.**	Assigned	**mm. below summer**
Winter North Atlantic	Not	**mm.**	Assigned	**mm. below summer**
Timber Tropical fresh water	—	**mm.**	—	**mm. above timber summer**
Timber fresh water	—	**mm.**	—	**mm. above timber summer**
Timber Tropical	—	**m m.**	—	**mm. above timber summer**
Timber Summer	—	**mm.**	—	**mm. above centre of ring**
Timber Winter	—	**mm.**	—	**mm. below timber summer**
Timber Winter North Atlantic	—	**mm.**	—	**mm. below timber summer**

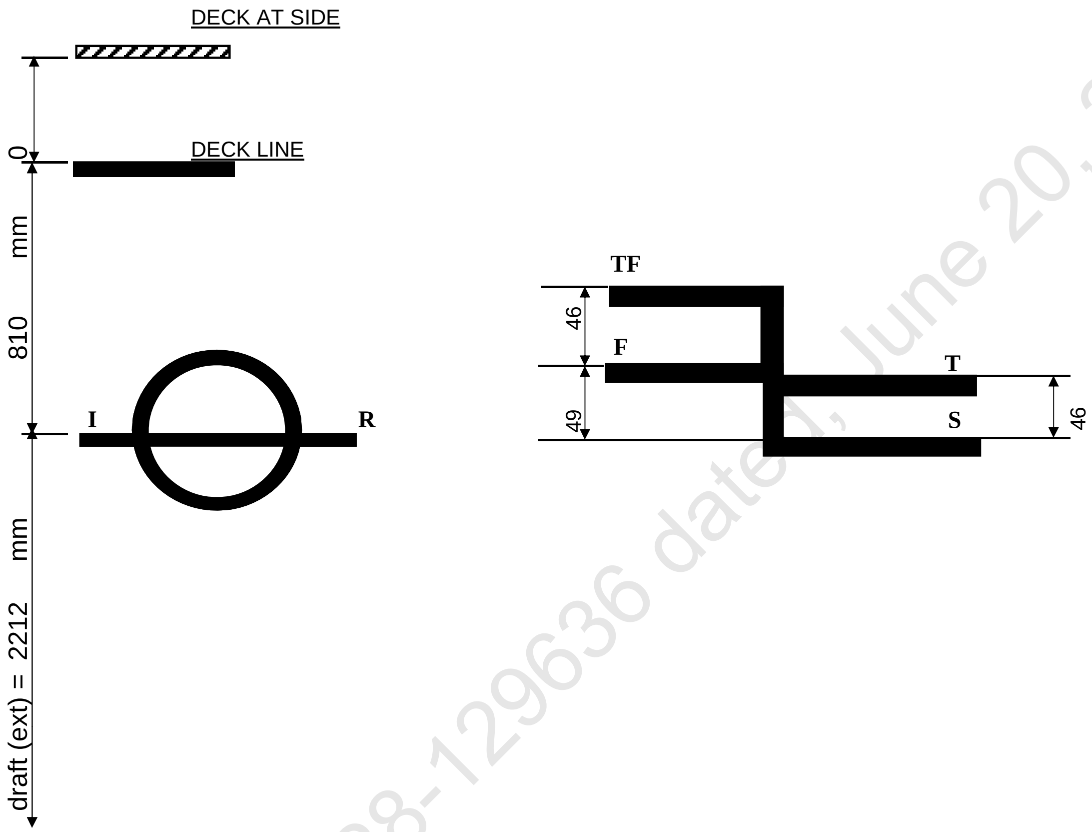
Valid only for unmanned operation while under tow along Indian Coast, during the course of which the vessel does not go more than 20 nautical miles from the nearest land, and may cross gulfs or similar features recognised by the Administration.



At Fr. 52

Form No. 4111 (07-83)

TABLE OF FREEBOARDS



LOADLINE	FREEBOARD (mm)	DRAFT (ext) KL (m)	DISPLACEMENT (ext) T	DEADWEIGHT T
SUMMER	810	2.212	1679.61	1290.22
FRESH WATER	761	2.261	1680.01	1290.62
TROPICAL	764	2.258	1719.41	1330.02
TROPICAL FRESH WATER	715	2.307	1718.84	1329.45

Deck Plate Thickness = 10 mm
Keel Plate Thickness = 12 mm